

Designing a Stablecoin Settlement Platform

for Existing Real-Time Payment Networks

A Reference Architecture Using Zelle as a Public Case Study

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Evidence disclaimer. Zelle is used only as a public case study. This paper does not describe or infer confidential Early Warning Services architecture, vendors, data stores, message contracts, controls, service levels, or launch mechanics. Public facts, documented technology behavior, inferred logical capabilities, proposed architecture, operational assumptions, and unknowns are explicitly distinguished.

Abstract

Existing real-time payment networks create a customer expectation of immediate, familiar money movement, while stablecoin settlement introduces asynchronous issuers, custodians, signing systems, blockchains, foreign-exchange providers, and payout rails. This paper asks how a bank-distributed network can add that settlement capability without turning its customer experience into a cryptocurrency wallet or treating a blockchain receipt as complete financial truth.

The proposed answer is a regulated control plane centered on an internal double-entry ledger and a durable, evidence-gated payment state machine. Participant institutions retain authentication, account posting, and customer support. The control plane accepts idempotent payment intents; coordinates identity, policy, fraud, sanctions, quotes, routing, and liquidity; and delegates chain-native work through explicit settlement, custody, node, issuer, foreign-exchange, and payout adapters. Transactional outbox and inbox patterns, independent observation, reconciliation, and compensating entries construct exactly-once business effects across systems that cannot share one transaction. Blockchain finality, legal settlement finality, customer-visible completion, and accounting finality remain separate policy decisions.

Zelle provides a public case study because its bank distribution, scale, and announced international stablecoin direction make these tensions concrete. The paper neither knows nor claims its private implementation. Public facts, documented technology behavior, inferred logical roles, proposed design, operational assumptions, and unresolved questions are kept distinct.

The recommended pilot uses one approved regulated asset, one approved public network, one corridor, bounded participants, prefunded institutional liquidity, no consumer wallet, and no bridge. Java and Spring fit the enterprise control plane; smart contracts, node clients, and qualified signing remain in their strongest supported ecosystems. Stablecoin settlement is therefore an additional governed rail, not the operational ledger of record.

Keywords. stablecoin settlement; real-time payments; double-entry ledger; payment state machine; finality; Java; reconciliation; Zelle.

Visual evidence statement. Figures use labels, badges, borders, and lane names to distinguish public facts, documented technology behavior, inferred logical capabilities, proposed architecture, assumptions, and unknowns. Figure-specific captions state only the takeaway; no proposed box represents disclosed Early Warning Services topology.

How to read this paper. Readers seeking the recommendation and boundary should read sections 1, 4 and 19. Payment and reliability specialists should continue through sections 6, 7, 11 and 12. Technology decision makers should use sections 15 to 17, then consult the detailed decision appendices. Section H summarizes the evidence method; the complete claim ledger is a separately generated companion artifact in the release package.

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1. Introduction: a settlement problem, not a wallet feature

Decision resolved. Preserve the participant institution as the customer channel and place stablecoin settlement behind a new institutional control plane. **PROPOSED**

Real-time payment products compress the customer-visible portion of money movement into seconds. Stablecoin networks can move tokenized value continuously. Neither fact makes the combined system atomic. A participant bank, network operator, issuer, custodian, signing service, blockchain, foreign-exchange (FX) provider, and destination payout rail each owns a different state, clock, failure domain, and legal obligation. The engineering problem is therefore to coordinate accountable financial effects under uncertainty—not simply to call a token contract.

Zelle makes the problem concrete without supplying a private implementation blueprint. **PUBLIC** EWS reports that Zelle is owned and operated by EWS, reached more than 2,300 financial institutions, and processed more than \$1.2 trillion across 4.2 billion transactions during 2025 (Early Warning Services 2026b; Early Warning Services 2026c). **PUBLIC** Consumers primarily use Zelle inside a participating bank or credit union experience and address recipients through an enrolled email address or U.S. mobile number (Zelle 2026c; Early Warning Services 2026d). Those facts establish a participant-distributed channel; they do not establish service topology, databases, contracts, queues, or recovery objectives.

Two announcements establish direction, with an important boundary. On 2025-10-24 EWS announced an initiative to use stablecoins for international Zelle money movement (Early Warning Services 2025b). On 2026-06-11 EWS named India as the first announced international destination and separately announced ZelleUSD for possible future capabilities in other markets (Early Warning Services 2026a). **PUBLIC** The latter announcement does not state that the India route uses ZelleUSD. It also does not disclose the issuer, authorization, reserves, redemption, network, contracts, custody, keys, controls, partners, or operating date of that asset.

1.1. Method and evidence boundary

Material statements use one evidence class: **PUBLIC** validated public fact, **CODE** official source-code behavior, **DOC** official documentation, **INFERRED** logical decomposition, **PROPOSED** target architecture, **ASSUMPTION** operating posture to approve, or **UNKNOWN** fact requiring validation. Section H gives the exact mapping and summary; the complete claim ledger is delivered as a separate release companion. Technology documentation proves what a technology says or does, not that EWS deploys it. Proposed components are recommendations, not inferred private boxes.

The method is obligation-first. The paper identifies customer, participant, accounting, legal, security, and operating obligations; assigns authority for each; defines evidence-gated transitions; then evaluates implementation choices. This ordering prevents a familiar technology from becoming the architecture by default.

1.2. Consumer wallet versus invisible settlement

The invisible rail is not “crypto hidden from the customer.” It is a deliberate allocation of duties: the participant remains accountable for its channel and account; the platform remains accountable for payment state, internal obligations, policy evidence, recovery, and participant truth; external providers remain accountable within explicit contracts and trust boundaries.

Table 1: The paper selects invisible institutional settlement for the pilot; a visible wallet is a different product and risk decision.

Dimension	Consumer-wallet framing	Invisible institutional-settlement framing
Customer instrument	Customer holds or controls tokens and addresses	Customer sees the institution's existing account and payment contract
Key and gas burden	Product must explain custody, recovery, fees, and transaction authorization	Qualified institutional custody and treasury absorb chain mechanics
Completion promise	Often coupled to chain status	Defined independently through product, payout, legal, and accounting gates
Support and disputes	New wallet and irreversible-transfer operating model	Existing participant support retained, with route-specific case evidence
Pilot value	Tests a new consumer proposition	Tests settlement economics, controls, recovery, and corridor operations

2. Problem context and obligation map

Decision resolved. Treat international stablecoin settlement as an additional set of systems of record and legal actors, not as a replacement for current payment obligations. **PROPOSED**

Public Zelle material shows participant institutions, enrolled aliases, a network role, support responsibilities, risk controls, and domestic settlement options. An EWS product brief depicts sending and receiving institutions settling through ACH or RTP, but does not prove that every payment follows one route (Early Warning Services 2025c). The RTP network describes final interbank settlement, while FedNow describes continuous processing and final settlement for eligible institutions (The Clearing House 2026; Federal Reserve Financial Services 2026a; Federal Reserve Financial Services 2026b). Nacha permits ACH reversals only for defined conditions and timeframes (Nacha 2026). These distinct behaviors already show why “complete” cannot be one universal status.

As reviewed on 2026-07-14, Zelle's public material describes an institution-mediated, bank-distributed network: consumers and eligible small businesses use participating institutions' experiences; institutions may work with named processor partners or reseller channels, or contact EWS; and corporate or government disbursers are directed to a treasury bank that receives the public flow's payment file (Zelle 2026d; Zelle 2026b; Zelle 2026f). No general-purpose EWS/Zelle developer API for arbitrary independent applications was identified in the reviewed public documentation; Jack Henry, IBM, and U.S. Bank instead document participant-side or treasury-bank products, which demonstrate possible integration patterns but neither rule out private or restricted EWS interfaces nor disclose EWS's private service topology, contracts, systems of record, or operating thresholds (Jack Henry 2021b; IBM 2026a; U.S. Bank 2026). **PUBLIC INFERRED**

INFERRED A logical domestic path includes a customer channel, participant authentication and account posting, alias resolution, network coordination, fraud/risk decisions, a domestic settlement rail, notification, and support. This decomposition explains obligations; it is not a claim that these are independent deployed services. A stablecoin path adds issuer and redemption rights, custody and signing authority, network and node evidence, transaction fees, contract controls, chain analytics, FX, treasury inventory, liquidity, a destination payout provider, and jurisdiction-specific compliance.

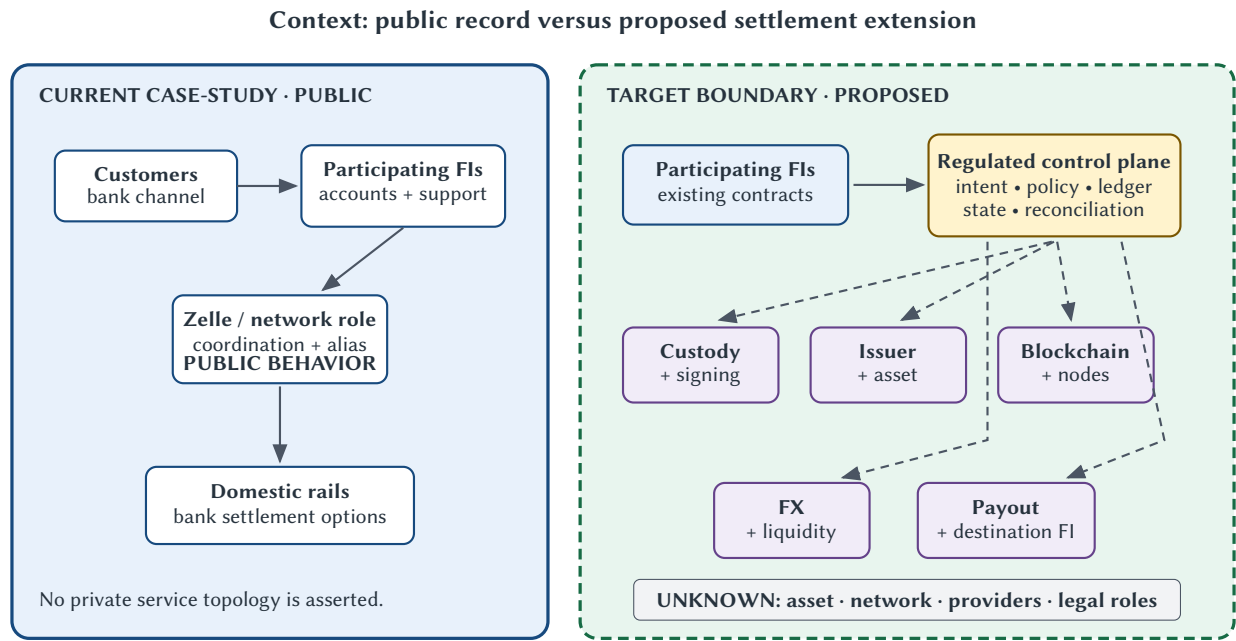


Figure 1: Public case-study context and the proposed target boundary. The PUBLIC region records only documented roles; all internal service boxes are explicitly PROPOSED.

Table 2. Role and obligation map. External evidence must be translated into platform state, never accepted as unqualified business truth.

System or role	Owner class	Authoritative evidence	Principal exposure	Evidence
Participant chan- nel/account	Financial insti- tution	Authenticated request, account posting, partici- pant case record	Unauthorized payment, insufficient funds, inconsis- tent customer status	PUBLIC
Alias/network coordination	Network role	Enrollment/recipient resolution and network operation record	Misrouting, stale enroll- ment, duplicate intent	PUBLIC / INFERRED
Domestic settle- ment rail	Rail and mem- ber banks	Rail acknowledgement and settlement record	Cutoff, return/reversal constraints, participant position	PUBLIC
Control plane	Proposed opera- tor	Intent, policy decision, workflow transition, ledger journal	Premature completion, duplicate effect, control bypass	PROPOSED
Stablecoin issuer	External regu- lated entity	Terms, reserve/redemp- tion records, contract events	Depeg, freeze, redemption denial, insolvency, upgrade	UN- KNOWN until se- lected

System or role	Owner class	Authoritative evidence	Principal exposure	Evidence
Custody and signer	Approved institutional provider/owner; legal classification separate	Policy decision, signed bytes, key audit, transaction identifier	Key compromise, unauthorized or unavailable signing	PROPOSED boundary
Blockchain and nodes	Net-work/provider	Independently observed transaction, receipt, block and finality evidence	Reorganization, censorship, congestion, inconsistent nodes	DOC / selection unknown
FX and treasury	Approved providers/operator	Quote, fill, inventory, settlement and exposure records	Expiry, slippage, shortfall, counterparty concentration	PROPOSED
Destination payout	Payout provider/institution	Idempotent payout operation and recipient-credit evidence	Final chain success but failed or delayed local credit	PROPOSED
Compliance/case operations	Accountable legal and operations roles	Versioned policy decision, screening inputs, case disposition	False positive/negative, post-payment designation, hold handling	PUBLIC obligation / PROPOSED control

The domestic route and the stablecoin route can share intent, policy, ledger, state, notification, audit, and case semantics. They cannot share an assumption that all settlement providers behave alike. The stablecoin route adds non-atomic value movement and an observation problem: the submission call, provider callback, node view, issuer record, custody balance, and internal journal can disagree temporarily or permanently. The architecture must preserve that disagreement long enough to resolve it safely.

Consequence. Domestic payment availability must not depend on digital-asset services. Stablecoin-provider failure disables or degrades only routes that need that provider. Existing participant, fraud, support, and settlement duties remain in force.

3. Goals, non-goals, and design principles

3.1. Testable goals

Decision resolved. Define success as safe, recoverable, and reconcilable payment obligations with a familiar participant experience—not merely successful token transfer. **PROPOSED**

Table 3. Goals are measurable at a decision gate even when an approved numeric target remains a route-approval unknown.

ID	Goal	Verification signal or gate	Owner
G1	Preserve bank-owned authentication, account, support, and customer status contracts.	Contract tests show no wallet, key, gas, or chain vocabulary in the participant API; route failure does not disable domestic payment.	Product / participant API
G2	Make every accepted request idempotent.	Same key plus same canonical payload returns the same intent; same key plus conflicting payload is rejected; concurrency tests show one effect.	Payment intent
G3	Preserve double-entry integrity.	Every journal balances per asset/currency; no direct balance mutation; operation IDs are unique; corrections are compensating entries.	Ledger / accounting
G4	Gate external effects through durable state and policy evidence.	Each effect has a recorded operation, expected version, authorization, timeout, and inquiry path.	Workflow / risk
G5	Isolate asset, issuer, chain, custody, FX, and payout providers.	Provider-specific adapters expose capabilities and risk; kill switches contain a provider without corrupting shared state.	Integration / SRE
G6	Construct exactly-once business effects across at-least-once delivery.	Duplicate, late, reordered, and replayed events cannot duplicate a posting, signature, submission, payout, or notification.	Platform / service owners
G7	Reconcile internal obligations to independent external evidence.	Every external balance/mutation has owner, cadence, tolerance, break state, case workflow, and aging alert.	Reconciliation / treasury
G8	Separate four finalities.	Product/corridor policy independently records chain, legal, customer, and accounting decisions and their evidence.	Product / legal / accounting / chain risk
G9	Bound operational and financial exposure.	Limits, liquidity buffers, pause scopes, degraded modes, and drills pass approved pilot gates.	Treasury / operations / security
G10	Retain asset/network exit.	Contract versions and data model can stop a route, reconcile it, and introduce a replacement without rewriting participant contracts.	Architecture / product
G11	Produce auditable evidence and telemetry.	Every transition and operator action carries causation, policy/configuration versions, business operation ID, and tamper-evident audit linkage.	Security / observability
G12	Learn measurably in the pilot.	Approved success, stop, exposure, reconciliation, recovery, customer, cost, and reliability measures determine phase promotion.	Program governance

Speed and certainty are intentionally separate. The system may expose an earlier customer status when an approved party absorbs the interval risk, but that policy does not alter chain, legal, or accounting evidence. Likewise, adapters improve portability only while preserving provider-specific powers such as freeze, pause, upgrade, fee, and finality behavior.

3.2. Non-goals

The initial design does not:

- expose a consumer wallet, seed phrase, address book, gas token, or self-custody recovery process;
- require a customer to receive or hold a stablecoin;
- replace the existing domestic Zelle experience or make it depend on a blockchain;
- preselect ZelleUSD, another stablecoin, a chain, proprietary infrastructure, or Hyperledger Besu;
- build a bridge, new blockchain, or permissioned network as the default pilot;
- claim universal asset, chain, bank, or corridor interoperability;
- solve every jurisdiction, data-residency rule, disclosure, refund, tax, or legal-finality question;
- provide legal advice or describe the current private EWS implementation; or
- implement production software or smart contracts.

These are scope boundaries, not permanent prohibitions. They isolate the value of the first experiment: can a regulated operator move institutional settlement value through one selected route while keeping accounting, policy, recovery, and customer obligations coherent? A later phase may change a boundary only through evidence and a separately owned product/architecture decision.

3.3. Principles that constrain the design

Table 4. Design principles have concrete consequences and prevent named failure modes.

Principle	Consequence	Prevented failure	Applied in
Preserve participant contracts.	Add route/version fields behind compatible payment-intent semantics; keep customer status provider-neutral.	Chain details leak into every institution and make exit impractical.	\$3, \$6
Operational ledger is off-chain and double-entry.	Every reservation, fee, FX, pay-out, custody, and suspense obligation is journaled internally.	Token ownership is mistaken for customer or participant entitlement.	\$5
Separate intent, authorization, execution, observation, reconciliation.	Each stage has distinct evidence, owner, operation ID, and retry policy.	A timeout or callback silently repeats value movement.	\$4–8
External mutation is uncertain.	Timeouts enter inquiry/manual-review states; blind resubmission is prohibited.	Both original and retry succeed.	\$6, \$8

Principle	Consequence	Prevented failure	Applied in
Construct business-effect exactly once.	Combine idempotency, ACID, versions, outbox/inbox, inquiry, compensation, and reconciliation.	Delivery guarantees are overstated across systems.	§7
Four finalities never collapse.	Product policy records chain, legal, customer, and accounting gates separately.	A confirmation count becomes an unsupported legal or customer promise.	§8
Adapters expose material differences.	A stable interface includes capabilities, evidence semantics, error classification, and control powers.	Least-common-denominator abstraction hides freeze, reorg, or redemption risk.	§3–4
Minimize on-chain data.	Use opaque intent references; keep personal and policy data off-chain.	Irreversible disclosure, correlation, and deletion conflicts.	§9
Bound exposure by default.	Prefund, limit, allowlist, rate-limit, and pause by participant/corridor/asset/provider.	A novel failure scales before recovery is proven.	§8, §12
Manual action is governed software behavior.	Cases require roles, reason codes, four-eyes approval where needed, and immutable evidence.	Ad hoc repair bypasses segregation and audit.	§8–9
Design for issuer, asset, and chain exit.	Version contracts and persist neutral operations plus provider-specific evidence.	Concentration becomes technical lock-in.	§11–12
Use the strongest supported ecosystem.	Java owns enterprise control; Solidity/Rust/Go/native services own chain-specific work where appropriate.	Language preference chooses a weaker node, signer, or contract toolchain.	§10
Promote only evidence-backed claims.	Unknowns remain gates and operational assumptions remain labeled.	A plausible design is misreported as deployed fact.	all chapters

Consequence. The architecture accepts coordination and reconciliation work because that work makes obligation, uncertainty, and authority visible. Safety does not emerge from a fictional end-to-end transaction.

4. Reference architecture

Decision resolved. Place ledger and workflow authority inside a regulated control plane; treat chains and providers as external execution and evidence domains. **PROPOSED**

PROPOSED

The architecture has two interacting planes. The *control plane* accepts intent, makes policy and routing decisions, reserves funds, records obligations, authorizes effects, advances state, communicates participant truth, and reconciles. The *settlement plane* signs and submits transactions, moves token or fiat value, observes

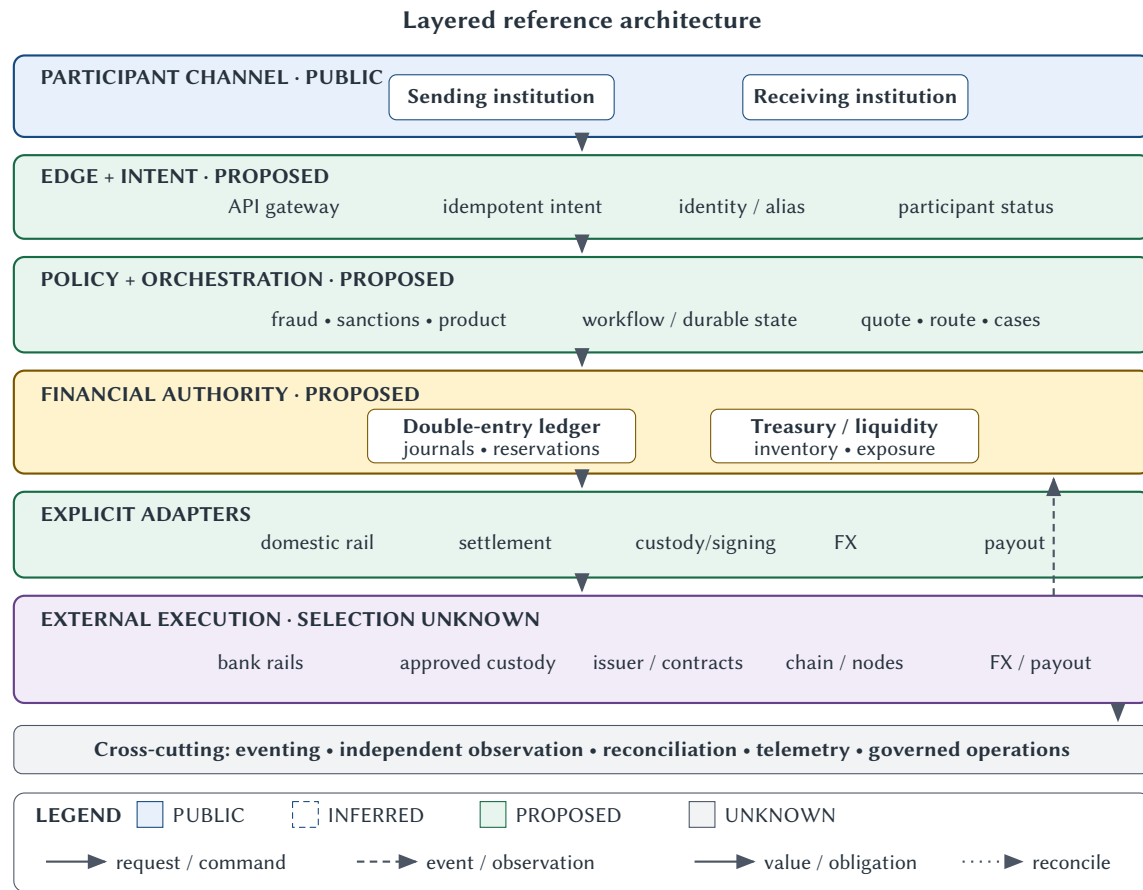


Figure 2: The layered reference architecture. The ledger and workflow are the operational center; external rails execute bounded effects and return evidence. All green components are proposed, not deployed EWS topology.

networks, obtains issuer/FX/payout evidence, and returns that evidence. A settlement provider never promotes business state directly; it produces an observation that the authoritative workflow evaluates against route and policy versions.

4.1. Layers and authority

Table 5. Architecture layers separate authority from execution.

Layer	Responsibility / owned state	Inputs and outputs	Prohibited responsibility	Evidence
Participant channel	Authentication, customer account, consent, presentation, support	Payment intent/status; account reservation/posting	Holding platform keys or deriving chain finality	PUBLIC context
Edge and intent	Contract version, participant identity, idempotency record, canonical request	Authenticated request; stable intent ID and receipt	Making settlement finality or accounting decisions	PROPOSED
Policy and case	Fraud, sanctions, product, limit, corridor, and manual disposition evidence	Versioned facts; decision and obligations	Editing ledger history or silently overriding signer policy	PROPOSED
Orchestration	Payment aggregate, transaction events, operation plan, timeouts	Commands and observations; durable states	Owning account balances or treating provider status as canonical	PROPOSED
Ledger and treasury	Journals, accounts, reservations, positions, fees, exposure	Authorized operation; balanced postings and balance version	Calling chains/payout providers inside a ledger transaction	PROPOSED
Adapters	Provider-specific command, inquiry, evidence normalization, capability version	Operation ID; provider request/observation	Collapsing freeze, reorg, issuer, or payout semantics into generic success	PROPOSED
Custody/signing	Key policy, approval, canonical-byte verification, signature audit	Authorized unsigned transaction; signed result/refusal	Accepting raw destination/amount outside approved intent	PROPOSED / vendor DOC
External rails	Token/fiat movement and provider records	Submission; receipts, balances, fills, credits	Declaring participant payment complete	EXTERNAL / selection unknown
Evidence/operations	Independent observation, reconciliation, telemetry, cases, controls	Internal/external records; breaks, alerts, evidence packages	Mutating source records to force agreement	PROPOSED

The ledger is single-writer by account partition, not necessarily one process or database instance. A partition owner serializes conflicting reservations and postings for its accounts; a journal spanning partitions requires a deterministic coordinator or a posting topology that keeps an atomic journal within one ownership boundary. The exact partition strategy is an implementation ADR after workload and accounting requirements are known. Read models may scale independently but cannot authorize money movement from stale balances.

Domestic and stablecoin routes share edge, intent, policy, ledger, workflow, messaging, audit, and participant status. The domestic adapter does not call the digital-asset plane. Therefore a signer, node, chain, issuer, or custody outage can disable affected routes while domestic service remains available, subject to its own dependencies.

4.2. Trust and deployment boundaries

1. **Participant boundary.** Mutual authentication, participant authorization, request signing, quotas, contract versions, and replay protection apply before intent acceptance.
2. **Regulated operator boundary.** Policy, ledger, workflow, configuration, audit, and case services run under controlled identities. Database ownership is service-scoped; shared-table integration is prohibited.
3. **Signing boundary.** Private keys remain within an approved HSM, multiparty-computation (MPC), or institutional custody/signing service. The signer independently verifies canonical transaction fields against an authorized intent. Whether a selected provider has any particular legal custody classification is a separate route-specific legal determination.
4. **Observation boundary.** Submission and observation are separable. A managed RPC endpoint may submit, but an independent node/provider and custody/issuer records are used to detect inconsistent evidence.
5. **Provider boundary.** Each issuer, FX, payout, chain, analytics, and custody provider has distinct identity, availability, data, legal, and exit obligations.
6. **Human authority boundary.** Operators act through case workflows with least privilege, reason codes, approval policy, immutable audit, and time-bounded elevation.

4.3. Control plane versus chain-centric design

A chain-centric design makes token transfer the primary state and attempts to reconstruct product obligations around it. That model cannot naturally represent participant reservations, fiat accounts, quote expiry, fees, pending payout, post-payment cases, customer status, or accounting close. The proposed model records those obligations before and after the external transaction. Chain evidence remains indispensable, but only for the portion of truth that the chain owns.

Provider adapters are likewise not generic “send money” interfaces. A common operation envelope supplies business operation ID, intent version, amount/asset, route, deadline, and authorization. Provider-specific capabilities report whether inquiry, replacement, nonce control, freeze, burn/mint, memo, finality proof, or payout cancellation exists. Portability comes from stable platform semantics plus explicit capability branches, not from hiding differences.

Consequence. Every later flow maps to one control-plane authority, one external mutation operation, and one evidence path. No callback, node, contract, or provider duplicates the business source of truth.

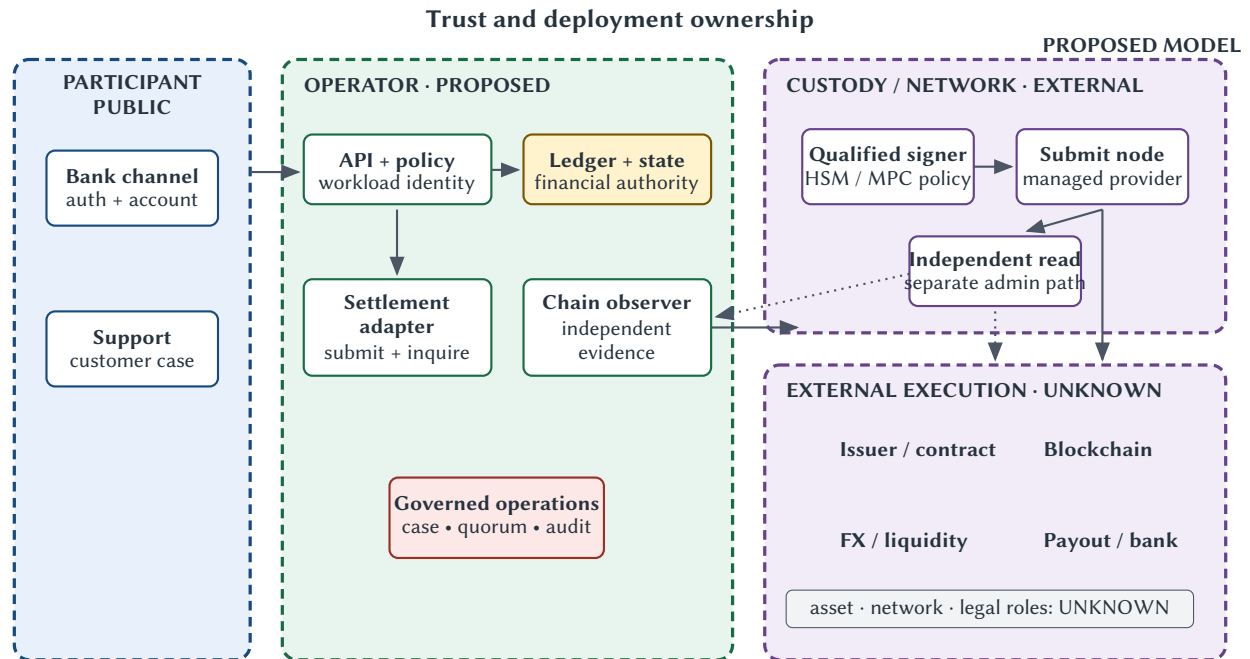


Figure 3: Deployment and trust ownership. No provider callback crosses directly into ledger mutation; custody and independent observation form separate authorization and evidence paths.

5. Component contracts and ownership

Decision resolved. Give each component one state authority, each financial effect one business operation ID, and each boundary an explicit ambiguity contract. **PROPOSED**

The proposed participant gateway is a stable logical contract, not an assumed EWS API. An institution may connect through a direct relationship, processor partner or reseller, core or digital-banking platform, or treasury-bank product; the adapter preserves those differences behind versioned capabilities and evidence rather than pretending they share one protocol (Zelle 2026d; Zelle 2026b; Zelle 2026f; Jack Henry 2021a).

An accepted participant request receives an immutable payment-intent ID. Every effect below it receives a business operation ID scoped by intent, operation type, and attempt policy—for example PAY-123/CHAIN-SUBMIT/1. The operation ID is stable across transport retries. A new attempt number is permitted only after route-specific evidence proves retry safety or an operator-authorized strategy deliberately creates a distinct replacement; absence is one possible input, not the rule itself.

Within the proposed boundary, the participant gateway authenticates and authorizes the institution, canonicalizes and deduplicates the request, and admits a versioned payment-intent contract. Durable asynchronous acceptance returns an immutable intent identity; the double-entry ledger, workflow state machine, policy obligations, and reconciliation records remain the authoritative control plane. A provider response or webhook supplies evidence but cannot directly mutate financial authority or declare completion.

The public-evidence panels in figure 5 are bounded by Zelle’s institution-mediated pages, participant-side vendor documentation, and an Early Warning role posting (Zelle 2026d; Zelle 2026b; Zelle 2026f; Jack Henry 2021b; IBM 2026a; U.S. Bank 2026; Early Warning 2026). The posting requires experience with “one or more”

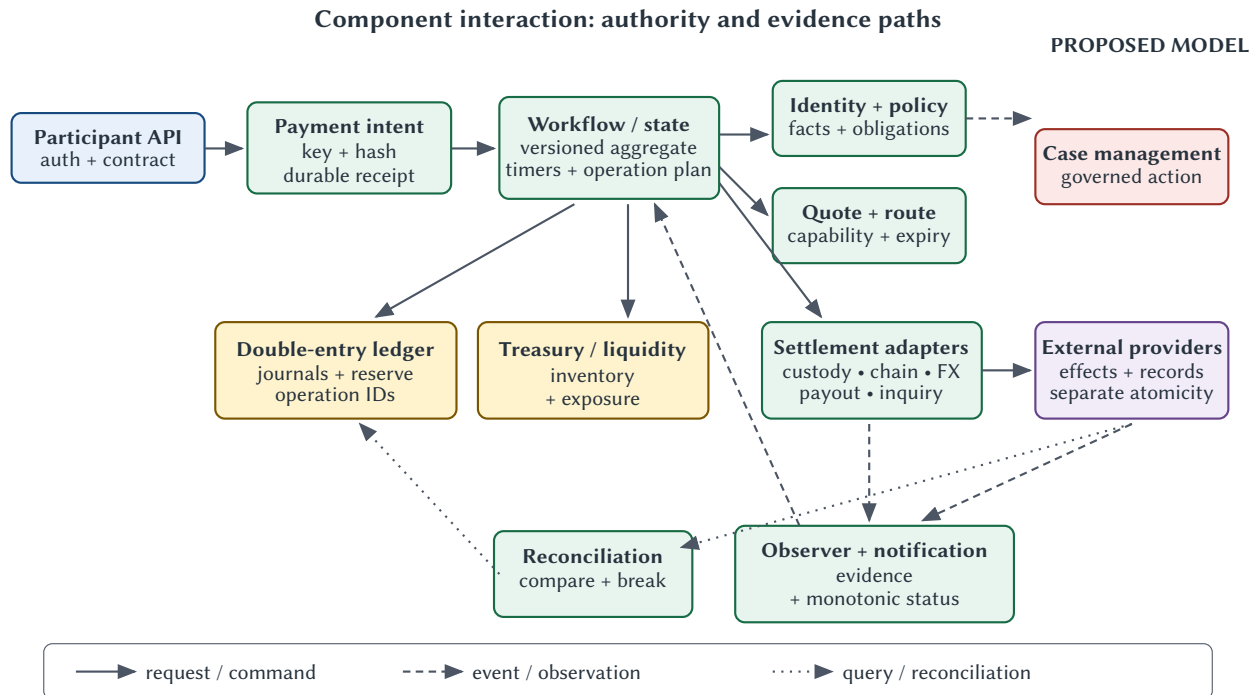


Figure 4: Bounded-context interaction. Synchronous validation ends at durable acceptance; financial execution continues through commands, observations, and reconciliation.

listed technologies; it is an organizational hiring signal, not a statement that every item is deployed or that the diagram represents private EWS topology. The complete overview is included in the source and release package; three portrait detail views preserve normal-zoom readability without creating edited raster copies.

Accessible reading order. Institutional clients enter through an authenticated participant boundary. The gateway authenticates, authorizes, canonicalizes, and deduplicates requests. The proposed payment-intent API returns durable asynchronous acceptance. The ledger and workflow control financial effects, and a transactional outbox publishes durable work. Provider-neutral adapters isolate external rails. Independent observation and reconciliation determine state advancement. Participants receive monotonic status through queries, events, or signed webhooks.

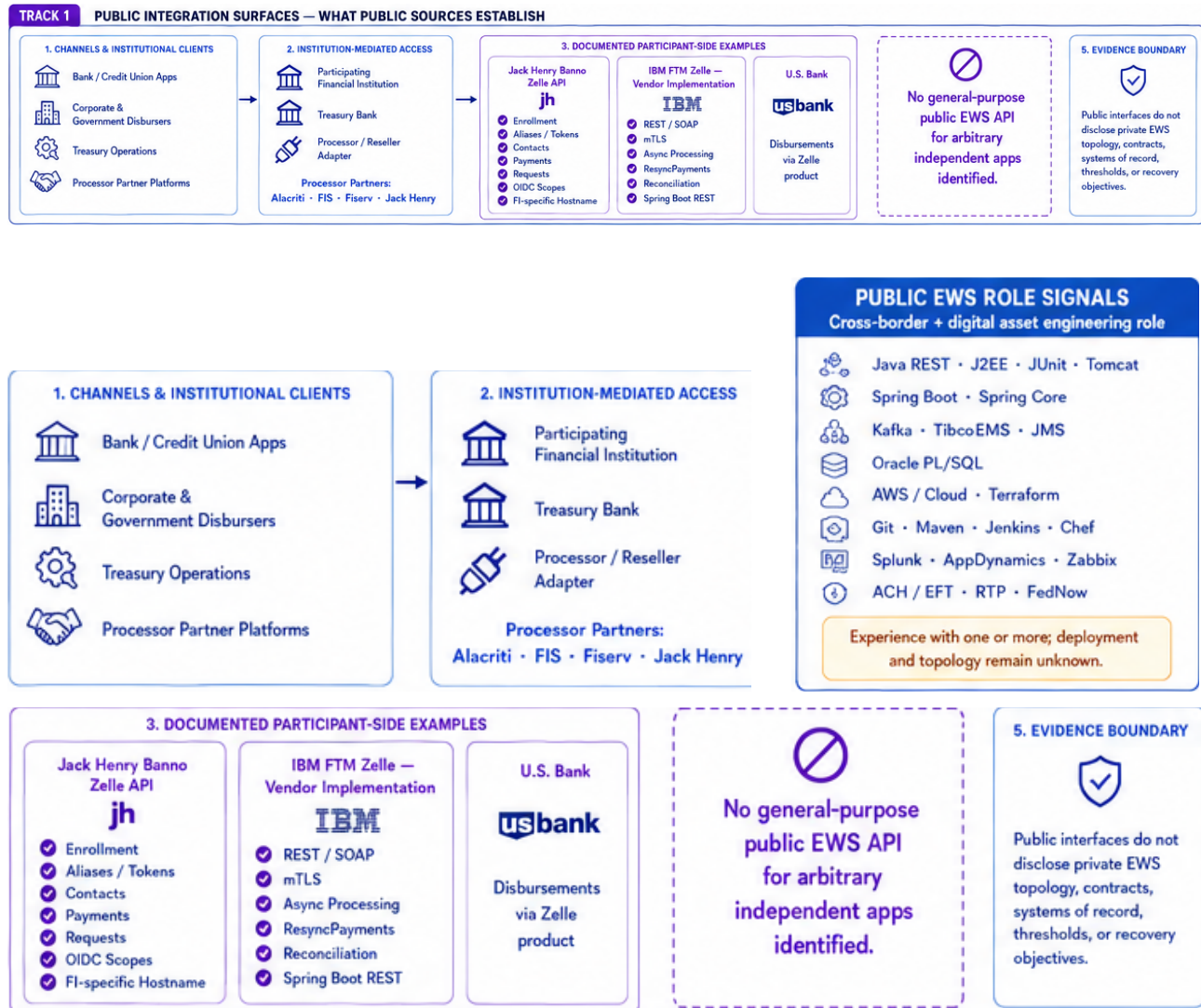


Figure 5: API-centric reference system design. Blue elements summarize public integration evidence and Early Warning role signals; purple elements are documented participant-side vendor examples; green elements are proposed target architecture; and amber elements remain route decisions or unknowns. The proposed payment-intent API, ledger, state machine, adapters, and target technology posture are recommendations and do not represent private Early Warning Services topology.

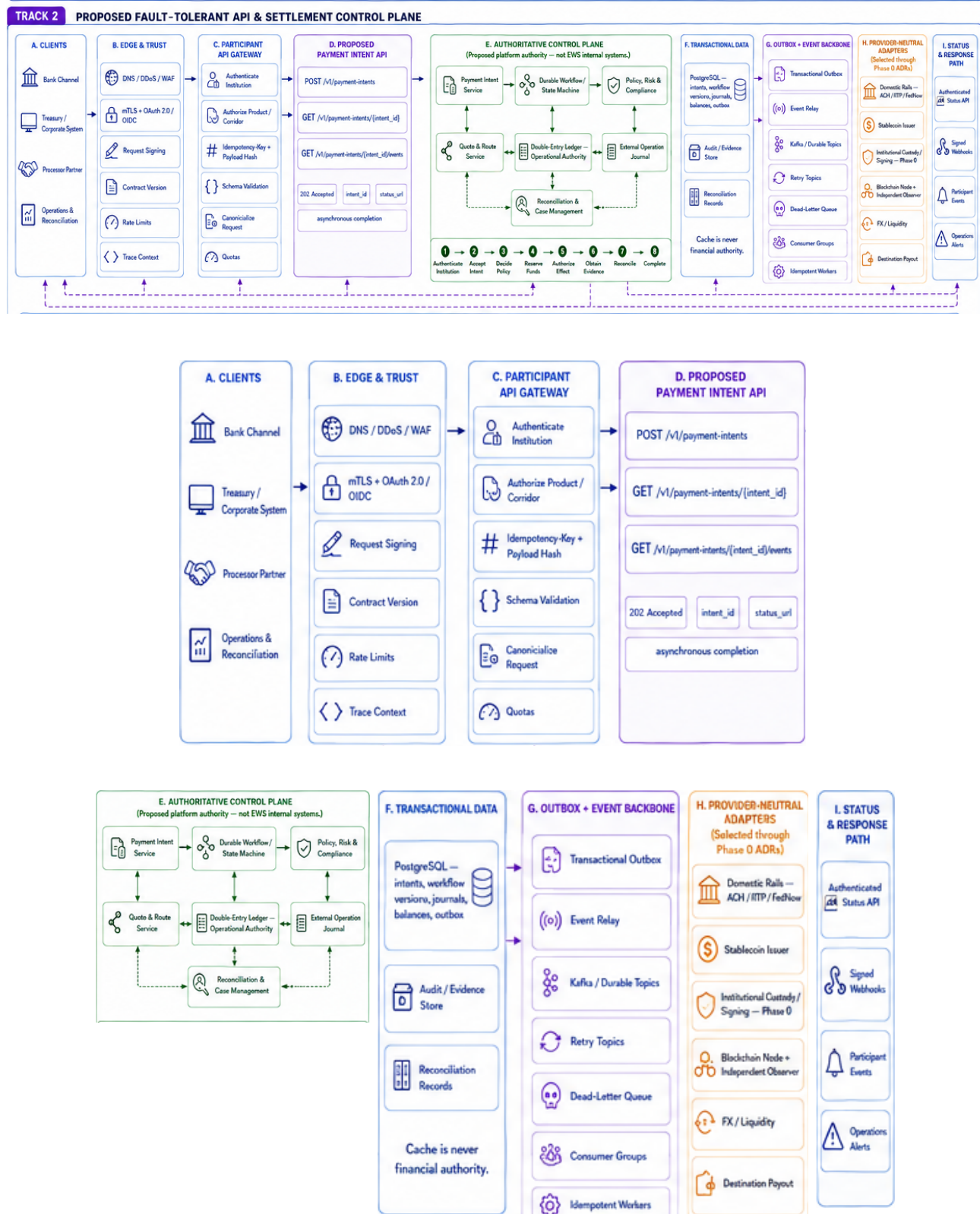


Figure 5, continued: proposed API and authoritative control plane.

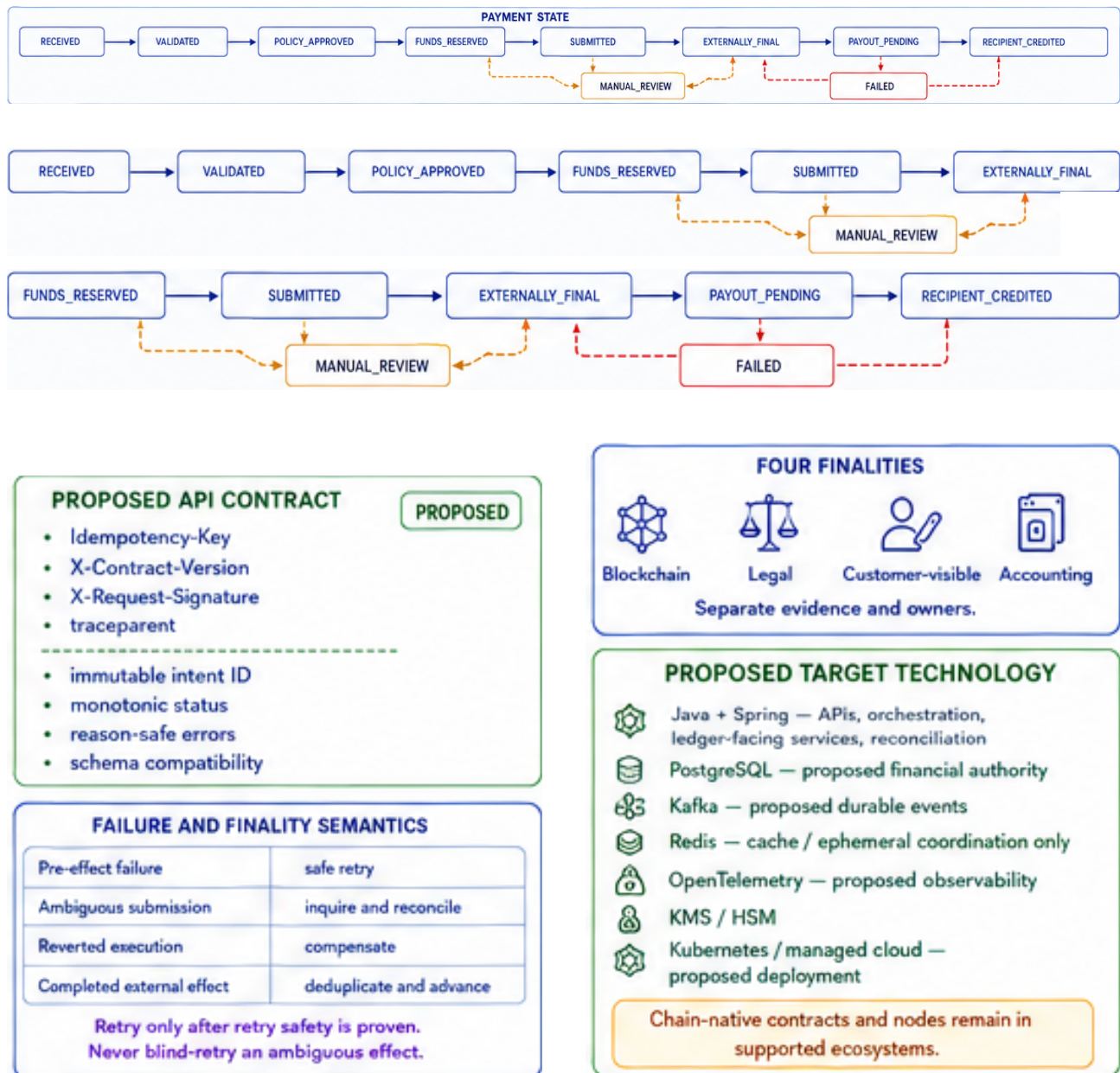


Figure 5, continued: failure semantics, finality, state, and target-technology classifications.

Detailed responsibility, input/output, authorization, consistency, and scaling contracts are collected in section A. The main narrative retains the authority model and the system-of-record decisions needed to evaluate it.

The complete system-of-record matrix is collected with the detailed component contracts in section A. In the primary narrative, the governing rule is sufficient: payment intent owns durable acceptance, ledger journals own obligations, workflow owns evidence-gated progression, independent observations establish external effects, and reconciliation owns close.

5.1. Contract and failure semantics

Participant APIs use synchronous validation only for checks that can complete within a bounded request budget. Durable acceptance returns an intent ID and status link; policy, funding, signing, submission, finality, and payout continue asynchronously. This avoids keeping a client connection open across systems whose completion may take minutes or require manual review.

Commands express requested effects: `ReserveFunds`, `AuthorizeSigning`, `SubmitTransfer`, or `InitiatePayout`. Events express durable facts: `FundsReserved`, `SignatureRefused`, `TransactionObserved`, or `RecipientCreditConfirmed`. Observations from providers are not events until validated, deduplicated, and durably recorded by the owning adapter or observer.

Versioned contracts use additive evolution by default. Envelopes include schema version, aggregate version, business operation ID, causation/correlation IDs, producer, occurred time, and policy/configuration version. Unknown future versions quarantine rather than guess; exact duplicates acknowledge without reapplying; a different payload at the same aggregate version becomes a conflict incident.

Combining ledger and workflow in one mutable record looks simpler but creates ambiguous authority: accounting corrections and payment-state recovery have different invariants, retention, and ownership. The design separates them and relates them through immutable operation IDs and atomic local commands where co-location is deliberately chosen. A workflow state never substitutes for a journal; a journal never declares a customer-visible state.

Consequence. Financial mutation authority is narrow. Provider uncertainty is retained, observable, and recoverable rather than normalized into a misleading Boolean success.

6. Internal ledger

Decision resolved. The internal double-entry ledger is authoritative for business obligations; blockchain balances are external assets to reconcile. **PROPOSED**

A chain can establish control of tokens at addresses. It cannot, by itself, represent a participant's reserved fiat, an unexpired quote, accrued fee, FX payable, recipient-credit obligation, sanctions hold, suspense investigation, or period close. Those obligations span multiple asset and legal domains. The platform therefore records immutable journals composed of balanced postings. A balance is a derived projection, not an independently editable field.

An account is keyed by legal entity, owner/participant, asset or currency, account purpose, and where needed provider/custody location. Quantity and reporting-currency valuation are separate. Reservations reduce available capacity without rewriting settled quantity. A journal correction references the original and posts a compensating journal; "rollback" means a local database transaction did not commit, never deletion of a committed financial fact.

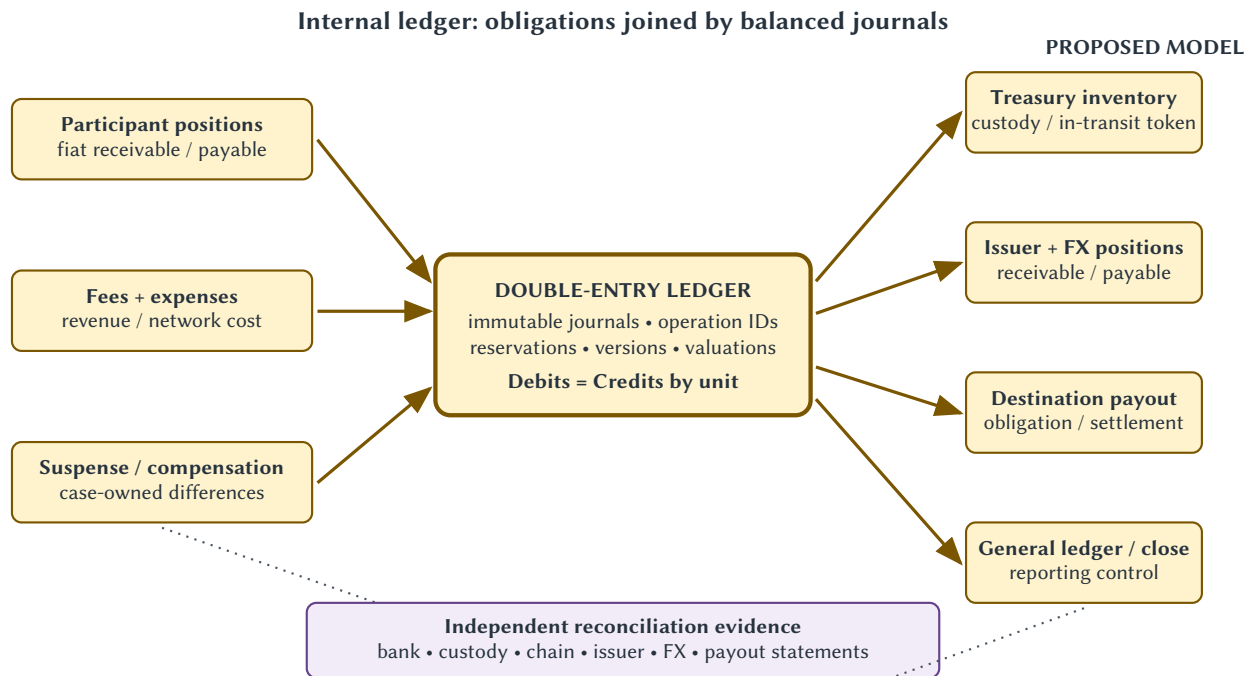


Figure 6: Ledger relationships. Participant positions, custody inventory, payout obligations, fees, FX, and suspense are separate accounts joined by balanced journals and reconciliation.

6.1. Illustrative chart of accounts

Table 6. Illustrative accounts. Final legal ownership and accounting treatment require accounting and counsel approval.

Account	Normal side	Dimension	Reconciles to	Purpose
Participant fiat receivable	Debit	Legal entity, participant, USD	Bank/participant statement	Funds due from sending participant
Participant settlement obligation	Credit	Legal entity, participant, USD	Participant settlement record	Amount due from or allocated to the participant settlement path
Participant funds available	Credit	Legal entity, participant, USD	Participant pre-funding/settlement account	Prefunded participant liability still eligible for reservation
Customer payment funds reserved	Credit	Participant, intent, USD	Participant authorization	Memo/control account for unavailable committed capacity
Treasury stablecoin inventory	Debit	Custodian, wallet class, asset, chain	Custody + independent chain + issuer	Controlled token asset
Stablecoin in transit	Debit	Operation, asset, chain	Chain transaction/contract events	Submitted but not yet economically final quantity
Stablecoin settlement consumption	Debit	Payment, route, asset, chain	Final chain effect plus approved valuation	Economically final token quantity delivered beyond platform control
Settlement cost / delivered value	Debit	Payment, corridor, reporting currency	Valuation record and general ledger	Reporting-currency value consumed to create the destination obligation
Issuer redemption receivable	Debit	Issuer, asset, jurisdiction	Issuer statement/redemption	Claim awaiting fiat redemption
FX receivable/payable	Debit/Credit	Provider, quote, currency pair	FX fill and settlement	FX principal and unsettled position
Destination payout payable	Credit	Provider, corridor, currency	Payout/institution statement	Fiat owed to recipient delivery path
Payout provider cash/settlement asset	Debit	Provider, corridor, currency	Provider and bank statement	Controlled cash or settlement asset consumed when payout is proven
Fee receivable	Debit	Participant, product, currency	Participant invoice/settlement	Contractual fee due
Fee revenue	Credit	Product, legal entity, currency	General ledger	Recognized platform fee
Network/custody fee expense	Debit	Provider, operation, asset/currency	Chain/custody invoice	External execution cost
Suspense—external break	Debit/Credit	Source, case, asset/currency	Reconciliation evidence	Controlled temporary difference

Account	Normal side	Dimension	Reconciles to	Purpose
Compensation expense/recovery	Debit/Credit	Incident, responsible party, currency	Case and recovery records	Loss absorption and recovery

6.2. Balanced posting examples

The examples use a USD 100 principal, INR 8,300 destination amount at an illustrative booked rate of INR 83 per USD, a USD 1 participant fee, and an illustrative network cost valued at USD 0.10. They demonstrate mechanics rather than a legal or accounting conclusion. Every journal balances in its native unit; quantity and reporting valuation remain distinct.

Table 7: J1A—accept the participant settlement obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J1A.1	Participant fiat receivable	FI-A / PAY-123	100.00	—
J1A.2	Participant settlement obligation	FI-A / PAY-123	—	100.00
	Total		100.00	100.00

Table 8: J1B—reserve prefunded participant capacity for the accepted intent (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J1B.1	Participant funds available	FI-A / USD	100.00	—
J1B.2	Customer payment funds reserved	FI-A / PAY-123	—	100.00
	Total		100.00	100.00

J1B is the committed ledger evidence for FUNDS_RESERVED: it reduces available prefunded capacity in the same balanced transaction that records the reservation operation and outbox. A credit-capacity product may instead use a ledger-native encumbrance against an approved limit, but it must enforce the same atomic available-capacity reduction. The accounting model and exact account names are route and accounting decisions; workflow never creates the reservation by itself.

Table 9: J2—designate 100.00 units of stablecoin inventory for an authorized chain operation.

Entry	Account	Dimension	Debit token	Credit token
J2.1	Stablecoin in transit	CHAIN-OP-1	100.00	—
J2.2	Treasury stablecoin inventory	CUST-A / ASSET-X	—	100.00
	Total		100.00	100.00

J2 is posted when the approved accounting policy changes inventory classification, not merely because an RPC returned HTTP 200. If submission is ambiguous, the in-transit classification and workflow ambiguity remain until inquiry resolves them.

Table 10: J3A—economic chain finality clears stablecoin inventory in transit (100.00 token units).

Entry	Account	Dimension	Debit token	Credit token
J3A.1	Stablecoin settlement consumption	PAY-123 / ASSET-X	100.00	—
J3A.2	Stablecoin in transit	CHAIN-OP-1	—	100.00
Total			100.00	100.00

J3A is the required disposition of J2: once the approved chain-finality policy proves delivery, the token quantity can no longer remain in transit. The valuation record links this token-unit journal to the separately balanced reporting and payout-currency journal below.

Table 11: J3B–J5N—concise multi-currency settlement example. Each row represents a balanced debit/credit pair in the named native unit.

Leg	Debit account	Credit account	Native amount	Valuation / evidence
J3B Fund- ing applied	Customer payment funds reserved	FX settlement funding control	USD 100.00	Committed reservation
J3C Ac- ceptance closed	Participant settlement obligation	Participant fiat receivable	USD 100.00	Acceptance-control close
J4A FX acquired	Destination-currency inventory	FX conversion control	INR 8,300.00	FX-VAL-1: USD 100.00 at INR 83/USD
J4B Payout recognized	FX conversion control	Destination payout payable	INR 8,300.00	Same FX-VAL-1; payout operation recorded
J4C Recipi- ent credited	Destination payout payable	Payout-provider settle- ment asset	INR 8,300.00	Contractually sufficient credit evidence
J5 Fee rec- ognized	Fee receivable	Fee revenue	USD 1.00	Approved participant fee schedule
J5N Net- work cost	Network/custody fee expense	Treasury gas-asset inven- tory	ETH 0.00004	NET-VAL-1: USD 0.10 at execu- tion time

J3B consumes the exact reservation created by J1B; it does not create a second source of value. FX-VAL-1 links the separately balanced USD funding and INR quantity journals without pretending that unlike units balance against each other. The destination payable remains open until recipient-credit evidence supports J4C. J5 and J5N separate platform revenue from external network cost, and NET-VAL-1 preserves the gas-asset quantity, price source, and timestamp. A controller and counsel must approve the production chart, recognition, valuation, and legal-entity treatment.

Table 12: J6A—compensate a failed route by releasing the participant obligation (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J6A.1	Participant settlement obligation	FI-A / PAY-123	100.00	—
J6A.2	Participant fiat receivable	FI-A / PAY-123	—	100.00
Total			100.00	100.00

Table 13: J6B—after proof of no chain effect, restore the in-transit token classification (100.00 token units).

Entry	Account	Dimension	Debit token	Credit token
J6B.1	Treasury stablecoin inventory	CUST-A / ASSET-X	100.00	—
J6B.2	Stablecoin in transit	CHAIN-OP-1	—	100.00
Total			100.00	100.00

J6B is permitted only when independent inquiry proves that no chain transfer occurred, or that a reverted transaction left the platform-controlled inventory intact. If the chain transfer actually succeeded, neither J6A nor J6B is safe: the platform would release participant funding or recreate inventory while retaining an external loss. That case requires loss allocation and perhaps a compensation-expense/recovery journal, not a mechanical reversal.

Table 14: J6C—release a reservation only after the route is terminated without consuming participant funding (USD 100.00).

Entry	Account	Dimension	Debit USD	Credit USD
J6C.1	Customer payment funds reserved	FI-A / PAY-123	100.00	—
J6C.2	Participant funds available	FI-A / USD	—	100.00
Total			100.00	100.00

J6C is the exact failure disposition of J1B. It is forbidden after J3B has consumed the reservation unless a separately authorized compensation journal funds the loss. This prevents a successful external effect from both consuming treasury value and restoring the participant's spendable prefunding.

Table 15: Worked happy-path closeout. Operation-scoped control balances return to zero; economic assets decrease by the delivered amount.

Account	Opening / created	Closing entry	End-state meaning
Participant fiat receivable	J1A debit 100	J3C credit 100	Zero for PAY-123
Participant settlement obligation	J1A credit 100	J3C debit 100	Zero for PAY-123
Customer payment funds reserved	J1B credit 100	J3B debit 100	Zero; not still reserved
Participant funds available	J1B debit 100	None	Available prefunding reduced by 100
Stablecoin in transit	J2 debit 100 token	J3A credit 100 token	Zero for CHAIN-OP-1
Destination payout payable	J3B credit 100	J4 debit 100	Zero after proven recipient credit
Treasury inventory / settlement consumption	J2 credit / J3A debit	None	Controlled token decreases; delivered quantity is retained for valuation and audit
Payout provider cash/settlement asset	Existing debit balance	J4 credit 100	Controlled payout asset decreases by 100

Table 16: J7—record an approved unresolved custody reconciliation shortage in suspense (1.00 token unit).

Entry	Account	Dimension	Debit token	Credit token
J7.1	Suspense—external break	CASE-77 / ASSET-X	1.00	—
J7.2	Treasury stablecoin inventory	CUST-A / ASSET-X	—	1.00
Total			1.00	1.00

J7 is not posted merely because two sources differ. It requires accounting authorization and evidence that the external controlled quantity is lower than the ledger quantity at the comparison cutoff. The open case retains both source snapshots. Recovery reverses J7; a proven loss clears suspense to the approved compensation/loss account through a separate journal.

6.3. Invariants and atomicity

Table 17. Testable ledger invariants.

ID	Invariant	Enforcement/evidence
INV-001	For every journal and unit, total debits equal total credits before commit.	Database constraint/service invariant plus property test
INV-002	A business operation ID posts at most one effective journal of a given type.	Unique constraint including legal entity and operation type
INV-003	Committed postings are immutable; corrections reference original postings.	Append-only permissions and audit reconciliation
INV-004	Available balance equals policy-eligible settled balance minus active reservations and restrictions; it is never read from an unauthoritative replica to approve spend.	Partition-owner check in the posting transaction
INV-005	An account has one writer/partition version at a time; expected-version conflict cannot silently overwrite.	Pessimistic lock or serializable/optimistic version strategy proven by contention tests
INV-006	Asset quantity and reporting-currency valuation are distinct dimensions with a timestamped valuation source.	Schema constraint and valuation journal
INV-007	Each external controlled address/account maps to ledger accounts and a reconciliation owner.	Configuration registry plus coverage check
INV-008	Workflow completion cannot bypass required journal and reconciliation gates.	Transition guard queries authoritative operation evidence

Postings for one journal, their outbox record, account versions, and the ledger-operation result commit in one local ACID transaction. PostgreSQL documents transaction isolation within such a database boundary, not across external providers (PostgreSQL Global Development Group 2026). **DOC** No database transaction spans the signer, chain, issuer, FX provider, payout provider, or participant. That limitation is a design input, not an implementation defect.

Table 18: External-to-ledger reconciliation map.

External source	Internal accounts	Comparison	Break owner/posture
Participant/bank statement	Receivables, payables, prefunding	Amount, currency, value date, operation/reference	Participant settlement; suspense while investigated
Custody statement	Treasury inventory, fees	Asset, address class, available/restricted quantity	Treasury/custody; route limit or pause on unexplained deficit
Independent chain data	Inventory in transit, network fees	Transaction, canonical block, logs, nonce, amount, contract	Chain operations; ambiguity/manual review
Issuer record	Inventory/redemption receivable	Supply/holdings, mint/burn, freezes, redemption status	Treasury/legal; asset kill switch
FX confirmation	FX receivable/payable	Quote/fill IDs, amounts, rate, fee, settlement	Treasury; revalue or suspense
Payout report	Payout payable/provider cash	Operation, recipient-credit status, amount, currency	Payout operations; retry/inquiry/case
General ledger/close	All control/subledger totals	Trial balance, valuation, cut-off, aged breaks	Accounting; no close while material break unresolved

7. Durable payment state machine

Decision resolved. A versioned, evidence-gated state machine governs payment truth; provider statuses are observations mapped into it. **PROPOSED**

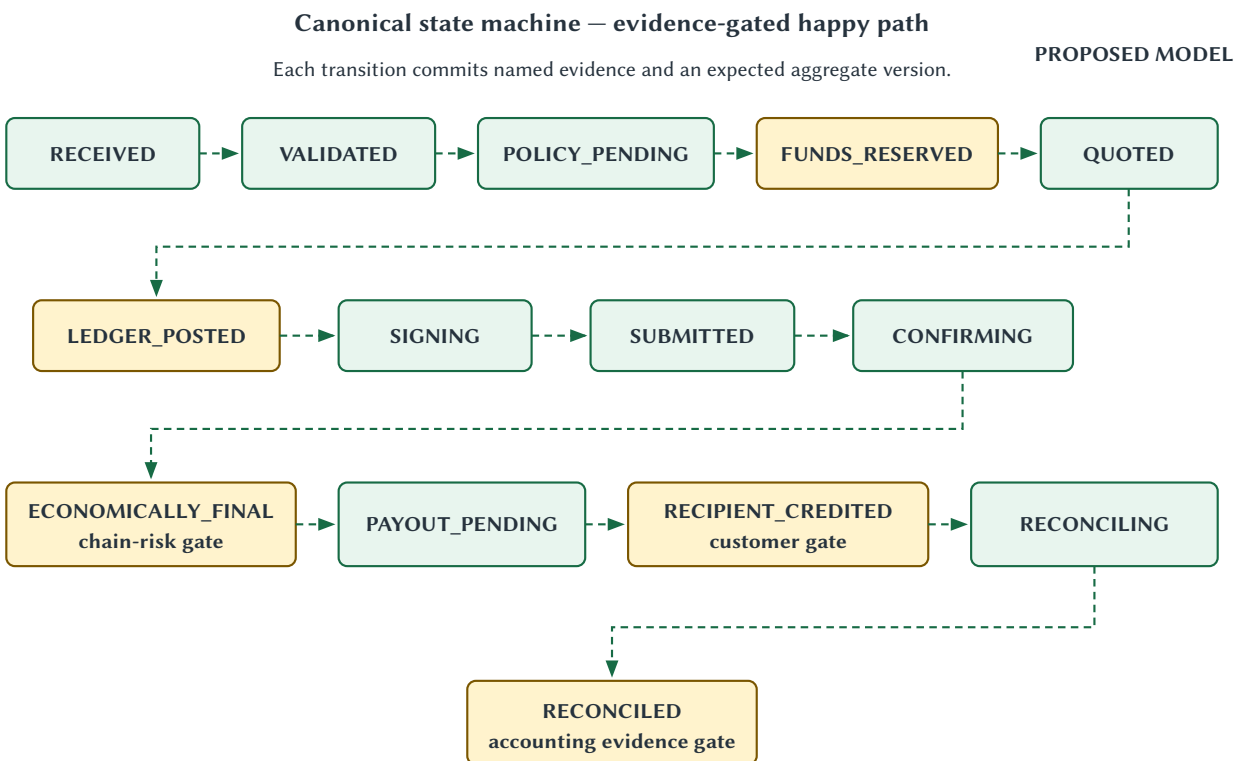


Figure 7: Happy-path state model. Routes may skip inapplicable states, but each transition requires named evidence and an expected aggregate version.

Each payment aggregate stores state name, schema/version, route version, policy/configuration versions, last transition evidence, next permitted action, timers, and optimistic aggregate version. A transition appends an immutable event and updates the current projection in one local transaction. A handler supplies the expected version; a conflict reloads and reevaluates instead of overwriting.

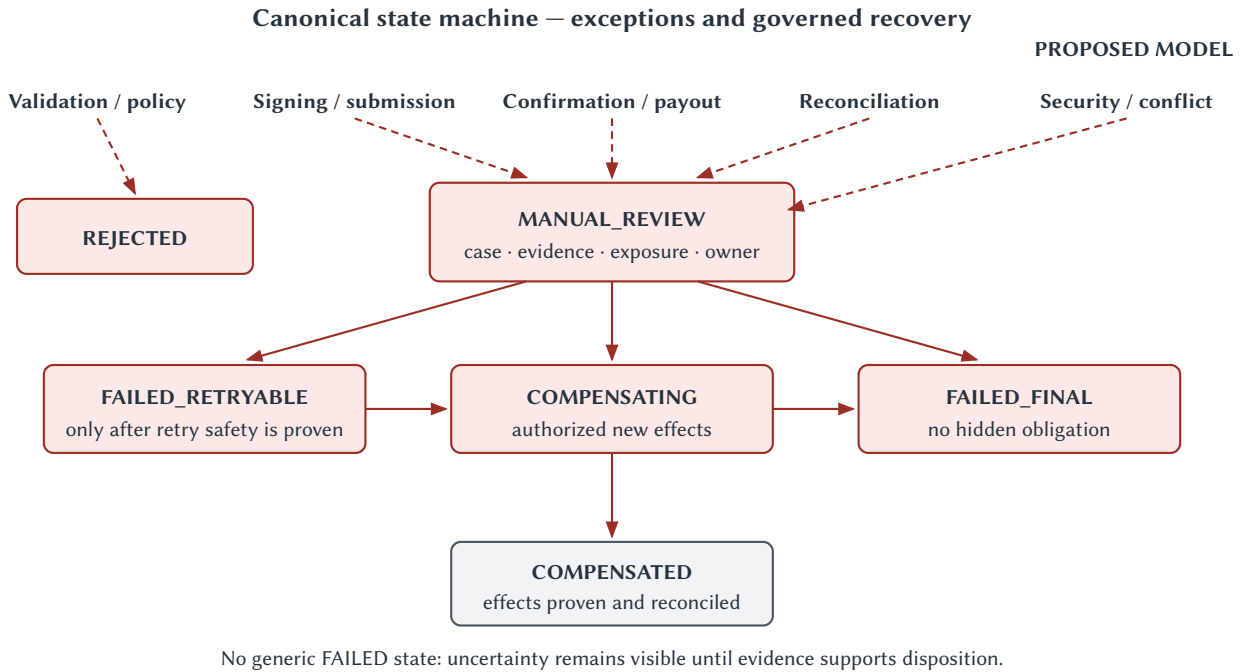


Figure 8: Exception state model. Ambiguity and governed manual review are stable states; retry and compensation are explicit transitions, not hidden loops.

Table 19. Canonical state dictionary. Customer wording is illustrative and route policy controls visibility.

State	Entry evidence / guard	Owner	Customer view	Timeout / terminality
RECEIVED	Durable intent, canonical hash, participant authorization	Intent	Received	Short validation timer
VALIDATED	Contract, identity, route prerequisites valid	Workflow	Processing	Policy timer
POLICY_PENDING	Versioned facts submitted for decisions	Policy	Processing/under review	SLA to hold/reject/review
FUNDS_RESERVED	Ledger reservation operation committed	Ledger/workflow	Processing	Reservation expiry
QUOTED	Accepted unexpired quote and route snapshot	Quote/router	Amount/terms confirmed	Quote expiry
LEDGER_POSTED	Required authorization journal committed	Ledger/workflow	Processing	Ready for execution
SIGNING	Authorized canonical transaction and signer operation recorded	Custody/signing	Processing	Outage/refusal to retry/review
SUBMITTED	Submission evidence or transaction identity sufficient for inquiry	Settlement adapter	Sent/processing	Never blind retry after timeout

State	Entry evidence / guard	Owner	Customer view	Timeout / terminality
CONFIRMING	Independent canonical-chain observation below finality threshold	Chain observer/workflow	Processing	Route-specific delay escalation
ECONOMICALLY_FINAL	Chain-risk policy satisfied against independent evidence	Workflow/chain risk	Processing or sent	Not legal/accounting final by implication
PAYOUT_PENDING	Authorized payout operation recorded	Payout adapter	Delivering	Inquiry/repair SLA
RECIPIENT_CREDITED	Contractually sufficient recipient-credit evidence	Workflow/product	Completed (policy-dependent)	Reconciliation remains
RECONCILING	External effects complete enough for comparison	Reconciliation	Completed/processing	Break aging thresholds
RECONCILED	All required comparisons within tolerance or approved disposition	Accounting/reconciliation	Completed	Terminal for route period
MANUAL_REVIEW	Ambiguous, policy, security, or material break case opened	Case owner	Under review	Governed SLA; nonterminal
COMPENSATING	Recovery plan authorized and operations recorded	Workflow/ledger	Correcting	Each effect separately idempotent
COMPENSATED	Compensating journals/effects proven and reconciled as required	Workflow/accounting	Canceled/refunded	Terminal for failed route
FAILED_RETRYABLE	Proven transient failure with safe retry conditions	Owning adapter/workflow	Delayed	Bounded attempts/backoff
FAILED_FINAL	Permanent failure with no authorized compensation pending	Workflow/case	Failed	Terminal
REJECTED	Pre-effect policy/validation decision	Policy/intent	Not sent	Terminal

Business state and provider state are intentionally different. “Provider completed” may map to SUBMITTED, CONFIRMING, or RECIPIENT_CREDITED depending on the provider contract and independent evidence. FAILED_RETRYABLE means the platform has evidence that retrying a particular operation is safe; it does not mean a generic transport error occurred. RECIPIENT_CREDITED may permit a completed customer message, but RECONCILED remains a separate accounting/evidence gate.

Consequence. A late callback can add evidence but cannot move a stale aggregate past its guards. A timeout preserves uncertainty rather than manufacturing a duplicate financial effect.

Table 20: Representative transition guards. The complete transition matrix appears in table 55.

Transition	Required guard	Side effect after commit
POLICY_PENDING → FUNDS_RESERVED	All decisions allow; unique reservation operation committed; expected aggregate version	Request quote against the committed reservation
QUOTED → LEDGER_POSTED	Quote unexpired; capacity and policy rechecked; balanced authorization journal committed	Publish signing or route-execution command through the journal outbox
SIGNING → SUBMITTED	Signer verified canonical bytes; signed transaction identity recorded; submission policy valid	Idempotent provider submit command
SUBMITTED → CONFIRMING	Independent node observes expected transaction/contract/amount	Schedule finality observations
CONFIRMING → ECONOMICAL_FINAL	Chain-specific threshold and conflict checks pass	Ledger classification and payout authorization evaluation
PAYOUT_PENDING → RECEIPT_CREDITED	Destination evidence satisfies corridor product policy	Versioned participant notification
Any external ambiguity → MANUAL_REVIEW	Inquiry cannot prove absent/success, or evidence conflicts	Case with exposure, evidence snapshot, and action limits
Recoverable failure → COMPENSATING	Original effects known; plan authorized; loss and legal constraints evaluated	Individually idempotent compensation operations

8. Payment lifecycle and flow catalog

Decision resolved. All products use one intent, ledger, state, evidence, and recovery model while selecting route-specific postings and finality gates. **PROPOSED**

The common lifecycle is: authenticate participant; resolve the idempotency key; validate the canonical intent; resolve recipient/identity; obtain versioned fraud, sanctions, product, and corridor decisions; reserve funding; quote and select a route; post authorization journals; construct and simulate any chain transaction; authorize signing; submit once; observe independently; apply finality policy; initiate destination payout where required; publish monotonic participant status; reconcile every external position; and route exceptions to bounded retry, compensation, or governed case work.

Each flow below names the distinguishing funds path and completion evidence. All inherit the same participant authorization, policy, operation-ID, audit, notification, and reconciliation requirements. Detailed concern coverage appears in table 42.

8.1. Six business flows

Table 21. Business-flow catalog. “Complete” is always a product-policy outcome, not a provider status.

Flow	Initiator and funds path	Ledger and external effects	Finality / notification	Recovery focus
FLOW-01 Existing domestic	Sending participant; source account to recipient institution through approved domestic route	Reserve/post participant position; domestic rail operation; no chain service	Rail/product evidence governs customer status; accounting close after domestic reconciliation	Duplicate rail request, return/reversal constraints, participant mismatch
FLOW-02 Stablecoin funded, fiat delivered	Institution supplies approved stablecoin; recipient receives destination fiat	Custody receipt/inventory, FX/redemption or sale, payout payable/settlement	Source token economic finality plus destination credit; legal/accounting gates separate	Wrong asset/network, incoming reorg, issuer/freeze, FX and payout failure
FLOW-03 Fiat invisible stablecoin settlement	Sending participant supplies fiat; treasury token settles behind the product	Participant receivable/reservation, treasury inventory/in-transit, destination payout payable	Customer sees conventional payment; chain threshold and payout evidence are internal gates	Ambiguous submit, inventory shortfall, finalized transfer but payout failure
FLOW-04 Cross-border with FX and payout	U.S. participant funds USD; recipient receives destination currency	USD obligation, quote/fill, token transfer if selected, destination-currency payout	Quote expiry, chain/FX/payout evidence and corridor completion policy	Slippage, FX settlement mismatch, payout delay, sanctions hold
FLOW-05 Stablecoin to stablecoin	Approved institutional sender and recipient use supported assets/networks	Source custody/token obligation, transfer or issuer-mediated conversion, destination token obligation	Both source and destination evidence; no assumption that bridge message is final	Asset mismatch, contract/admin powers, cross-network ambiguity
FLOW-06 Tokenized deposit	Bank-issued deposit token interfaces with platform/settlement asset	Bank deposit liability/claim mapped separately from stablecoin inventory and participant obligation	Issuing-bank rules, network evidence, redemption/convertibility and accounting gates	Singleness/conversion, bank/network availability, legal claim mismatch

For FLOW-04 the participant request identifies source amount/currency, recipient, destination currency, corridor, and terms; it never supplies an arbitrary token contract or destination wallet. Policy produces an allow/hold/reject decision with obligations. Treasury reserves both participant capacity and route inventory. The quote record fixes rate, fees, expiry, provider, tolerance, and re-quote policy. A transaction builder derives the approved contract, destination, amount, chain ID, nonce and fee cap from the route, then simulates. The signer independently checks those fields and binds the signature to the intent/policy versions. Chain observation and finality authorize the next economic step but do not prove foreign exchange or destination

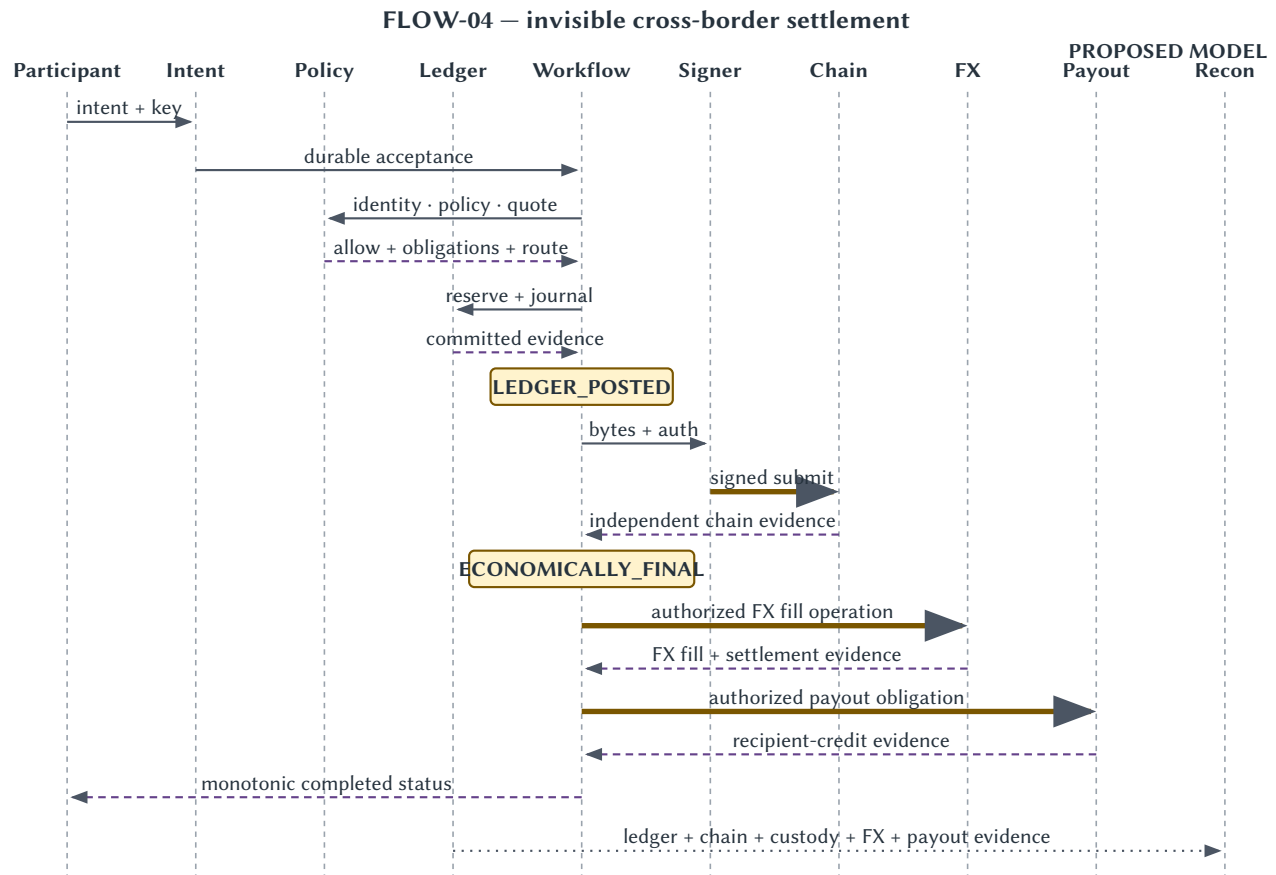


Figure 9: FLOW-04 invisible cross-border settlement. Ledger authorization precedes chain execution; payout and reconciliation remain separate after economic chain finality.

payout. A separately authorized FX operation records quote/fill identity, principal, rate, fee, expiry and settlement evidence before the payout obligation is released. The payout adapter then uses its own operation ID and source-of-truth inquiry. Recipient-credit evidence advances customer status; custody, chain, issuer, FX, payout, bank, and ledger records later reconcile.

FLOW-01 is deliberately important: it proves the common control plane is not synonymous with blockchain. Its route omits SIGNING, SUBMITTED, and CONFIRMING, maps domestic-rail evidence into the canonical model, and remains available while digital-asset routes are paused.

FLOW-02 validates incoming-asset controls. A custody notification is insufficient: the observer verifies supported network and contract, amount, destination ownership, transaction provenance required by policy, and finality threshold before inventory is recognized as available. FLOW-05 adds a strict rule against generic bridge use. An issuer-mediated burn/mint protocol such as CCTP illustrates how source finality, attestation, and destination mint are separate evidence steps; it is a USDC-specific documented mechanism, not a neutral architecture fact (Circle 2026a). [DOC](#)

FLOW-06 keeps a tokenized commercial-bank deposit distinct from a stablecoin. BIS analysis highlights different issuer claims, singleness, liquidity, and settlement implications (Bank for International Settlements 2023; Bank for International Settlements 2026). [DOC](#) Interoperability therefore needs a conversion and legal/accounting contract, not just token-interface compatibility.

8.2. Seven failure flows

Table 22. Failure-flow catalog. Inquiry precedes retry whenever an external effect may exist.

Flow	Trigger	State / ledger posture	Recovery, evidence, and audit
FLOW-07 Chain fail- ure/com- pensation	Pre-effect validation failure; route-safe absence; reverted execution	FAILED_RETRYABLE or COMPENSATING; reservation/in-transit handled according to proven effect	Correct route defect or release/compensate with a retry-safe new operation; retain pre-effect, absence, or revert evidence
FLOW-08 Reorg or delayed finality	Observed transaction loses canonicity or threshold stalls	Stay CONFIRMING; if previously risk-accepted, case and exposure record	Observe independent nodes; do not payout unless approved risk policy allows; reorg drill and loss path
FLOW-09 Compliance hold	New/changed fact before payout or post-payment concern	POLICY_PENDING or MANUAL_REVIEW; funds restricted, not erased	Versioned screening evidence, blocked/rejected property procedure, legal disposition, disclosure controls
FLOW-10 Liquidity shortfall	Required token, gas, fiat, FX, or payout capacity unavailable	Before external effect: FAILED_RETRYABLE/queued/rejected; retain reservation policy	Rebalance separately; never borrow implicitly or reroute after quote without reauthorization
FLOW-11 Duplicate re- quest/de- livery	Same participant key, event, callback, or payout retry arrives	Same intent/operation result; no duplicate journal/effect	Compare canonical hash; inbox/unique constraint; return monotonic state; conflicting payload is incident

Flow	Trigger	State / ledger posture	Recovery, evidence, and audit
FLOW-12 Recon- ciliation mismatch	Internal and chain/cus- tody/issuer/FX/pay- out/bank evidence dif- fer	RECONCILING or MANUAL_R EVIEW; suspense only by authorized journal	Snapshot both sources, quantify expo- sure, pause scope, case disposition and compensating entry if approved
FLOW-13 Sign- ing/key incident	Signer unavailable, refuses policy, or key compromise suspected	SIGNING, FAILED_RETRYABLE, or security MANUAL_REVIEW; no submission	Disable affected key/route, invoke quorum/recovery, identify signed set, rotate/migrate, independently reconcile

Ambiguous submission is the canonical distributed-systems failure. If the provider accepted a signed transaction but the response was lost, the database may show “command outstanding” while the chain later includes it. The workflow must persist the signed transaction hash and relevant nonce before submission. After a timeout it queries independent nodes, custody, and where applicable the provider by operation ID. Observation resumes if found. Another submission is allowed only when route-specific evidence establishes retry safety, including transaction identity, nonce or replacement semantics, canonical execution status, and inventory posture; otherwise the payment remains in manual review. A participant retry returns the same intent because its idempotency record is independent of the submit response.

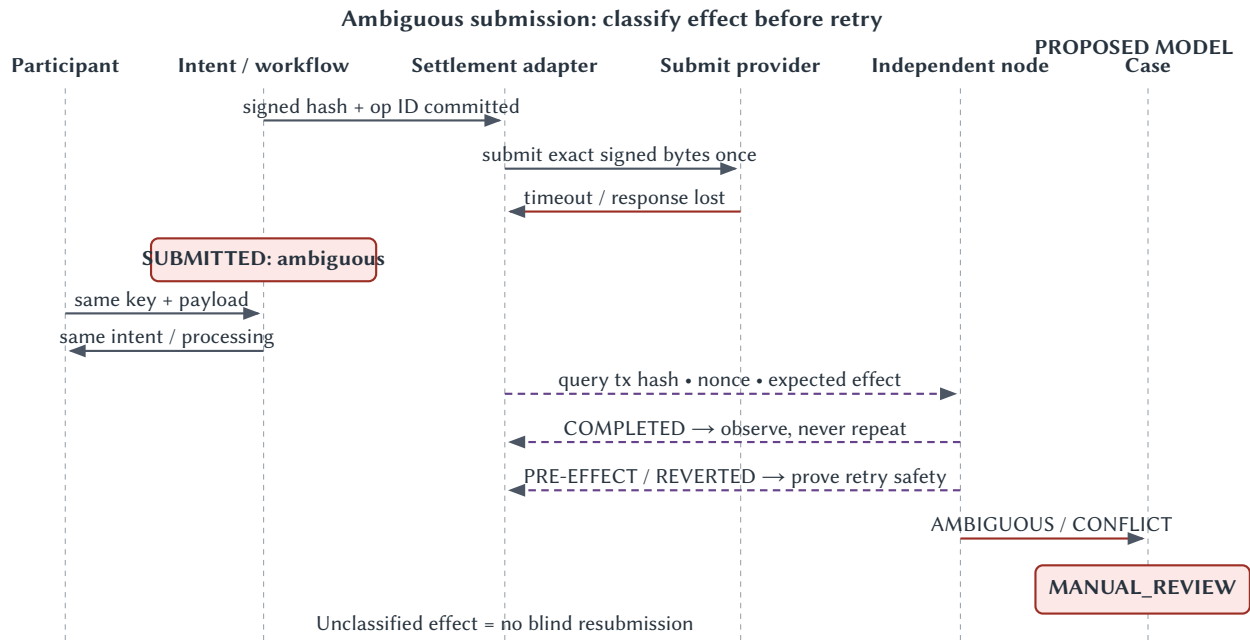


Figure 10: FLOW-07/11 ambiguous submission and duplicate retry. A timeout leads to inquiry by stable transaction identity; another attempt is allowed only after route-specific retry safety is established.

Finalized-chain/payout failure proves why a chain receipt cannot define payment completion. The platform still owes destination value or a contractually approved remedy. It preserves the payout payable, suspends any route that would increase unbounded exposure, inquires by payout operation ID, and follows corridor policy for safe retry, alternative payout, refund/redemption, or compensation. Customer messaging reflects what is known. An immutable case connects the chain transaction, payout evidence, ledger posture, disclosure, loss owner, and final disposition.

8.3. Flow coverage

The detailed concern-by-flow records are collected in section B; the six business flows and seven failure flows above remain the operational narrative.

Consequence. Product variants change funding, settlement, payout, and policy gates. They do not create alternative sources of truth or bypass the shared recovery model.

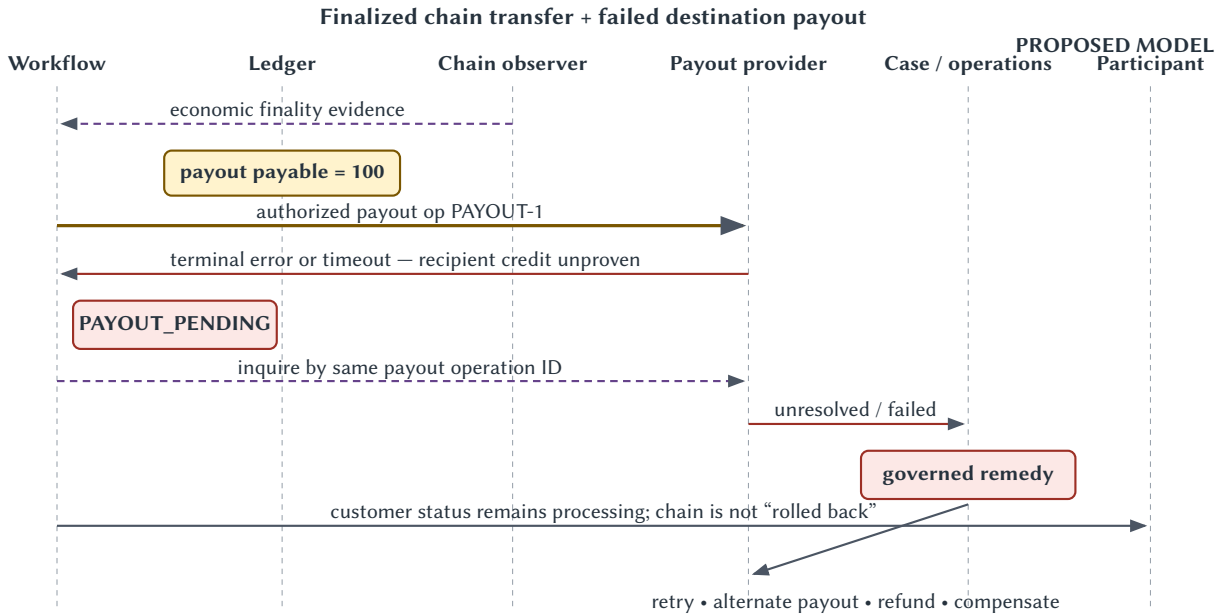


Figure 11: FLOW-04 failure branch: finalized token movement with failed destination payout. The chain effect is not rolled back; the payout obligation remains, with controlled retry, reroute, refund, or compensation.

9. Consistency: narrow atomicity, end-to-end recoverability

Decision resolved. Promise exactly-once business effects through composable controls; do not claim distributed exactly-once delivery. **PROPOSED**

No transaction manager can atomically commit the ledger database, message broker, signer, chain, issuer, FX provider, payout provider, and participant account. The architecture makes each local boundary strong and every cross-boundary gap durable, observable, and reconcilable.

Exactly-once business effect is constructed as follows:

1. participant-scoped idempotency key and canonical payload hash produce one intent;
2. a deterministic business operation ID names each financial effect;
3. local ACID transactions commit an aggregate version, balanced journal or provider-operation record, and an outbox row together;
4. a publisher may send more than once; consumers use an inbox/unique effect constraint and expected aggregate version;
5. external requests reuse the operation ID or stable transaction identity, and adapters inquire before uncertain retry;
6. reconciliation compares independent evidence and opens a controlled break when records disagree; and
7. compensation uses new authorized, idempotent operations and journals rather than deleting history.

AWS's idempotent-API guidance emphasizes token recording and semantic equivalence; an idempotency key alone does not make arbitrary retries safe (Featonby 2026). Debezium documents the transactional-outbox pattern, whose publication remains asynchronous and whose consumers still require idempotent handling

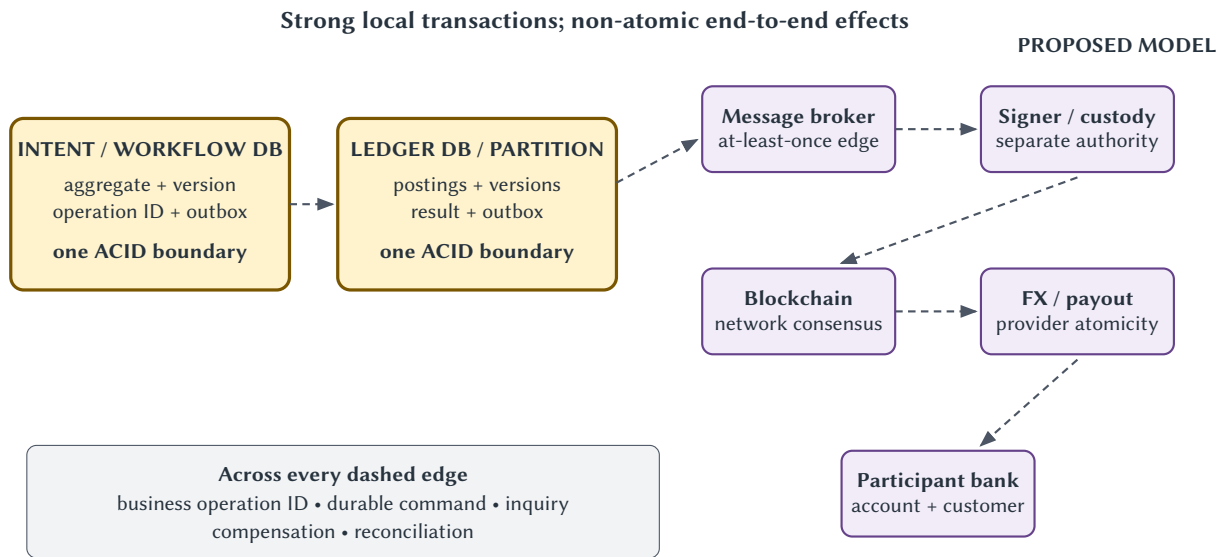


Figure 12: Transaction boundaries. The ledger/workflow database can commit business state and outbox locally; every provider edge is asynchronous and potentially ambiguous.

(Debezium 2026). Spring Kafka’s exactly-once semantics apply to Kafka read-process-write transactions; they do not include a ledger database or external rail (Spring 2026). [DOC](#)

IBM’s version-specific institutional product independently illustrates the operational problem: it documents asynchronous Zelle processing, communication-fault states whose outcome may be unknown, resynchronization-oriented retries, state-change events, and end-of-day reconciliation (IBM 2026a; IBM 2026b; IBM 2026c). [DOC](#) This corroborates the need for durable ambiguity and reconciliation; it is not EWS architecture and does not set the target design’s retry rule.

Table 23. Guarantee matrix. Strong guarantees are deliberately local.

Operation	Consistency scope	Duplicate control	Ambiguity resolution	Reconciliation
Intent acceptance	One intent store transaction	Participant/key unique + payload hash	Query by key/intent	Signed ingress audit
Ledger posting	One journal/account partition transaction	Ledger operation unique + balanced journal	Transaction result by operation ID	Trial balance/external sources
State transition	One aggregate transaction	Expected version + transition/effect key	Reload events/current projection	Operation/state invariant scan
Outbox publication	State + outbox local commit; broker separate	Event ID; publisher retries	Broker acknowledgment plus outbox state	Outbox age and broker offsets
Consumer effect	Consumer-owned local transaction	Inbox event ID + effect constraint	Query consumer result	Event/effect coverage
Signing	Custody boundary	Sign operation + canonical bytes hash	Provider inquiry/audit; never reconstruct a different payload	Signer versus authorized set

Operation	Consistency scope	Duplicate control	Ambiguity resolution	Reconciliation
Chain submission	Chain/provider boundary	Stable signed tx hash, nonce, op ID	Independent node/customer inquiry	Chain/custody/ledger
Payout	Provider boundary	Payout operation ID	Provider/destination inquiry before retry	Payout report/bank/ledger
Notification	Notification store/provider	Intent + audience + status version	Participant status query authoritative	Delivery status versus state

Account mutations require a partition owner, serializable behavior, or equivalent explicit locking proven by contention tests. A quote is accepted only while its expiry, route version, available capacity, and policy remain valid in the authorization transaction. Stale read models can display progress but cannot release a reservation, choose a nonce, or authorize spend. Participant status is monotonic in product semantics even when underlying evidence regresses: a reorg after a risk-accepted customer credit becomes an incident/compensation state, not a deceptive backward UI transition.

The word *saga* is useful only if it does not imply that financial effects can always be undone. A finalized token transfer may be irreversible at the protocol layer. Recovery might create a new payout, redemption, refund, or compensation expense, each with its own legal and liquidity consequences. The design therefore names compensating operations explicitly.

10. Messaging and event contracts

Decision resolved. Messaging transports durable facts and commands; it never replaces ledger authority, state guards, or source-of-truth inquiry. **PROPOSED**

10.1. Envelope and ordering

CloudEvents supplies a useful interoperable envelope vocabulary but does not define business authority, ordering, or delivery guarantees (Cloud Native Computing Foundation 2026). **DOC** The platform envelope contains:

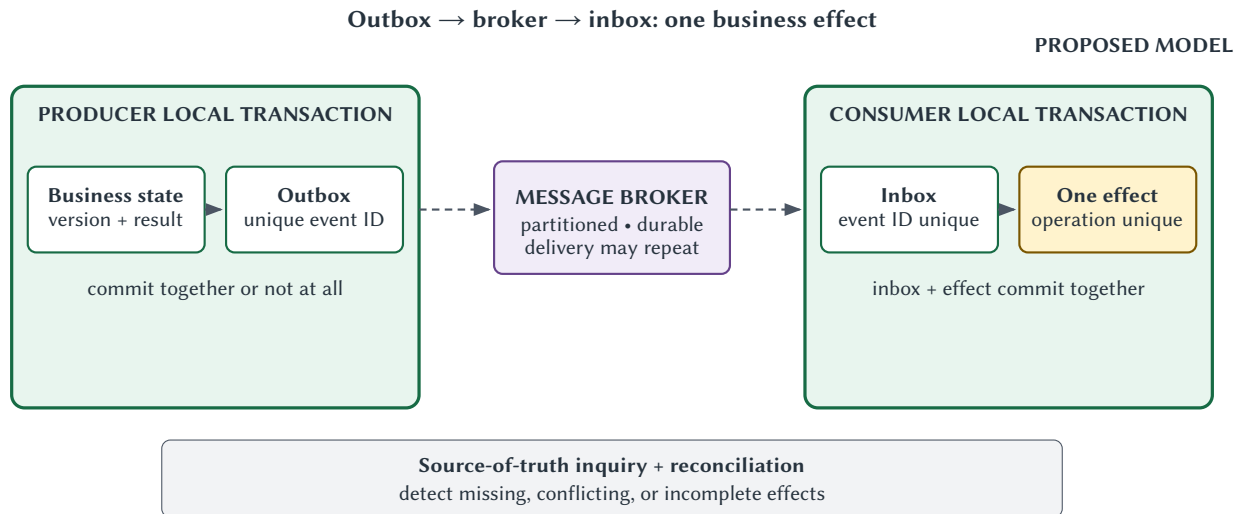


Figure 13: Outbox/broker/inbox path. At-least-once transport is converted into one local business effect through unique event and operation identities.

Payment events partition by payment-intent ID because aggregate transitions require order. Ledger commands partition by account owner or shard because balance conflicts, not payment affinity, determine safe serialization. A journal involving multiple account partitions follows the ledger's approved coordination design. Consumers never mutate another service's table.

10.2. Retry, replay, and poison handling

Table 24: Required message envelope.

Field	Constraint	Purpose
event ID	Globally unique, immutable	Inbox deduplication and audit
aggregate ID/version	Payment/account identity; monotonic version	Ordering and conflict detection
business operation ID	Stable across delivery retry	Exactly-once effect identity
causation/correlation IDs	Required ancestry and end-to-end trace	Reconstruct decision/effect chain
schema version/type	Registered, compatibility-governed	Safe evolution and quarantine
occurred time/recorded time	Source time plus durable-ingest time	Ordering diagnosis and latency
producer	Authenticated workload identity	Ownership and authorization
policy/configuration version	Immutable approved snapshot IDs	Decision reproducibility
evidence references	Opaque internal links, no secrets/PII payload	Trace external observation safely

Table 25. Retry and replay policy separates transport recovery from business authority.

Condition	Automated action	Business posture	Operator evidence
Transient broker/consumer outage	Bounded exponential back-off with jitter; preserve partition order	Effect not assumed; source state queryable	Attempt/lag/age metrics
Exact duplicate	Inbox acknowledges existing effect/result	No new transition, journal, or provider call	Duplicate count and original link
Late lower aggregate version	Retain/audit, no regression; query owner if needed	Monotonic business state preserved	Late-event diagnostic
Same version, different payload	Quarantine and security/data-integrity incident	No guessed winner	Both hashes/producers
Unknown future schema	Quarantine; do not dead-letter into oblivion	Owner upgrades or explicitly rejects	Schema, producer, payload retention policy
Poison/corrupt message	Stop affected partition or isolate per policy	No unbounded retry storm	Sanitized payload, parser error, owner
Technical dead letter	Controlled queue with owner/SLA, then source query	Not a business rejection	Retry history and disposition
Business exception	Case/event in business exception workflow	Funds/state remain under explicit posture	Decision, limits, exposure
Authorized replay	New replay job ID; original event IDs preserved	Inbox prevents duplicates; missing effects may execute	Approver, scope, before/after counts

Retention must exceed the longest approved recovery, dispute, replay, and audit window, subject to privacy and legal policy. The exact duration is **UNKNOWN** until product/corridor requirements are approved. Replay requires scoped authorization and dry-run reporting; it is not an unrestricted broker console action.

Participant webhooks carry intent ID, monotonic status version, occurred time, event ID, contract version, and a query link. They are signed with rotated participant-specific credentials, deduplicated by event ID, retried on an approved

schedule, and do not expose policy reasons or sensitive chain data beyond contract need. A webhook is notification; the authenticated status API is the participant source of truth.

Consequence. Duplicate, late, reordered, incompatible, and poison messages each have deterministic handling. At-least-once transport remains visible rather than being renamed exactly once.

11. Failure model and containment

Decision resolved. Uncertainty triggers inquiry and containment; it never triggers blind repetition of a financial effect.

PROPOSED

Failures are classified by what is known, not by transport exception class. A *pre-effect failure* occurred before an external mutation could escape and may retry the same authorized operation within bounds. An *ambiguous submission* may already have an effect and requires inquiry; another attempt is prohibited until route-specific evidence establishes retry safety. A *reverted execution* proves a particular transaction did not complete, but a corrected attempt still requires current nonce, authorization, inventory, quote, and policy checks. A *completed external effect* is never repeated; the platform observes, records, reconciles, and if necessary performs a separately authorized downstream recovery. Business, permanent, poison/corrupt, and security failures retain their distinct authorization and containment rules.

Table 26. Failure-domain taxonomy maps every lifecycle boundary to a class, posture, containment, and owner.

Domain	Likely class/state	Evidence and containment	Primary owner
Validation/contract	Permanent or poison; REJECTED	Deterministic schema/auth result; isolate corrupt parser input and version conflict	Participant API/intent
Policy/eligibility	Business retry, hold, or permanent; MANUAL_REVIEW/REJECTED	Preserve facts/rule versions and obligations; no downstream effect while unresolved	Risk/compliance
Funding/reservation	Safe transient or business retry; FAILED_RETRYABLE	Ledger operation result and account version; bound reservation age and exposure	Ledger/participant
Quote/route	Safe transient or business retry	Quote identity/expiry and capability snapshot; never substitute silently after acceptance	Router/treasury
Liquidity/treasury	Business retry or permanent for route	Position/reservation/limit versions; queue, rebalance, cap, or disable route	Treasury/risk
Signing/custody	Safe transient, ambiguous, permanent, or security incident	Signature identity, policy/quorum record, custody inquiry; disable key/route on uncertainty	Security/custody
Submission	Ambiguous until effect and retry safety are classified; SUBMITTED/MANUAL_REVIEW	Stable transaction/operation identity and independent inquiry; never blind replay	Settlement adapter
Confirmation/finality delay	Safe waiting or incident; CONFIRMING	Independent block/receipt lineage and network health; stop promotion, not observation	Chain risk/observer
Reorganization/conflict	Ambiguous or security/network incident	Preserve competing lineage; contain network/value band and quantify downstream exposure	Chain risk/incident command
Issuer/contract control	Permanent or security/legal incident	Admin/code/freeze/issuer evidence; pause asset and retain redemption/migration options	Treasury/security/legal

Domain	Likely class/state	Evidence and containment	Primary owner
FX and payout	Safe transient, ambiguous, business retry, or permanent; PAYOUT_PENDING/MANUAL_REVIEW	Quote/fill/payout operation and provider/destination inquiry; retain payable	Treasury/payout operations
Notification	Safe transient; business state unchanged	Monotonic status version and delivery receipt; query remains authoritative	Participant operations
Reconciliation	Ambiguous/conflict; RECONCILING/MANUAL_REVIEW	Freeze source snapshots, age/quantify break, suspense only by approved journal	Accounting/reconciliation
Region/platform	Safe transient or disaster; writes paused until fenced ownership	Fencing token, recovery point, invariants, backlog and provider exposure	SRE/ledger owner
Operator action	Security incident or governed business repair	Immutable proposal/approval/action evidence; revoke scope and reconstruct effects	Case/security/accounting

Transport handling for these classes is defined in table 25; that table is subordinate to this business classification and never authorizes an external effect by itself.

Table 27: Failure truth records. The database and blockchain success/failure quadrants are represented explicitly.

Local commit failed; external absent

State and ledger posture. Prior durable state; no journal/effect operation committed

Inquiry/evidence. Local transaction result, outbox absent, provider/chain evidence if a call could have escaped

Retry rule. Retry the deterministic command with the same operation ID only after pre-effect posture and retry safety are established

Customer status. Processing/delayed

Operator action/exposure. Alert if commit outcome itself uncertain; exposure zero only when no effect is established

Local commit failed; external succeeded (blockchain success/database failure)

State and ledger posture. MANUAL_REVIEW; create recovery journal/state only after external identity linked

Inquiry/evidence. Independent chain/custody/provider evidence, logs, signed hash/nonce

Retry rule. Never repeat external effect; repair internal record through governed recovery

Customer status. Processing; do not claim failure

Operator action/exposure. Contain route, quantify unrecorded asset movement, four-eyes repair

Local commit succeeded; external failed (database success/blockchain failure)

State and ledger posture. Reservation/journal and operation remain; FAILED_RETRYABLE or COMPENSATING

Inquiry/evidence. Provider error plus independent proof of absence/revert

Retry rule. Same signed transaction/operation only if safe; otherwise new authorized attempt or compensate

Customer status. Delayed/failed after policy

Operator action/exposure. Age reservation; release only with proven posture

Submitted; provider timeout

State and ledger posture. SUBMITTED or MANUAL_REVIEW; in-transit posture retained

Inquiry/evidence. Transaction hash/nonce, independent nodes, custody/provider inquiry

Retry rule. No blind resubmit; absence threshold and replacement policy required

Customer status. Processing

Operator action/exposure. Monitor ambiguity age and max exposure

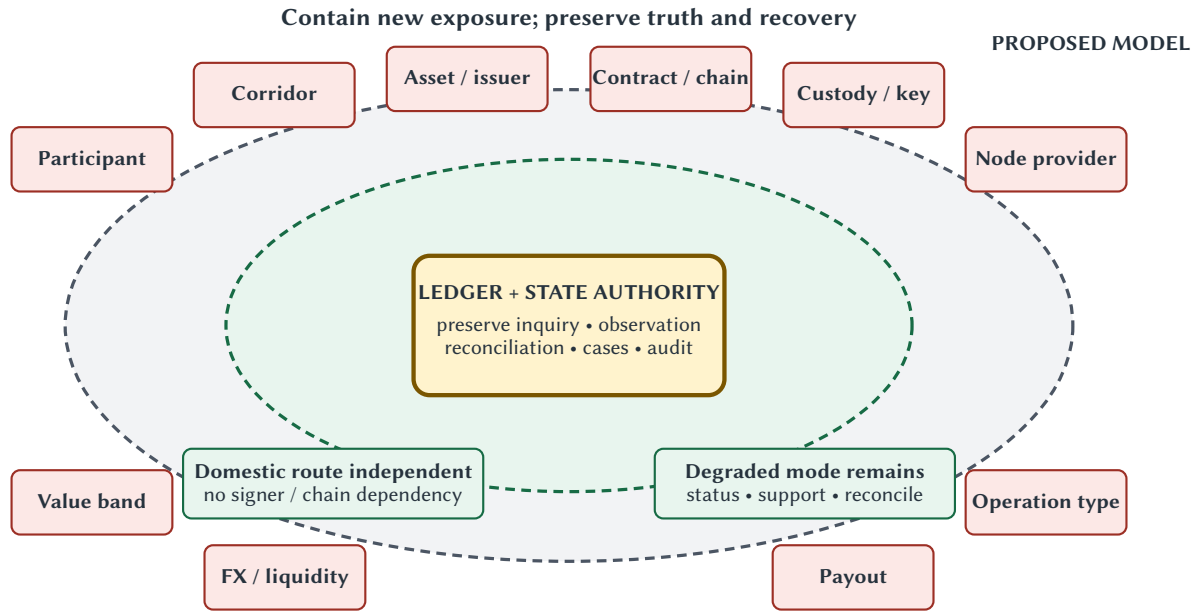


Figure 14: Failure containment and degraded modes. Kill switches are scoped by participant, corridor, asset, issuer, contract, chain, signer, provider, and operation; domestic service remains independent.

Confirmed then reorganized

State and ledger posture. CONFIRMING; or incident if risk-accepted payout occurred

Inquiry/evidence. Multiple node/block lineage observations and custody/contract state

Retry rule. Observe canonical outcome; do not replay blindly

Customer status. Processing; if already credited, preserve customer status and handle incident

Operator action/exposure. Freeze affected route/value, quantify at-risk payout

Economically final; payout failed

State and ledger posture. PAYOUT_PENDING/MANUAL_REVIEW; payout payable remains

Inquiry/evidence. Payout provider plus destination institution inquiry

Retry rule. Retry with same payout ID only if provider contract guarantees safety; else authorized alternative

Customer status. Processing/delayed; completed only if policy already absorbed risk

Operator action/exposure. Bound outstanding obligations; reroute/refund/compensate per corridor

Payout complete; later compliance concern

State and ledger posture. Recipient credit/journals remain; restricted case/recovery accounts as legally directed

Inquiry/evidence. Original facts, designation times, screening version, legal instructions

Retry rule. No automatic reversal; legal/case decision

Customer status. Do not misstate completed history; required communication only

Operator action/exposure. Sanctions/AML incident process, evidence preservation, reporting

Conflicting external evidence / reconciliation mismatch

State and ledger posture. RECONCILING or MANUAL_REVIEW; suspense only via balanced authorized entry

Inquiry/evidence. Independent node/provider/issuer/custody/bank sources with timestamps

Retry rule. No mutation retry until authority/conflict resolved

Customer status. Completed or processing per known customer gate

Operator action/exposure. Pause narrow scope, assign owner, age and quantify difference

The most dangerous quadrant is external success with missing local commit. It can occur when an application reaches a provider before durably recording the operation, or when a provider-side action succeeds while the acknowledgement and local recovery state are lost. The architecture reduces this risk by persisting authorization and stable transaction identity before submission, but cannot erase it. Recovery imports external truth through a case-controlled transition and balanced journal. Operators do not insert a success flag by hand.

Kill switches are versioned configuration with named approvers, expiry/review time, reason, scope, and audit. A global “off” is necessary but insufficient. Operators need narrower controls for participant, corridor, asset, issuer, contract, network, custodian, signer key, node provider, FX provider, payout provider, value band, and operation type. Every switch is exercised in simulation and pilot drills; an untested runbook is not a control.

Table 28: Degraded-mode capability. A safe mode narrows new exposure while preserving inquiry, evidence, and customer support.

Trigger	Disabled	Remains available	Exit evidence
Signer/custody outage	New affected signatures	Domestic route, status, observation, reconciliation, support	Provider recovery plus signing/inquiry drill
Node disagreement/chain incident	New submissions and finality promotion on network	Independent observation, cases, unaffected routes	Canonical convergence and chain-risk approval
Issuer/contract control incident	New asset use, mint/redeem/transfer as scoped	Reconciliation, existing-customer support, alternative approved routes	Issuer evidence, exposure reconciliation, legal/security approval
FX/payout outage	New affected corridor after capacity limit	Existing payout inquiry, status, alternative corridors if independently approved	Backlog reconciled and recovery test
Reconciliation integrity loss	Exposure-increasing operations for affected source/account	Read/query, domestic/unaffected routes, incident evidence	Source integrity restored and break set resolved
Policy service outage	New intents requiring unavailable decision	Query/support; routes with explicitly approved fail-safe policy only	Current policy service and decision audit healthy
Regional control-plane outage	New writes according to ledger ownership/failover policy	Read/status where consistent; alternate region only after fenced takeover	Fencing, recovery point, invariants and owner approval

12. Four independent finalities

Decision resolved. Blockchain, legal settlement, customer-visible completion, and accounting finality are four independently owned policy gates. **PROPOSED**

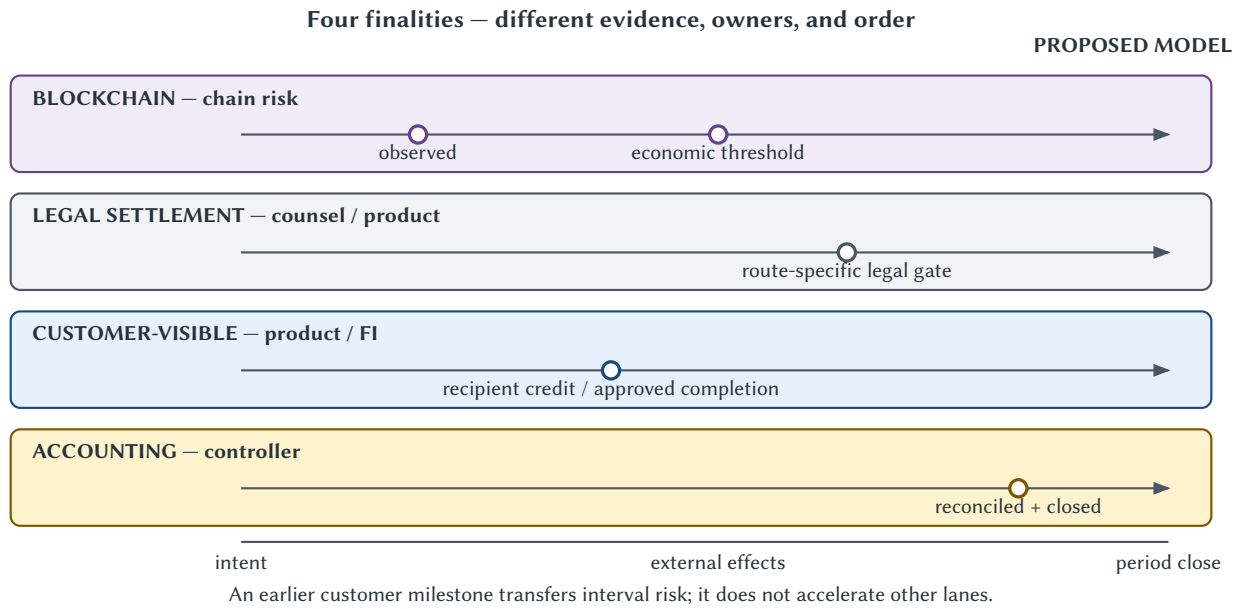


Figure 15: Four-finality timeline for a cross-border flow. The lanes can complete in different orders; a policy-approved early customer credit transfers interval risk to an accountable party.

Table 29. Finality decision matrix. Threshold values are corridor and route decisions, not universal constants.

Finality	Required evidence	Policy owner	Reversal/remediation	Customer consequence
Blockchain finality	Expected transaction/contract/log, canonical block lineage, independent node agreement, chain-specific threshold, no disqualifying contract/network incident	Chain risk / engineering	Protocol may reorganize before/under threshold; after threshold loss handled by route incident policy	Normally internal; may gate payout or status
Legal settlement finality	Product/corridor legal analysis of parties, instrument, rail, terms, insolvency, revocation and applicable law	Legal/product governance	Rights/remedies follow governing agreements and law; paper gives no opinion	Defines disclosures, rights, loss allocation and when obligation is legally discharged
Customer-visible completion	Product policy plus required source debit/reservation, recipient credit or approved risk-absorption evidence, disclosure state	Product/participant operations	Customer remedy/compensation may be owed even if chain is irreversible	Controls “completed,” “processing,” “failed,” and support response
Accounting finality	Balanced journals, external reconciliations, valuation, cutoff, approved break disposition and close	Controller/accounting	Later corrections are new dated journals; closed history not edited	Usually invisible; supports statements, financial reporting and audit

Ethereum documents proof-of-stake finality as depending on validator votes and notes that finality can be delayed when a blocking share does not vote (Ethereum Foundation 2026). **DOC** This is protocol evidence, not a legal conclusion.

“Twelve confirmations” or any other number is only a route policy parameter supported by chain-specific analysis, transaction value, reorg history, network health, contract behavior, and loss tolerance.

Two orderings illustrate the model:

1. **Conservative payout.** Chain threshold is met; destination payout succeeds; the recipient is credited and customer status becomes completed; reconciliation then establishes accounting finality. Legal finality conditions may be evaluated across more than one of these events.
2. **Risk-accepted early credit.** A destination institution credits the recipient before the source chain reaches the platform’s economic threshold. Product policy identifies the party absorbing reorg/default risk and caps exposure. Blockchain finality follows later; accounting close follows reconciliation. The earlier customer status does not accelerate the chain.

Conversely, a chain transfer can be economically final while payout is still pending, as figure 11 shows. Customer-visible completion must then remain “processing” unless an approved product contract defines another outcome. A payout can complete before all accounting sources reconcile; the customer should not be regressed because a back-office break opens, but the break remains an operational and accounting obligation.

Consequence. `ECONOMICALLY_FINAL`, `RECIPIENT_CREDITED`, and `RECONCILED` are distinct because they answer different questions. Legal finality remains an explicit corridor-specific unknown until qualified counsel and governing contracts decide it. `UNKNOWN`

Table 30: Product/corridor finality policy template. Every selected route must complete this record before production.

Field	Required decision
Identity	Product, corridor, source/destination institutions, legal entities, asset, issuer, network, contracts, custody, FX and payout route versions
Blockchain gate	Observation sources, threshold algorithm, value bands, network-health constraints, reorg handling and escalation
Legal gate	Counsel-approved rule, governing agreements, settlement moment, revocation/insolvency treatment, unknowns and review date
Customer gate	Status terms, recipient-credit evidence, disclosure, error/dispute path, early-credit risk owner and maximum exposure
Accounting gate	Journals, statements, valuation, tolerance, cutoff, break disposition, close owner and evidence retention
Pause/recovery	Kill-switch scope, safe modes, compensation authority, notification, drills and restart approvers

13. Security architecture

Decision resolved. Make authorization of canonical transaction bytes independent of application intent, and make key/custody authority a segregated trust boundary. **PROPOSED**

Threat actors include an external attacker, malicious or compromised participant, compromised application workload, provider insider, operator with excessive privilege, software-supply-chain attacker, chain/contract attacker, and colluding approvers. Protected assets include participant credentials, personal and screening data, ledger authority, route/configuration policy, treasury inventory, signing keys, contract administration, audit evidence, and recovery mechanisms.

13.1. Key roles and signer policy

Keys are separated by production/test environment, legal entity, network, asset, treasury/settlement purpose, fee payment, contract administration, and recovery. Treasury inventory does not share a key with contract administration; recovery authority does not perform daily settlement. Generation and use occur in an approved hardware security module (HSM), MPC system, or institutional custody/signing boundary. “Custody” describes a technical and operating responsibility here; any regulated or qualified-custodian classification requires separate legal approval for the selected entity, role, asset, and jurisdiction. Private key material never enters application memory, logs, CI/CD, tickets, backups, or operator workstations.

NIST FIPS 140-3 defines cryptographic-module security requirements, and NIST SP 800-57 provides lifecycle guidance (National Institute of Standards and Technology 2019; National Institute of Standards and Technology 2020). **DOC** Module validation is necessary evidence for a selected control posture but does not define custody governance, quorum, recovery, or transaction authorization.

As one provider-diligence example, Fireblocks documents role-based access, transaction policies, administrative quorum, MPC key shares, co-signers, and a recovery process (Fireblocks 2026b; Fireblocks 2026a). **DOC** These are documented product behaviors, not a provider recommendation or proof of independent assurance. A selected custodian must still prove contractual asset segregation, exact authorization semantics, key/share ownership, recovery and export, incident evidence, financial condition, jurisdiction, and tested exit.

For every signing operation the signer independently verifies:

- authorized business operation and immutable intent/policy/configuration versions;
- exact canonical unsigned bytes and their cryptographic hash;

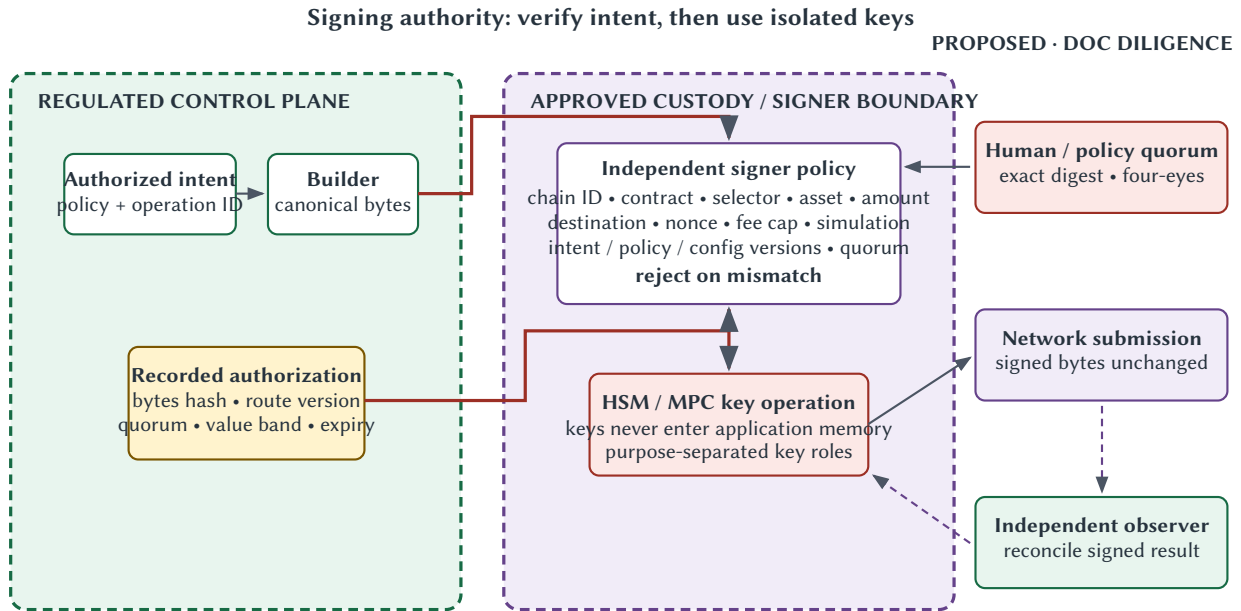


Figure 16: Key and custody authority. The application proposes canonical bytes; an independent policy boundary verifies intent, chain, contract, selector, amount, destination, nonce, fee, and approvals before any key operation.

- network/chain ID, transaction type, nonce, validity window, and replacement policy;
- allowlisted contract and code/version identity where available;
- function selector/instruction, asset, amount, decimals, destination, and memo/opaque intent reference;
- fee/gas ceiling and fee-payer authority;
- simulation result and prohibited state changes;
- value-band quorum, segregation of duties, rate/velocity limits, and kill-switch state; and
- an append-only authorization result usable for independent reconciliation.

The signer rejects caller-supplied fields that do not reproduce the canonical bytes. For higher value bands a second human or system authority approves the exact transaction digest, not a screenshot or mutable form. Recovery keys require a distinct quorum and time delay where supported; their use automatically pauses affected routes and triggers a full signed-transaction and balance reconciliation.

13.2. Contract, node, application, and data controls

ERC-20 standardizes token interface behavior but says nothing about reserves, redemption, custody, or legal rights (Vogelsteller and Buterin 2015). Upgrade patterns such as EIP-1967 make implementation/admin slots discoverable, while libraries such as OpenZeppelin document role and delayed-administration patterns (Worthington et al. 2019; OpenZeppelin 2026). **DOC** The platform inventories contract code hashes, proxy implementations, administrators, pause/freeze/denylist roles, upgrade events, decimals, and approved function selectors. A contract or administrator change pauses promotion until reviewed; a library name never substitutes for governance analysis.

Managed nodes use mutually authenticated or strongly authenticated channels, per-workload credentials, egress restrictions, rate limits, and response integrity monitoring. Independent observation uses a different failure and administrative

domain. Nodes and custody callbacks enter a validation queue; they do not reach ledger credentials. Application services use least-privilege workload identities, service-owned schemas, encrypted transport and storage, secret rotation, dependency pinning, signed artifacts, software bills of materials, provenance, and policy-as-code review. NIST's Secure Software Development Framework supplies a baseline, not a system-specific threat model (National Institute of Standards and Technology 2022). [DOC](#)

On-chain data is limited to what the approved transaction and protocol require. Personal identifiers, sanctions results, participant names, case reasons, and reusable customer aliases remain off-chain. An opaque reference is non-guessable, non-personal, scoped, and not a direct database key. Logs redact secrets and sensitive payloads; access to chain analytics and case data is purpose-bound and audited. Retention/deletion policy accounts for the impossibility of deleting public-chain data.

Table 31. Representative threat/control/evidence matrix.

Threat or abuse	Prevent/detect controls	Recovery/containment	Evidence
Participant replay or payload swap	Mutual auth, signed request, key+hash idempotency, quotas	Return original or reject conflict; credential containment	Ingress signature, canonical hash, idempotency record
Application requests wrong transfer	Route-derived bytes, simulation, signer independent field checks, allowlists	Sign refusal; disable work-load/route	Intent, policy, bytes hash, signer decision
Key compromise	HSM/MPC/custody isolation, quorum, purpose/value limits, anomaly monitoring	Pause key/asset/network; rotate/migrate; reconcile signed set	Key audit, all signed hashes, chain/custody state
Contract upgrade/admin abuse	Contract inventory, admin/upgrade event monitoring, code-hash policy, delay where available	Pause asset/contract; value containment; migration plan	Pre/post code/admin state and review
Node lies or is stale	Independent providers/nodes, block/receipt lineage, lag/divergence metrics	Stop finality promotion/submission as scoped	Signed/recorded observations and comparison
Operator bypass	RBAC/ABAC, just-in-time elevation, four-eyes actions, immutable case/audit	Revoke access, contain affected route, reconstruct actions	Identity, approvals, before/after state
Supply-chain compromise	Pinned dependencies, provenance, signed builds/SBOM, isolated CI, secret scanning	Roll back signed artifact, rotate credentials, incident response	Build provenance, manifests, deployment identity
Denial of service/fraud burst	Participant/corridor/value rate limits, queues, capacity reservations, backpressure	Degraded mode, priority inquiry/settlement queues	Rate/queue/exposure telemetry

Table 32. Privileged-operation approval matrix: authority and limits.

Action	Initiator	Authorization and quorum	Limits and automatic alerts
Settlement signing	Workflow service identity	Independent signer policy; value-band system/human quorum over exact digest	Asset, destination, function, value, velocity, fee, validity window; mismatch/refusal alert
Contract administration	Segregated contract-admin role	Named multisignature/quorum plus delayed change where supported; never treasury key	Approved code hash/function and maintenance window; any admin event pauses promotion

Action	Initiator	Authorization and quorum	Limits and automatic alerts
Key/share recovery	Offline recovery custodians	Distinct recovery quorum, time delay, dual control, incident authorization	No routine settlement permission; any invocation pauses key scope and pages security
Route or asset kill switch	On-call security/risk role	Pre-authorized narrow stop; dual approval to re-enable	Scope, reason, expiry, affected exposure; alert participants/operations
Compensating journal or state repair	Case owner proposes	Accounting plus product/legal/risk approval as required; ledger/workflow service executes	Case/value/corridor limits; invariant and customer-remedy alert
Payout reroute or beneficiary change	Payout case owner	Re-screened beneficiary and corridor-specific dual approval	Same economic obligation; no silent destination change; high-risk route alert
Configuration/policy release	Authorized proposer	Segregated reviewer/approver/deployer with signed version	Cohort, effective window, rollback, policy-diff and stale-version alerts
Bulk evidence/data export	Audit/compliance purpose owner	Purpose-bound role plus data-owner approval and logged destination	Class, period, subject count, region, expiry; anomalous export alert

Table 33. Privileged-operation audit and recovery requirements.

Action	Mandatory audit evidence	Recovery or containment
Settlement signing	Intent/policy/config versions, canonical bytes/hash, approvals, signature identity, signed result	Disable workload/key/route; reconcile every signed hash and controlled balance
Contract administration	Proposal, code/admin before/after state, approvers, delay, execution transaction	Pause asset/contract, revoke role where possible, migrate only through approved plan
Key/share recovery	Incident, custodians, recovered scope, ceremony transcript, new key/address mapping	Rotate/migrate, revoke old authority, reconcile historical and new controlled addresses
Route or asset kill switch	Actor, scope, reason, time, exposure snapshot, notifications, re-enable approval	Preserve inquiry/reconciliation; resume only after evidence-backed disposition
Compensating journal or state repair	Original effects, plan, legal/accounting rationale, journal IDs, state versions, customer action	Reverse only by new journals/events; retain loss/recovery and reconciliation case
Payout reroute or beneficiary change	Original/new destination tokens, screening, approvals, provider operation IDs	Stop duplicate path, inquire both providers, retain payable until one outcome is proven
Configuration/policy release	Signed artifact, diff, tests, approvals, deployment cohort, observed metrics	Roll back to signed prior version or disable route; preserve both decision versions
Bulk evidence/data export	Purpose, query, classes, region, recipient, approval, checksum, deletion/expiry proof	Revoke link/credential, contain destination, privacy/security incident process

Table 34: Signing technology and key-temperature comparisons.

Option	Strength for this design	Limitation and operating burden	Pilot posture
Hardware security module (HSM)	Mature hardware boundary, standardized module assurance, direct key control	High-availability clusters, backup/ceremony, firmware, integration and specialist staffing remain operator duties	Accept only inside approved custody/provider or proven institutional service
Multi-party computation (MPC)	Distributes signing shares and can express organizational quorum without reconstructing one key	Protocol/provider implementation, share governance, recovery/export, latency and assurance require independent diligence	Accept conditionally through an approved provider; do not treat “MPC” as assurance or a legal custody classification by itself
Hot operational key	Lowest signing latency for bounded routine settlement	Highest online exposure; requires narrow purpose, value/velocity limits, rapid disable and continuous monitoring	Only for tightly capped operational inventory
Warm approval key	Additional human/system quorum and delay reduce autonomous compromise impact	Higher latency and on-call/approval dependency can block time-sensitive settlement	Higher value bands, administrative actions and controlled replenishment
Offline/recovery key	Strong separation from routine workloads	Ceremony, availability, geographic custody, obsolescence and untested recovery risk	Recovery only; periodic witnessed exercise and automatic route pause on use

Security drills cover signer unavailable, key loss, suspected key compromise, malicious transaction construction, node inconsistency, contract upgrade, custody exit, and restoration from independently verified balances. Vendor exit must identify transaction/authorization export, address/key migration, record retention, recovery cooperation, and service continuity before onboarding.

14. Compliance as versioned decision evidence

Decision resolved. Compliance is an overlay across intent, funding, execution, payout, monitoring, and case handling—not a one-time screening API. **PROPOSED**

The architecture records a decision as facts considered, rules/model versions, timestamp, accountable service/role, outcome, reason category, obligations, expiry/re-screen trigger, and evidence references. It avoids putting sensitive source data in broad event payloads. Policy can allow, reject, hold, require enhanced review, cap value, constrain route, require information exchange, or mandate post-payment monitoring. A decision expiring before signing or payout is re-evaluated; the new result never rewrites the prior decision.

The GENIUS Act became U.S. Public Law 119-27 on 2025-07-18 and contains a payment-stablecoin issuer framework including permitted-issuer, reserve, redemption, disclosure, and Bank Secrecy Act provisions (United States Congress 2025). **PUBLIC** FinCEN’s analysis of convertible virtual-currency business models depends on the facts and roles involved (Financial Crimes Enforcement Network 2019). OFAC states that sanctions obligations apply to virtual-currency transactions as they do to fiat transactions (Office of Foreign Assets Control 2021). FATF guidance addresses risk-based controls and information sharing for virtual-asset transfers (Financial Action Task Force 2025). Regulation E establishes U.S. consumer electronic-fund-transfer protections for covered transfers (Consumer Financial Protection Bureau 2026). **PUBLIC** These sources motivate controls; none is a legal conclusion for a particular structure.

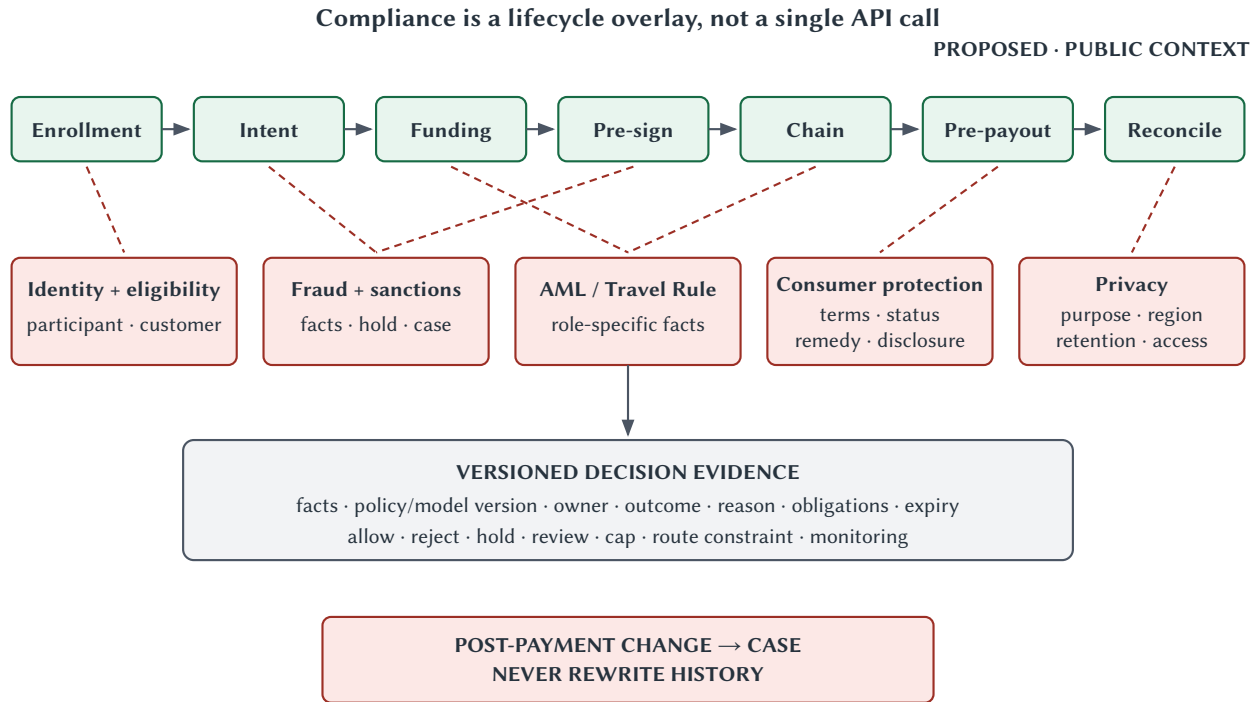


Figure 17: Compliance decision overlay. Identity, sanctions, fraud, transaction monitoring, Travel Rule analysis, consumer protection, privacy, and case obligations are evaluated at different lifecycle gates.

Table 35. Compliance responsibilities remain open until legal entities, roles, products, and corridors are selected.

Obligation	Lifecycle gates	Architecture evidence	Required owner decision	Status
Participant/customer identity	Enrollment, intent, recipient resolution	Identity assertions, assurance, participant source, expiry	Which party performs/relies on which checks	UNKNOWN / route gate
Sanctions screening	Intent, address/contract, pre-sign, pre-payout, designation change	List/version, facts, decision, hold/case, transaction linkage	Screening scope, false-positive, blocked/rejected handling	UNKNOWN / route gate
Fraud and scams	Enrollment, intent, velocity, recipient, post-event	Signals, warnings, participant/customer actions, model/rule version	Allocation, intervention, reimbursement, escalation	Existing context + new route design
AML monitoring/reporting	Funding, transaction, payout, linked behavior	Source/destination, asset/network, amounts, counterparties, alerts/cases	Regulated roles, filing/reporting and retention	UNKNOWN / route gate
Travel Rule/information exchange	Before qualifying transfer/payout	Counterparty classification, required fields, secure exchange/receipt	Jurisdiction, threshold, role and data minimization	UNKNOWN / route gate
Consumer protection	Terms, authorization, status, error/dispute, remedy	Consent, disclosures, status history, investigation and disposition	Product coverage, error rules, liability and timing	UNKNOWN / route gate

Obligation	Lifecycle gates	Architecture evidence	Required owner decision	Status
Privacy/data localization	Collection, processing, sharing, retention/deletion	Purpose, classification, region, access, provider and deletion record	Applicable law and cross-border transfer mechanism	UN-KNOWN / route gate
Blocked/restricted property	Hold, post-designation, case disposition	Restricted ledger posture, legal order, access controls, reports	Custodian/issuer/operator ownership and permissible action	UN-KNOWN / route gate

Table 36: Data-classification, residency, and public-chain exposure records.

Participant/customer identity and aliases

Region/residency decision. Participant and corridor law; approved processing region
Cross-boundary transfer. Tokenized/minimized fields only under approved role and mechanism
Retention and access. Product/legal schedule; participant support, identity, compliance roles
On-chain exposure. Prohibited; no names, email, phone, account, or reusable alias
Owner. Product/privacy/participant

Beneficiary and payout instructions

Region/residency decision. Destination corridor and provider location are route-approval gates
Cross-boundary transfer. Purpose-bound payout payload over approved channel
Retention and access. Through remedy/reconciliation plus legal schedule; tightly scoped payout/case access
On-chain exposure. Prohibited; destination chain address only when the approved rail requires it
Owner. Payout/privacy/legal

Screening, fraud, and case evidence

Region/residency decision. Region follows regulated-role and provider decisions
Cross-boundary transfer. Minimum facts under information-sharing and vendor terms
Retention and access. Regulatory/case schedule; need-to-know immutable access audit
On-chain exposure. Prohibited; no score, match, hold, or case reason
Owner. Compliance/fraud

Payment intent and participant status

Region/residency decision. Operator region plus participant contract
Cross-boundary transfer. Stable opaque intent/status contract; no unnecessary sensitive facts
Retention and access. Product, dispute, accounting, and audit schedule
On-chain exposure. Opaque non-personal reference only if required and approved
Owner. Product/operations

Ledger journals and valuations

Region/residency decision. Legal-entity accounting location and approved recovery region
Cross-boundary transfer. Controlled finance/participant reporting and disaster-recovery copies
Retention and access. Financial-record schedule; append-only finance/audit access
On-chain exposure. Never as the sole ledger; chain transaction identifier may be referenced
Owner. Accounting/treasury

Canonical transaction and signature evidence

Region/residency decision. Custody/network jurisdictions and operator evidence store
Cross-boundary transfer. Signer/custody/node interfaces; immutable hash/evidence export
Retention and access. Key/transaction/audit schedule; signer, security, reconciliation access

On-chain exposure. Protocol-required transaction fields are public and effectively permanent

Owner. Security/custody

Chain observations and analytics

Region/residency decision. Public data plus provider processing locations

Cross-boundary transfer. Node/analytics providers receive addresses and transaction context

Retention and access. Route/reconciliation schedule; restrict enrichment and model output

On-chain exposure. Intrinsically public for public-chain routes; never enrich on-chain

Owner. Chain risk/privacy

Telemetry and logs

Region/residency decision. Service region and approved observability platform

Cross-boundary transfer. Redacted structured signals only; no secrets or raw sensitive payloads

Retention and access. Short operational schedule unless promoted to audit evidence

On-chain exposure. None

Owner. Platform/SRE/privacy

Backups, exports, and recovery artifacts

Region/residency decision. Same or explicitly approved recovery jurisdiction

Cross-boundary transfer. Encrypted, inventoried, purpose-bound transfer

Retention and access. Tested expiry/deletion; break-glass access and export audit

On-chain exposure. None beyond already-public transaction data

Owner. Data owner/security

Address analytics is a risk signal, not identity or legal truth. The system records provider/model/version, input addresses and transaction context, result, timestamp, threshold, and human disposition. A post-payment concern cannot erase a completed journal or promise a reversal of finalized value. It invokes the approved legal/case process, restricts assets or routes where authorized, preserves evidence, and records any compensation or reporting separately.

Consequence. Legal, sanctions, consumer, privacy, and role decisions remain visible route-approval gates. Architecture makes those decisions executable and auditable; it does not make them on behalf of counsel or regulators.

Table 37: Customer-obligation comparison: visible token product versus invisible settlement rail.

Obligation	Customer-visible token product	Invisible settlement rail	Decision owner
Instrument disclosure	Explain issuer, token rights, custody/wallet, network, redemption, fees and admin powers	Explain cross-border payment, exchange/payout terms, status, fees and remedies without implying customer token ownership	Product/legal
Authorization and security	Wallet/key or custodial authorization, address risks, token-transfer consequences	Existing bank authentication plus payment intent; no seed phrase, gas, or direct address handling	Participant/product/security
Finality and status	Distinguish chain confirmation, legal rights, redemption and possible freeze/reorg conditions	Customer completion follows corridor policy and recipient-credit evidence, not chain status alone	Product/legal/risk
Error, dispute, and remedy	Define token/custody error paths and limits on reversing public-chain effects	Preserve bank support path and applicable transfer investigation/remedy while external settlement may be irrevocable	Participant/support/legal
Privacy	Disclose public address/transaction visibility and analytics	Keep personal data off-chain; disclose provider/cross-border processing appropriate to the product	Privacy/product
Exit and recovery	Customer redemption/transfer, wallet or custody migration, lost-access treatment	Operator route disable, payout remedy, reconciliation and provider migration remain behind stable participant contract	Treasury/operations/legal

15. Java architecture and language boundaries

Decision resolved. Use Java/Spring for the enterprise control plane; choose chain-native, custody, cryptographic, and node implementations by ecosystem and operating fit. **PROPOSED**

Java is appropriate where the dominant work is authenticated APIs, domain invariants, orchestration, relational transactions, policy integration, event handling, reconciliation, observability, and long-lived enterprise operations. It is not a reason to implement private-key cryptography, an execution or consensus client, or smart contracts in Java when the selected ecosystem has a stronger native implementation and security toolchain.

An Early Warning posting for Principal Engineer–Digital Assets requires hands-on design and development experience using “one or more” of Java REST services and related Java tooling, Spring, enterprise messaging frameworks, and Oracle PL/SQL; it separately names cloud platforms, delivery tooling, monitoring products, ACH/EFT, RTP, and FedNow (Early Warning 2026). **PUBLIC** Public role evidence therefore associates Early Warning’s cross-border and digital-asset engineering work with Java REST services, Spring, enterprise messaging, relational database technology, cloud platforms, and established payment rails. This supports Java as a credible organizational fit but does not disclose the deployed Zelle architecture.

Oracle PL/SQL, Kafka, Tibco EMS, JMS, Splunk, AppDynamics, and Zabbix are role signals rather than proof of one production database, broker, or monitoring stack. PostgreSQL and OpenTelemetry remain proposed target choices; Kubernetes or managed-cloud deployment likewise remains a proposed operating posture. The posting does not establish that any of those proposed choices is used by EWS.

IBM documents Spring Boot REST services in one participant-side Zelle integration product (IBM 2026a). **DOC** This is a concrete example of Java at an institutional integration boundary, not evidence that EWS uses Spring Boot, that every participant uses IBM, or that Java is required.

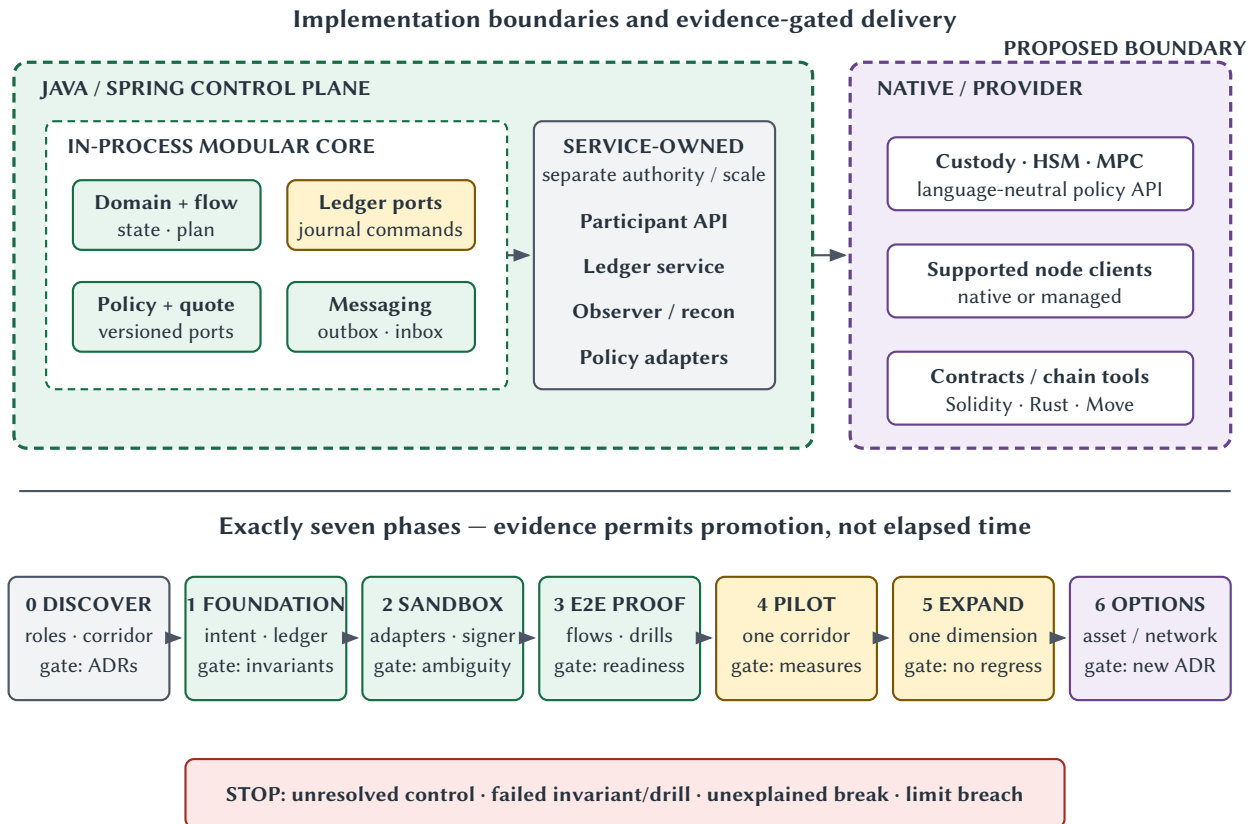


Figure 18: Java/native boundary and phased delivery. The upper panel assigns enterprise control to Java and chain-native execution to supported ecosystems; the lower panel makes promotion evidence-gated across exactly seven phases.

15.1. Module and dependency model

A practical starting point is a modular control-plane deployment, split only where ownership, availability, security, scaling, or data authority differs. Domain modules define payment intent, money/asset quantities, journal commands, state transitions, route capability, evidence, and operation IDs without depending on Spring or a chain SDK. Application modules implement command handlers and ports. Adapters implement HTTP/event/persistence/provider contracts. Deployment modules wire configuration, security, telemetry, and runtime concerns.

Ledger service, signer/custody boundary, and chain observer deserve separate security and availability ownership even if other modules initially share a deployable unit. A service split must not create distributed transactions between objects that require one invariant. Conversely, a modular monolith must not share tables or in-process bypasses that obscure future ownership.

The full language and build/buy responsibility matrix is collected in section C. The controlling boundary is concise: Java owns business authority and enterprise integration; chain-native languages and supported provider or node ecosystems own contracts, clients, and specialist cryptography.

Web3j can construct EVM transactions, bind contracts, call JSON-RPC, decode events, and integrate with HSM interfaces (LF Decentralized Trust Web3j Project 2026b; LF Decentralized Trust Web3j Project 2026a). **DOC CODE** It is appropriate inside a bounded EVM adapter when its version supports the selected network and transaction types and conformance tests compare encoded bytes with native tooling. It is not a node, signer policy, operational ledger, or end-to-end reliability mechanism.

Java virtual threads can support high-throughput thread-per-request code with compatible blocking libraries (Pressler and Bateman 2023). **DOC** They do not remove provider latency, database contention, connection-pool limits, or backpressure. Select conventional blocking plus virtual threads, reactive APIs, or a mix only after measuring workload and library behavior. Durable workflow, not an open thread, represents a payment waiting on finality or a case.

Testing layers include pure domain/property tests for balance and transitions; database integration and concurrency tests; contract/schema compatibility tests; golden transaction encoding and simulation vectors; provider sandbox/record-replay tests; fault injection for ambiguous effects; reconciliation fixtures; and end-to-end drills with duplicate, late, reorged, and conflicting evidence. OpenTelemetry provides standard traces, metrics, and logs (OpenTelemetry 2026); business audit remains a separate immutable record. **DOC**

16. Technology evaluation

Decision resolved. Select technologies after ownership, invariant, failure, regulatory, and exit obligations; a familiar implementation language is not a selection criterion. **PROPOSED**

All evaluations are as of 2026-07-13. “Performance” below identifies the scaling axis and required measurement; it does not repeat incomparable vendor benchmarks. Fatal constraints—custody/legal approval, data authority, recovery, licensing, or missing required protocol behavior—cannot be averaged away.

16.1. Enterprise control and data plane

The enterprise recommendation is Java/Spring for the control plane, an approved relational database that proves ledger invariants, a durable broker only as transport, a workflow product only if domain state remains authoritative, OpenTelemetry for operational signals rather than audit, and owned reconciliation semantics. The detailed control/data-plane comparison is collected in section C.

The relational database recommendation is a class, not an unconditional PostgreSQL procurement choice. PostgreSQL documentation establishes relevant transaction behavior (PostgreSQL Global Development Group 2026); the organization should prefer an already approved relational platform if it proves the unique constraints, isolation, partition ownership, recovery, audit, and performance model. Developer experience must include deterministic local tests and inspectable SQL. A workflow product is valuable only if it preserves domain-owned state semantics, deterministic versioning, export, and incident recovery.

16.2. Distributed-ledger and integration options

The DLT/SDK posture remains conditional: Web3j may serve a bounded EVM adapter; Besu and Hedera/Hiero are options to evaluate rather than defaults; Corda is a strategic permissioned alternative rather than an initial public-stablecoin rail; bitcoinj does not meet the initial stablecoin obligation; and a custom/private blockchain is rejected for the pilot. The complete comparison is collected in section C.

Besu is an Apache-licensed Ethereum execution client written in Java and can operate with public or private networks; public Ethereum operation also requires a consensus client (Besu Project 2026b; Besu Project 2026a). **DOC CODE** Those facts make Besu an option to evaluate if independent node operation becomes an evidence or concentration requirement. They do not make a permissioned network or self-operated node the default.

Hedera provides Java SDKs and EVM-compatible contracts while its consensus and mirror-node model remains distinct; the Hiero repository contains the Java consensus-node implementation (Hedera 2026; Hiero Project 2026). Corda documents a permissioned application-network model with identities, flows, UTXO options, and notary uniqueness (R3 2026; R3 2024). bitcoinj supplies Bitcoin protocol/wallet functionality, not a general stablecoin platform (bitcoinj Project 2026b; bitcoinj Project 2026a). **DOC CODE** Source language is never evidence of Zelle use or network suitability.

16.3. Comparable technology dossiers

Table 45 makes the twelve evaluation questions visible for every required technology. “Performance” names an axis to measure; no cell is a benchmark claim. Product selection remains conditional where the organization has not supplied its approved platform, support model, workload, data jurisdiction, or recovery objectives. Versions are the reviewed baselines, not an instruction to bypass enterprise patch governance.

The comparable twelve-question dossiers are collected in section C. The concise evaluations above preserve the Java boundary, pilot posture, alternatives, and decision triggers without carrying the full diligence record in the primary narrative.

16.4. Custody, nodes, and operating ownership

The pilot buys or integrates specialist custody/signing, node access, and analytics capabilities behind owned policy and evidence contracts; it evaluates an existing durable-workflow platform if suitable and always retains reconciliation semantics and cases. The detailed build/buy triggers and exit concerns appear in section C.

ASSUMPTION Managed node access plus an independent read path is the pilot posture, subject to provider failure-domain, data, integrity, and recovery tests. The platform records both observations and their provenance; “two providers agree” is not independent if both proxy the same upstream or share administration.

Consequence. The recommended stack is intentionally incomplete until route approval selects legal entities, asset, network, custody, node, FX, and payout contracts. Architecture contracts make those decisions substitutable; they do not pretend substitution is free.

17. Stablecoin and settlement-asset options

Decision resolved. Build issuer/network portability into contracts, but launch one approved combination; do not assume an announced proprietary asset is pilot-ready. **PROPOSED**

Asset selection is not a ticker-symbol choice. It selects an issuer and legal claim; reserve and redemption model; custody and contract; mint, burn, freeze, denylist, pause, and upgrade authorities; supported networks; liquidity and market makers; accounting; privacy; incident response; and an exit path.

Circle describes USDC as multichain, backed by highly liquid assets, and redeemable 1:1 under its product terms; Paxos publishes issuer and reserve information for PYUSD (Circle 2026b; Paxos 2026). **DOC** These facts make external regulated

assets diligence candidates, not automatic selections. Eligibility, direct redemption, legal rights, freezes, supported networks, custody, liquidity, jurisdiction, and contract/version still require approval.

PUBLIC EWS has announced ZelleUSD for potential future capabilities in markets other than the first announced India route (Early Warning Services 2026a). **UNKNOWN** Public evidence does not establish the asset's issuer entity, authorization, reserves, redemption, network, contracts, custody, keys, controls, partners, circulation, or operating schedule. Accordingly the paper evaluates "future ZelleUSD" as a strategic option whose diligence cells remain open; it neither rejects nor recommends it for the initial pilot.

The bounded pilot uses one externally issued, approved asset as an invisible settlement rail. That posture concentrates issuer risk but keeps state, accounting, custody, reconciliation, and incident handling testable. Multiple external assets are deferred because they multiply route-dependent contracts, liquidity, accounting, and support. Bank-issued tokenized deposits remain a distinct interoperability path; a future ZelleUSD remains a strategic option pending primary operating evidence; and a permissioned token or network is not recommended unless an existing approved consortium proves corridor fit, governance, liquidity, and exit. Detailed option records appear in section C.

Stablecoins and tokenized deposits must remain distinct in the data model. A tokenized deposit can represent a commercial-bank liability subject to that bank/network arrangement; a stablecoin represents a claim and control model defined by its issuer and applicable framework. Both may share an adapter vocabulary, but their legal owner, redemption, insolvency, reserve, liquidity, and finality policy differ.

17.1. Per-option control dossiers

The summary recommendation is not a substitute for diligence. Table 49 exposes every required decision dimension for all six options. "Unknown" is an intentional stop condition, especially for a future ZelleUSD; it is not a neutral score.

The full option dossiers are collected in section C; the primary narrative retains the option categories and initial recommendation.

17.2. Asset due-diligence gate

The mandatory asset diligence questions and stop signals appear in section C. No production route may proceed until each required answer has approved evidence or an explicit stop decision.

18. Tradeoffs and alternatives

Decision resolved. Accept explicit coordination and reconciliation complexity to retain accountable financial control across heterogeneous rails. **PROPOSED**

The full posture, operating-burden, reversibility, change-trigger, and owner records are collected in section D.

A chain-as-source-of-truth advocate correctly notes that public ledgers provide shared, independently observable transaction history. The chosen posture retains that evidence and adds internal obligation truth rather than discarding it. A permissioned-network advocate correctly values identity, controlled membership, and privacy; the countervailing cost is governance, operation, liquidity, and interoperability. A proprietary-asset advocate correctly values strategic control; the countervailing fact is that issuance creates a regulated product and operating system, not merely a contract deployment.

Similarly, managed services reduce time to first learning but create concentration and exit work. Self-operation is not inherently safer: it transfers patching, key recovery, node integrity, incident response, and staffing to the operator. The decision changes only when those obligations can be performed better and proven, not when ownership feels architecturally pure.

Consequence. The initial posture is intentionally bounded and reversible. It optimizes learning about controlled settlement, not breadth of asset marketing or infrastructure ownership.

19. Evidence-gated roadmap

Decision resolved. Increase real-money exposure only after the preceding phase proves ownership, invariants, reconciliation, recovery, and operational control. **PROPOSED**

The roadmap has exactly seven delivery phases, numbered 0 through 6. These delivery phases are distinct from the publication series' Phase 0 foundation. The roadmap is not a calendar: a phase can stop, repeat, or reduce scope, and elapsed time never substitutes for exit evidence. Numeric limits, recovery objectives, and service measures are **ASSUMPTION** values to approve from workload, loss, liquidity, customer, and regulatory data.

The delivery sequence is: Phase 0 resolves legal, asset, route, provider, accounting, and control facts; Phase 1 proves the ledger and control plane; Phase 2 proves sandbox adapters and ambiguity handling; Phase 3 exercises all normal and failure flows with bounded internal value; Phase 4 runs one tightly limited production pilot; Phase 5 expands one dimension at a time; and Phase 6 evaluates strategic options independently. Detailed deliverables, exit evidence, exposure limits, and stop conditions appear in section D.

19.1. Decision records and risk burn-down

The program maintains at least: ADR-001 customer/participant boundary; ADR-002 ledger authority and accounting model; ADR-003 state vocabulary and consistency model; ADR-004 corridor/legal/finality policy; ADR-005 asset/issuer; ADR-006 network/finality; ADR-007 custody/signing/key roles; ADR-008 node/observation; ADR-009 FX/liquidity; ADR-010 payout; ADR-011 data/privacy/compliance roles; ADR-012 Java/module/deployment; ADR-013 messaging/workflow; ADR-014 reconciliation/close; ADR-015 limits/degraded modes; ADR-016 migration/exit. An ADR records owner, evidence, options, decision, dissent, assumptions, operational obligations, test, review date, and change trigger.

The detailed risk burn-down record is collected in section D. The governing order is unchanged: prove ledger and state correctness first, then provider ambiguity, finality, key/custody, liquidity/payout, legal/customer, operational-load, and exit controls before customer exposure expands.

Route disable never deletes obligations. It blocks new exposure, permits inquiry/observation/reconciliation, and allows only explicitly authorized recovery effects. Promotion and rollback preserve the route, policy, schema, contract, and configuration versions that governed every existing payment.

20. Conclusion

The central architecture choice is not a blockchain or token. It is the decision to preserve a regulated payment system's own accountabilities while adding another settlement rail. Participant institutions retain their customer channel and account relationship. The proposed control plane retains intent, policy, internal ledger, durable payment state, participant truth, recovery, and reconciliation. Issuers, custody, signing, chains, FX, and payout providers contribute bounded effects and evidence through explicit trust boundaries.

That allocation makes hard cases representable. A blockchain can succeed while a database write fails; a database can commit while chain submission fails; a transaction can reorganize or remain unfinalized; a finalized token movement can be followed by payout failure; a sanctions fact can change after payment; delivery can duplicate; and external records can disagree. None is repaired by changing a status field or "rolling back the chain." Stable operation requires inquiry, immutable journals, versioned states, compensation, independent observation, cases, exposure limits, and reconciliation.

Java and Spring are strong choices for enterprise APIs, orchestration, ledger-facing services, policy integration, messaging, reconciliation, and operations. They are not a mandate to select a weaker node, reproduce custody cryptography, or move smart contracts out of their native ecosystem. Besu and other DLT technologies remain evaluated options, not defaults. The pilot selects one approved asset and public network through separate decisions, with an approved institutional custody/signing arrangement whose legal classification is separately confirmed, managed node access plus independent observation, one corridor, bounded participants, prefunding, no consumer wallet, and no bridge.

The unresolved facts are material: private EWS topology and contracts; ZelleUSD issuer, reserves, redemption, network and controls; corridor legal roles and finality; selected providers; customer remedies; workload and recovery objectives; liquidity and loss thresholds. They belong in delivery discovery and governed ADRs, not in plausible-sounding prose.

The general lesson extends beyond Zelle: successful stablecoin integration is disciplined payment infrastructure with an additional asynchronous rail. It succeeds when every obligation has an owner, every transition has evidence, every external effect can be inquired and reconciled, and every increase in exposure is earned.

21. Annotated bibliography

Each source appears once. Category tags show the evidence domains it supports, and each annotation states both relevance and limitation. Official statements validate public product facts; standards and regulations validate their text; documentation validates described technology behavior; source repositories validate code and release behavior. None proves private Early Warning Services deployment or final legal applicability.

Access dates use ISO year-month-day format. Long addresses may wrap at punctuation without changing their link targets. The complete claim-level mapping is retained in the separate evidence companion; section H summarizes the evidence classes and totals used in this paper.

Apache Software Foundation (2026). *Apache Kafka 4.1 Documentation*. URL: <https://kafka.apache.org/41/documentation.html> (visited on 2026-07-13)

Categories: Enterprise Integration, Infrastructure, Reliability

Relevance and limitation: Official documentation for the event-streaming reference implementation used in the broker-class evaluation. Kafka delivery and transactions do not make external financial effects atomic.

Bank for International Settlements (2023). *Blueprint for the Future Monetary System: Improving the Old, Enabling the New*. BIS Bulletin 73. URL: <https://www.bis.org/publ/bisbull73.htm> (visited on 2026-07-13)

Categories: Stablecoins, Payment Networks

Relevance and limitation: Policy analysis comparing tokenized deposits, stablecoins, and unified-ledger concepts. It is not a mandate or deployment fact.

– (2026). *Annual Economic Report 2026, Chapter III*. URL: <https://www.bis.org/publ/arpdf/ar2026e3.htm> (visited on 2026-07-13)

Categories: Stablecoins, Payment Networks

Relevance and limitation: Current BIS analysis of tokenisation and monetary-system design; useful for tradeoffs, not evidence of EWS choices.

Besu Project (July 13, 2026a). *Besu Ethereum Client Source*. Version 26.7.0. URL: <https://github.com/besu-eth/besu/releases/tag/26.7.0> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols, Infrastructure

Relevance and limitation: Canonical source repository and release history. Source language and activity do not prove deployment or fit.

– (2026b). *Besu Documentation*. URL: <https://docs.besu-eth.org/> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols, Infrastructure

Relevance and limitation: Official documentation for Besu’s execution-client role, public/private operation, JSON-RPC, and key boundary. It does not establish EWS use.

bitcoinj Project (May 4, 2026a). *bitcoinj Source*. Version 0.17.1. URL: <https://github.com/bitcoinj/bitcoinj/releases/tag/v0.17.1> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Canonical Java library source and releases. Bitcoin support does not supply issuer or EVM functionality.

– (2026b). *bitcoinj*. URL: <https://bitcoinj.org/> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Official library documentation used to distinguish a protocol integration library from a general blockchain platform.

Circle (2026a). *Cross-Chain Transfer Protocol Technical Guide*. URL: <https://developers.circle.com/ccdp/references/technical-guide> (visited on 2026-07-13)

- Categories:* Stablecoins, Blockchain Protocols
Relevance and limitation: Official documentation for issuer-mediated burn, attestation, and mint flows. It is specific to Circle and USDC.
- (2026b). *What Is USDC?* URL: <https://developers.circle.com/stablecoins/what-is-usdc> (visited on 2026-07-13)
Categories: Stablecoins
Relevance and limitation: Issuer documentation for reserves, redemption framing, and multichain availability. Contractual rights and approved access require diligence.
- Cloud Native Computing Foundation (2026). *CloudEvents Specification*. URL: <https://github.com/cloudevents/spec/blob/main/cloudevents/spec.md> (visited on 2026-07-13)
Categories: Enterprise Integration, Standards
Relevance and limitation: Open event-envelope specification used for metadata vocabulary. It does not define business authority or delivery guarantees.
- Consumer Financial Protection Bureau (2026). *12 CFR Part 1005 - Electronic Fund Transfers (Regulation E)*. URL: <https://www.ecfr.gov/current/title-12/chapter-X/part-1005> (visited on 2026-07-13)
Categories: Compliance, Standards
Relevance and limitation: Authoritative electronic-fund-transfer regulation. Coverage of a specific stablecoin or cross-border product requires legal analysis.
- Debezium (2026). *Outbox Event Router*. URL: <https://debezium.io/documentation/reference/stable/transformations/outbox-event-router.html> (visited on 2026-07-13)
Categories: Enterprise Integration, Reliability
Relevance and limitation: Official implementation documentation for the transactional outbox pattern. Publication and consumption remain asynchronous.
- Early Warning (2026). *Principal Engineer - Digital Assets*. LinkedIn Jobs. URL: <https://www.linkedin.com/jobs/view/principal-engineer-digital-assets-at-early-warning-4372794663> (visited on 2026-07-14)
Categories: Organizational Evidence, Java, Payment Networks
Relevance and limitation: Public Early Warning role posting for cross-border money movement and digital-asset settlement. It requires experience with one or more listed technologies and separately names cloud, monitoring, ACH/EFT, RTP, and FedNow; it does not establish that every item is deployed, that they form one stack, or that the posting describes Zelle production topology.
- Early Warning Services (Sept. 18, 2025a). *Actions to Address Payments Fraud RFI Response*. URL: <https://www.earlywarning.com/sites/default/files/2025-09/EWS%20Actions%20to%20Address%20Payments%20Fraud%20RFI%20%28September%2018%202025%29.pdf> (visited on 2026-07-13)
Categories: Payment Networks, Security, Compliance
Relevance and limitation: Primary EWS description of fraud signals, warnings, enrollment controls, and participant responsibilities. Confidential rules and models remain undisclosed.
- (Oct. 24, 2025b). *Zelle Goes International: Early Warning Expands One-Trillion-Dollar Payments Network with Stablecoin Initiative*. URL: <https://www.earlywarning.com/press-release/zelle-goes-international-early-warning-expands-1t-payments-network-stablecoin> (visited on 2026-07-13)
Categories: Payment Networks, Stablecoins
Relevance and limitation: Primary announcement of EWS’s cross-border stablecoin initiative. It establishes direction, not chain, issuer, custody, reserve, contract, ledger, or API design.
 - (Oct. 2025c). *Zelle P2P Product Brief, Q4 2025*. URL: <https://www.earlywarning.com/sites/default/files/2025-10/Zelle-P2P-Product-Brief-Q425.pdf> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official product brief depicting participating financial institutions and ACH or RTP settlement context. It does not prove one universal route.

Early Warning Services (June 11, 2026a). *Zelle Heads to India, Unveils ZelleUSD Stablecoin for Other Markets*. URL: <https://www.earlywarning.com/press-release/zelle-heads-india-unveils-zelleusd-stablecoin-other-markets> (visited on 2026-07-13)

Categories: Payment Networks, Stablecoins

Relevance and limitation: Primary source for India as the first announced destination and ZelleUSD as a future option in other markets. It does not state that India uses ZelleUSD or disclose implementation details.

– (Feb. 11, 2026b). *Zelle Posts 20 Percent Growth with 1.2 Trillion Dollars Sent*. URL: <https://www.earlywarning.com/press-release/zelle-posts-20-growth-12-trillion-sent-far-outpacing-consumer-spending-and-cementing> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Primary source for EWS-reported 2025 volume, transaction count, participant scale, and network ownership. Annual totals do not establish peak load or SLOs.

– (Jan. 27, 2026c). *Zelle Widens the On-Ramp for Community Banks and Credit Unions*. URL: <https://www.earlywarning.com/press-release/zelle-widens-ramp-community-banks-and-credit-unions-337-new-bank-and-credit-unions> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Primary source for participant reach and the variety of financial-institution integration contexts. It does not enumerate current interfaces or versions.

– (2026d). *Brands: Zelle*. URL: <https://www.earlywarning.com/brands> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official organizational description of Zelle. It provides product context rather than implementation details.

Ethereum Foundation (2026). *Proof-of-Stake*. URL: <https://ethereum.org/developers/docs/consensus-mechanisms/pos/> (visited on 2026-07-13)

Categories: Blockchain Protocols, Reliability

Relevance and limitation: Official educational documentation for validator voting and finality behavior. Route policy still requires network-specific risk decisions.

Featonby, M. (2026). *Making Retries Safe with Idempotent APIs*. Amazon Web Services. URL: <https://aws.amazon.com/builders-library/making-retries-safe-with-idempotent-APIs/> (visited on 2026-07-13)

Categories: Enterprise Integration, Reliability

Relevance and limitation: Primary AWS engineering guidance for idempotency-token contracts and retry ambiguity. It does not replace financial reconciliation.

Federal Reserve Financial Services (2026a). *About the FedNow Service*. URL: <https://www.frbfinancial-services.org/financial-services/fednow/about.html> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official overview of continuous instant-payment processing and eligible participants. Institution-specific product behavior remains outside its scope.

– (2026b). *FedNow Service Product Sheet*. URL: <https://www.frbfinancial-services.org/binaries/content/assets/crsocms/financial-services/fednow/fednow-product-sheet.pdf> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official feature and settlement description used to distinguish rail finality from customer and legal outcomes.

Financial Action Task Force (2025). *Targeted Update on Implementation of the FATF Standards on Virtual Assets and VASPs*. URL: <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2025.html> (visited on 2026-07-13)

Categories: Compliance, Standards

Relevance and limitation: Primary international policy source for risk-based AML/CFT and transfer-information expectations. Jurisdictional implementation varies.

Financial Crimes Enforcement Network (May 9, 2019). *Application of FinCEN's Regulations to Certain Business Models Involving Convertible Virtual Currencies*. FIN-2019-G001. URL: <https://www.fincen.gov/resources/statutes-regulations/guidance/application-fincens-regulations-certain-business-models> (visited on 2026-07-13)

Categories: Compliance

Relevance and limitation: Primary guidance for role- and facts-based money-transmission analysis. It does not decide the proposed product's legal status.

Fireblocks (2026a). *Disaster Recovery Service Process Overview*. URL: <https://developers.fireblocks.com/docs/perform-drs-process> (visited on 2026-07-13)

Categories: Custody, Security, Reliability

Relevance and limitation: Provider documentation for its key-share and recovery model. Contractual availability, implementation quality, export, jurisdiction, and tested recovery still require independent diligence.

– (2026b). *Fireblocks Key Features and Capabilities*. URL: <https://developers.fireblocks.com/docs/capabilities> (visited on 2026-07-13)

Categories: Custody, Security, Infrastructure

Relevance and limitation: Provider documentation for roles, administrative quorum, policy controls, vaults, MPC key shares, and transaction authorization. It is a diligence example, not evidence of EWS use, independent assurance, or selection.

Hedera (2026). *Hedera Documentation*. URL: <https://docs.hedera.com/> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Official SDK, consensus, mirror-node, and smart-contract documentation. A Java SDK does not resolve governance or network fit.

Hiero Project (May 21, 2026). *Hiero Consensus Node Source*. Version 0.74.0. URL: <https://github.com/hiero-ledger/hiero-consensus-node/releases/tag/v0.74.0> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Canonical Java consensus-node source. Source language does not establish operating-model suitability.

IBM (2026a). *Financial Transaction Manager Zelle*. Version 4.0.9. URL: <https://www.ibm.com/docs/en/ftmfm/4.0.9?topic=financial-transaction-manager-zelle> (visited on 2026-07-14)

Categories: Payment Networks, Enterprise Integration, Java, Reliability

Relevance and limitation: Official documentation for an institutional Zelle integration product using RESTful services, mutual TLS, Spring Boot, state-change events, asynchronous processing, retries, resynchronization, and reconciliation. It is IBM-version-specific and does not prove EWS internal architecture, universal participant interfaces, or EWS use of Java.

- IBM (2026b). *Zelle Out-of-Sync Payments*. Version 4.0.9. URL: <https://www.ibm.com/docs/en/ftmf/4.0.9?topic=task-zelle-out-sync-payments> (visited on 2026-07-14)
Categories: Payment Networks, Enterprise Integration, Reliability
Relevance and limitation: Official product documentation showing uncertain state after communication faults, history and alert records, and resynchronization-oriented retry handling. These are IBM product behaviors, not universal EWS recovery semantics or authority for the target retry policy.
- (2026c). *Zelle Reconciliation Task Configuration Parameters*. Version 4.0.9. URL: <https://www.ibm.com/docs/en/ftmf/4.0.9?topic=parameters-zelle-reconciliation-task-configuration> (visited on 2026-07-14)
Categories: Payment Networks, Enterprise Integration, Reliability
Relevance and limitation: Official product documentation for comparing daily Zelle transactions with end-of-day records and retaining task evidence. The implementation and its time windows are IBM-version-specific and do not establish EWS’s systems of record.
- Jack Henry (May 27, 2021a). *Zelle API Details*. URL: <https://jackhenry.dev/open-api-docs/consumer-api/api-reference/v0/zelle/details/> (visited on 2026-07-14)
Categories: Payment Networks, Enterprise Integration
Relevance and limitation: Official participant-side documentation exposing account-entitlement, disclosure, enrollment, alias, contact, activity, payment, request, authorization, and error surfaces. The version is dated 2021; endpoint names and behaviors must not be generalized to EWS or current production integrations.
- (May 27, 2021b). *Zelle API Overview*. URL: <https://jackhenry.dev/open-api-docs/consumer-api/api-reference/v0/zelle/> (visited on 2026-07-14)
Categories: Payment Networks, Enterprise Integration
Relevance and limitation: Official overview of a Zelle-related interface in Jack Henry’s Banno Consumer API, useful as a participant-side vendor example. Its 2021 age limits representativeness, and it is neither a general-purpose EWS API nor evidence of EWS internal contracts.
- LF Decentralized Trust Web3j Project (June 29, 2026a). *Web3j Source and Releases*. Version 6.0.0. URL: <https://github.com/LFDT-web3j/web3j/releases/tag/v6.0.0> (visited on 2026-07-13)
Categories: Java, Blockchain Protocols
Relevance and limitation: Canonical source repository identifying the evaluated release. Version support must be pinned during implementation.
- (2026b). *Web3j Documentation*. URL: <https://docs.web3j.io/latest/> (visited on 2026-07-13)
Categories: Java, Blockchain Protocols
Relevance and limitation: Official Java integration-library documentation for RPC, contract wrappers, events, and HSM support. It is not a node or custody policy.
- Nacha (2026). *Reversals and Enforcement*. URL: <https://www.nacha.org/rules/reversals-and-enforcement> (visited on 2026-07-13)
Categories: Payment Networks
Relevance and limitation: Official summary of defined ACH reversal reasons and timing. Current rules and counsel govern specific cases.
- National Institute of Standards and Technology (Mar. 22, 2019). *Security Requirements for Cryptographic Modules*. FIPS 140-3. URL: <https://csrc.nist.gov/pubs/fips/140-3/final> (visited on 2026-07-13)
Categories: Security, Standards
Relevance and limitation: Normative cryptographic-module security baseline. Validation alone does not define custody or authorization policy.

- (May 2020). *Recommendation for Key Management: Part 1, General*. NIST SP 800-57 Part 1 Revision 5. URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final> (visited on 2026-07-13)
Categories: Security, Standards
Relevance and limitation: Authoritative key-management lifecycle guidance used for custody, compromise, recovery, and cryptoperiod design.
 - (Feb. 2022). *Secure Software Development Framework Version 1.1*. NIST SP 800-218. URL: <https://csrc.nist.gov/pubs/sp/800/218/final> (visited on 2026-07-13)
Categories: Security, Standards
Relevance and limitation: Authoritative software-development control framework. It complements rather than replaces system threat modeling.
- Office of Foreign Assets Control (Oct. 2021). *Sanctions Compliance Guidance for the Virtual Currency Industry*. URL: https://ofac.treasury.gov/system/files/126/virtual_currency_guidance_brochure.pdf (visited on 2026-07-13)
Categories: Compliance, Security
Relevance and limitation: Authoritative sanctions guidance establishing that virtual-currency activity remains subject to sanctions obligations.
- OpenTelemetry (2026). *OpenTelemetry Specification*. URL: <https://opentelemetry.io/docs/specs/otel/> (visited on 2026-07-13)
Categories: Observability, Standards
Relevance and limitation: Official telemetry standard for traces, metrics, and logs. Audit and business evidence remain application responsibilities.
- OpenZeppelin (2026). *Access Control*. URL: <https://docs.openzeppelin.com/contracts/access-control> (visited on 2026-07-13)
Categories: Smart Contracts, Security
Relevance and limitation: Official smart-contract library documentation for roles and delayed administration. Governance and monitoring remain design obligations.
- Oracle (2025). *Java Platform, Standard Edition 25 Specifications*. URL: <https://docs.oracle.com/javase/specs/> (visited on 2026-07-13)
Categories: Java, Standards
Relevance and limitation: Normative platform specifications for the evaluated Java 25 runtime. They establish language and VM behavior, not organizational supportability or workload performance.
- Paxos (2026). *PayPal USD*. URL: <https://docs.paxos.com/guides/stablecoin/pyusd> (visited on 2026-07-13)
Categories: Stablecoins
Relevance and limitation: Issuer documentation for PYUSD and its reserve materials. Product, chain, redemption, and jurisdictional details remain decision inputs.
- PostgreSQL Global Development Group (2026). *Transaction Isolation*. URL: <https://www.postgresql.org/docs/current/transaction-iso.html> (visited on 2026-07-13)
Categories: Infrastructure, Reliability
Relevance and limitation: Official relational-database transaction and isolation documentation. Its guarantees stop at the database boundary.
- Pressler, R. and A. Bateman (2023). *JEP 444: Virtual Threads*. URL: <https://openjdk.org/jeps/444> (visited on 2026-07-13)
Categories: Java

Relevance and limitation: Official Java platform proposal for virtual threads. It is relevant only when workload and library behavior support the model.

R3 (Nov. 28, 2024). *Corda Runtime Open Source*. Version 5.2.2. URL: <https://github.com/corda/corda-runtime-os/releases/tag/release-5.2.2.0> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Canonical Corda runtime source. Licensing, support, and application-network governance remain separate decisions.

– (2026). *Corda 5.2 Documentation*. Version 5.2. URL: <https://docs.r3.com/en/platform/corda/5.2.html> (visited on 2026-07-13)

Categories: Java, Blockchain Protocols

Relevance and limitation: Official documentation for permissioned application networks, flows, UTXO ledgers, identities, and notaries. It is not a public stablecoin chain.

Spring (2026). *Spring for Apache Kafka: Exactly Once Semantics*. URL: <https://docs.spring.io/spring-kafka/reference/kafka/exactly-once.html> (visited on 2026-07-13)

Categories: Java, Enterprise Integration

Relevance and limitation: Official scope for Kafka read-process-write transaction semantics. It does not extend atomicity to external financial systems.

Spring Project (Apr. 23, 2026). *Spring Boot 4.0.6*. Version 4.0.6. URL: <https://github.com/spring-projects/spring-boot/releases/tag/v4.0.6> (visited on 2026-07-13)

Categories: Java, Infrastructure

Relevance and limitation: Immutable release page for the evaluated Spring Boot baseline. It documents a framework release and dependencies, not fitness for this architecture without compatibility and workload testing.

Temporal Technologies (2026). *Temporal Platform Documentation*. URL: <https://docs.temporal.io/> (visited on 2026-07-13)

Categories: Enterprise Integration, Infrastructure, Reliability

Relevance and limitation: Official documentation for a representative durable-workflow product considered during class evaluation. It does not preselect Temporal or make workflow history a financial ledger.

The Clearing House (2026). *RTP for Financial Institutions*. URL: <https://www.theclearinghouse.org/payment-systems/rtp/institution> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official RTP description of real-time final settlement and irrevocable credit transfers. Product recourse remains separate.

U.S. Bank (2026). *Disbursements via Zelle*. Version v1. URL: <https://developer.usbank.com/products/disbursements-via-zelle/v1> (visited on 2026-07-14)

Categories: Payment Networks, Enterprise Integration

Relevance and limitation: Official evidence that U.S. Bank offers an institutional developer product for Zelle disbursements using an email address or United States mobile number. Product-page features are bank-specific and do not establish a general EWS API, universal status semantics, or another treasury bank's capabilities.

United States Congress (July 18, 2025). *S.1582 - GENIUS Act, Public Law 119-27*. URL: <https://www.congress.gov/bill/119th-congress/senate-bill/1582> (visited on 2026-07-13)

Categories: Compliance, Stablecoins

Relevance and limitation: Authoritative legislative record for the United States payment-stablecoin issuer framework. Application and implementing rules require legal review.

Vogelsteller, F. and V. Buterin (Nov. 19, 2015). *ERC-20: Token Standard*. URL: <https://eips.ethereum.org/EIPS/eip-20> (visited on 2026-07-13)

Categories: Smart Contracts, Standards

Relevance and limitation: Normative token interface used to explain why interface compatibility does not establish economic equivalence.

Worthington, S. et al. (Apr. 24, 2019). *ERC-1967: Proxy Storage Slots*. URL: <https://eips.ethereum.org/EIPS/eip-1967> (visited on 2026-07-13)

Categories: Smart Contracts, Standards

Relevance and limitation: Normative proxy storage-slot standard used to identify upgrade and admin-control monitoring requirements.

Zelle (2026a). *Contact Support*. URL: <https://www.zellepay.com/support/contact-support> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Official support routing that preserves the participating institution's customer-support role. It does not define loss allocation.

– (2026b). *Financial Institutions*. URL: <https://www.zelle.com/financial-institutions> (visited on 2026-07-14)

Categories: Payment Networks, Enterprise Integration

Relevance and limitation: Primary evidence that Zelle is integrated into participating institutions' digital-banking experiences and that institutions may use named reseller paths or contact Early Warning Services. It does not disclose reseller contracts, protocols, certification, private topology, or current endpoint specifications.

– (2026c). *How It Works*. URL: <https://www.zellepay.com/how-it-works> (visited on 2026-07-13)

Categories: Payment Networks

Relevance and limitation: Public product description of bank-channel access and email or mobile-number enrollment. It is not topology documentation.

– (2026d). *Join the Zelle Network: Partners*. URL: <https://www.zelle.com/join-zelle-network/partners> (visited on 2026-07-14)

Categories: Payment Networks, Enterprise Integration

Relevance and limitation: Primary evidence for institution-mediated distribution, participating-institution customer experiences, supported payment categories, and the named processor partners Alacriti, FIS, Fiserv, and Jack Henry. It does not establish processor contracts, uniform interfaces, institution assignments, private EWS topology, or international-stablecoin participation.

– (2026e). *Security*. URL: <https://www.zellepay.com/security> (visited on 2026-07-13)

Categories: Payment Networks, Security

Relevance and limitation: Official customer guidance on trusted recipients and cancellation limits. It is not a complete legal or dispute rulebook.

– (2026f). *Zelle Disbursements*. URL: <https://www.zelle.com/zelle-disbursements> (visited on 2026-07-14)

Categories: Payment Networks, Enterprise Integration

Relevance and limitation: Primary evidence that corporate and government disbursers are directed to treasury banks and that the public flow sends a payment file containing an email address or United States mobile number to a member bank. The page does not establish one treasury-bank API or common operating model.

A. Detailed component contracts

These records expand the component authority model in section 5. They are implementation-review detail, not a claim about deployed Early Warning Services components.

Table 38. Responsibility matrix. Each row defines authority, duplicate control, and failure posture.

Component	Owned data / responsibility	Idempotency or order key	Failure behavior and observability	Prohibited shortcut
Participant API gateway	Participant identity, authorization context, contract/rate policy	Participant + request key	Reject before acceptance; trace authentication and quota decision	Reusing a key across participants; trusting body identity
Payment-intent service	Canonical payload hash, intent receipt, participant idempotency	Participant + idempotency key	Same payload returns same intent; conflicting hash is a deterministic conflict	Starting side effects before durable acceptance
Identity/alias adapter	Versioned resolution request/evidence cache	Intent + resolution version	Unknown/stale result blocks route or revalidates; logs source/time	Claiming directory data ownership
Policy/risk service	Versioned facts, decisions, obligations, expiry	Intent + policy stage + version	Hold/reject/manual review; reason-coded metrics	Mutable opaque “risk passed” flag
Quote/router	Quote, route candidates, capability and expiry snapshot	Intent + quote request	Expired quote requires reauthorization and exposure recheck	Silent substitution after acceptance
Treasury/liquidity	Inventory view, reservations, limits, rebalance requests	Asset/account + operation	Shortfall queues or rejects; exposure and aging alerts	Spending from an eventually consistent balance
Ledger service	Accounts, journals, postings, reservations, account versions	Ledger operation ID; account partition	Atomic balanced post or no post; invariant metrics and audit	Direct balance updates or external call in transaction
Workflow/state service	Aggregate version, transitions, timers, operation plan	Intent ID + expected version	Conflict reloads deterministically; ambiguity enters explicit state	Provider callback writing status directly
Settlement adapter	Provider request map, capability/version, observations	Chain operation ID	Classifies absent/ambiguous/confirmed; exposes inquiry	Blind timeout retry or generic success
Signer/custody adapter	Authorization reference, canonical bytes hash, signature audit	Signing operation ID + bytes hash	Refusal/outage is explicit; quorum and latency telemetry	Private key in application or signer trusting caller fields
Chain observer	Node observations, block/receipt lineage, confidence/finality facts	Network + transaction/block	Conflicts create evidence incident; lag and divergence alarms	Advancing business state itself
Payout adapter	Idempotent payout request, provider mapping, credit evidence	Payout operation ID	Inquiry before retry; terminal/ambiguous codes preserved	Treating API 200 as recipient credit
Reconciliation	Comparison runs, breaks, evidence snapshots, disposition	Source + account + period	Break remains open/aged until evidence-backed disposition	Editing source data to clear a break

Component	Owned data / responsibility	Idempotency or order key	Failure behavior and observability	Prohibited shortcut
Notification	Participant/customer status delivery and receipt	Intent + status version + audience	At-least-once with monotonic versions; query remains authoritative	Notification success defining payment completion
Case management	Case, assigned role, action, approval, evidence links	Case/action ID	SLA, aging and segregation alerts	Unlogged repair or direct data mutation
Configuration/policy registry	Approved versioned route, limits, thresholds, kill switches	Configuration version	Safe default disables undefined route; change audit/rollout metrics	Mutable environment flags without approval history

The responsibility table is complemented by the following contract views. Together, the three rows for each component define all fields required for implementation review; the matrices are separated to preserve a readable print size.

Table 39. Component contract inputs and outputs.

Component	API or event inputs	Outputs
Participant API gateway	Authenticated participant request, contract version, idempotency key	Authorized canonical request or reason-safe rejection; trace context
Payment-intent service	Authorized request, canonical payload, participant key	Durable intent receipt, payload hash, validation command, status URI
Identity/alias adapter	Versioned resolution command, alias token, corridor context	Source-stamped resolution observation or explicit unknown/stale result
Policy/risk service	Versioned payment facts, party/address signals, route stage	Allow, reject, hold, obligation, expiry, and reason-coded decision evidence
Quote/router	Eligible routes, asset/network capability, amount, expiry policy	Versioned quote, selected route snapshot, alternatives, or no-route result
Treasury/liquidity	Reservation/release/rebalance command, position and limit version	Reservation evidence, exposure update, shortfall, or rebalance operation
Ledger service	Authorized balanced journal command, account versions, operation ID	Committed journal/postings and outbox fact, or invariant/version conflict
Workflow/state service	Intent commands; ledger, policy, provider, timer, and case evidence	Versioned transition event, next operation plan, timer, participant status fact
Settlement adapter	Authorized operation, canonical provider payload, inquiry command	Provider request identity and normalized absent/ambiguous/observed evidence
Signer/custody adapter	Authorized canonical bytes/hash, key-policy reference, operation ID	Signature/refusal, quorum evidence, custody operation and audit references
Chain observer	Network/contract allowlist, transaction/block identity, finality policy	Source-stamped receipt/log/canonicity observations and confidence conflicts
Payout adapter	Authorized payout operation, beneficiary token, amount/currency	Provider request identity, inquiry result, recipient-credit or ambiguity evidence
Reconciliation	Versioned statements/observations, ledger snapshot, rule set, period	Match, break, aged exposure, evidence bundle, and governed disposition request
Notification	Monotonic participant status fact, audience and template version	Delivery attempt/receipt evidence; query link remains authoritative
Case management	Evidence snapshot, case type, assigned role, proposed action	Approved/rejected governed action, reason, approvers, expiry, audit event

Component	API or event inputs	Outputs
Configuration/policy registry	Signed change proposal, approvals, effective window	Immutable approved version, rollout/kill-switch event, read-only snapshot

Table 40. Component authorization, consistency, and scaling contracts.

Component	Authorization boundary	Consistency expectation	Scaling dimension
Participant API gateway	Mutual participant identity plus contract/role/rate policy	Authentication and request admission before durable acceptance	Participant, endpoint, request rate
Payment-intent service	Only gateway principal may create; owner participant scopes reads	Key/hash uniqueness and intent commit are one local transaction	Participant/key partition and creation rate
Identity/alias adapter	Workflow-approved purpose and minimized alias token	Versioned observation; staleness blocks or forces revalidation	Directory/provider quota and alias partition
Policy/risk service	Policy role controls facts; only governed service signs decisions	Decision binds exact fact and policy versions; no mutable pass flag	Stage, participant, model/provider latency
Quote/router	Workflow requests; treasury/risk gate eligible candidates	Quote and route capability snapshot are immutable through expiry	Corridor, asset/network pair, quote rate
Treasury/liquidity	Treasury policy authorizes limits and rebalance; workflow requests reserve	Single writer/version per inventory partition; no overspend from replica	Asset, account, custody location, corridor
Ledger service	Narrow journal-command roles; segregation for correction/close	Serializable or locked account partition; balanced atomic commit	Account partition, contention, journal volume
Workflow/state service	Domain handler plus expected aggregate version; manual actions are role-approved	One versioned aggregate transition per commit; evidence may arrive at least once	Intent partition, timers, history, active cases
Settlement adapter	Route-specific service identity; command must reference approved operation	Local request-map commit precedes effect; observations are deduplicated	Provider/network quota, operation/inquiry rate
Signer/custody adapter	Independent key policy/quorum validates canonical bytes and intent binding	Signature identity immutable; no assumption of atomicity with workflow	Key class, approval queue, signer latency
Chain observer	Read-only network credentials and approved contract/network registry	Multiple provenance-preserving observations; canonicity can change before threshold	Network, block/log rate, reorg window
Payout adapter	Corridor/provider role and approved beneficiary/operation binding	Request identity durable before send; credit requires contractual evidence	Corridor/provider quota and payout/inquiry rate
Reconciliation	Read-only source access; disposition requires accounting/case role	Rerunnable versioned comparison over frozen source snapshots	Source, account, period, history volume
Notification	Audience-scoped delivery credential; cannot authorize payment state	At-least-once delivery with monotonic status version	Audience, channel, delivery rate

Component	Authorization boundary	Consistency expectation	Scaling dimension
Case management	Assigned operator role, dual approval for material action	Append-only actions; case projection may lag but approval evidence may not	Case type, queue, SLA, analyst capacity
Configuration/policy registry	Segregated proposer/approver/deployer roles	Immutable signed version and effective-time ordering	Route/version count, rollout cohort, read fan-out

A.1. System-of-record matrix

Table 41: Each question has one authoritative source and named corroborating evidence.

Question	Authority	Corroboration / recovery source
Was the participant request accepted?	Payment-intent record and canonical hash	Signed ingress/audit record
May the effect execute?	Versioned workflow transition plus policy obligations	Configuration snapshot and ledger reservation
What does the platform owe?	Balanced ledger journals and reservations	Participant account posting, treasury subledger
Was a transaction authorized?	Signer/custody authorization record and bytes hash	Workflow operation and policy version
Was it submitted?	Independent chain inquiry by transaction hash/nonce and custody record	Submission provider response
What happened on-chain?	Finality policy applied to independent block/receipt observations	Managed provider, explorer only as tertiary evidence
Was the recipient credited?	Destination institution/payout evidence under contract	Provider inquiry and participant confirmation
What status may the customer see?	Product/corridor completion policy applied by workflow	Underlying ledger and external evidence
Is the period/account closed?	Reconciliation disposition and accounting close	Chain, custody, issuer, payout, bank, FX statements

B. Detailed flow coverage

The records below map each normal and failure flow to its funding, ledger, external-effect, finality, payout, duplicate-control, recovery, and audit concerns.

Table 42: Flow coverage records. Every flow inherits intent authentication, versioned policy, audit, participant query, and reconciliation; fields name the distinguishing concern.

01 Domestic

Funding. Bank or participant
Ledger. Position + fee
External mutation. Domestic rail
Chain gate. Not applicable
Payout. Recipient FI
Duplicate control. Intent + rail op
Recovery. Return or case
Audit evidence. Rail + bank

02 Token to fiat

Funding. Incoming token
Ledger. Custody + FX + payable
External mutation. Custody and FX
Chain gate. Incoming finality
Payout. Fiat provider
Duplicate control. Deposit + payout ops
Recovery. Hold or redeem
Audit evidence. Chain, issuer, payout

03 Fiat, token rail

Funding. Fiat + treasury
Ledger. Receivable + inventory
External mutation. Chain transfer
Chain gate. Economic
Payout. Fiat provider
Duplicate control. Intent, chain, payout
Recovery. Inquiry or compensate
Audit evidence. Full evidence set

04 Cross-border

Funding. Fiat + quote
Ledger. FX + inventory + payable
External mutation. FX, chain, payout
Chain gate. Route-specific
Payout. Destination
Duplicate control. Per-leg ops
Recovery. Requote or case
Audit evidence. Quote, fill, chain, credit

05 Token to token

Funding. Source token
Ledger. Per-asset positions
External mutation. Transfer or conversion

Chain gate. Source + destination
Payout. Token delivery
Duplicate control. Per-network ops
Recovery. Contain or migrate
Audit evidence. Both chains + issuer

06 Tokenized deposit

Funding. Bank claim
Ledger. Deposit + conversion
External mutation. Bank network
Chain gate. Network + legal
Payout. Bank or recipient
Duplicate control. Issuer ops
Recovery. Convert or suspend
Audit evidence. Bank + network

07 Chain failure

Funding. Reserved
Ledger. In-transit or suspense
External mutation. Submit or revert
Chain gate. Absence or revert
Payout. Not started
Duplicate control. Stable tx or op ID
Recovery. Retry or compensate
Audit evidence. Inquiry + revert

08 Reorg or delay

Funding. Reserved, in transit
Ledger. No premature release
External mutation. Observation
Chain gate. Canonicity
Payout. Held or risk-accepted
Duplicate control. Block or tx identity
Recovery. Wait, case, loss
Audit evidence. Independent nodes

09 Compliance hold

Funding. Restricted
Ledger. Reservation or restriction
External mutation. None or contained
Chain gate. Evidence retained
Payout. Held
Duplicate control. Decision version
Recovery. Legal case
Audit evidence. Facts, decision, actions

10 Liquidity shortfall

Funding. Insufficient inventory
Ledger. Reservation retained or released
External mutation. Rebalance separate
Chain gate. Not applicable
Payout. Not started

Duplicate control. Rebalance op
Recovery. Queue or reject
Audit evidence. Inventory + limits

11 Duplicate

Funding. Original funding
Ledger. One journal
External mutation. One external op
Chain gate. Original evidence
Payout. One payout
Duplicate control. Key, hash, inbox
Recovery. Return original
Audit evidence. Duplicate + conflict log

12 Recon break

Funding. Existing
Ledger. Suspense if approved
External mutation. Inquiry only
Chain gate. Re-observe
Payout. Reconcile
Duplicate control. Run or break ID
Recovery. Case or entry
Audit evidence. Source snapshots

13 Key incident

Funding. Frozen capacity
Ledger. No new external post
External mutation. Signer disabled
Chain gate. Observe signed set
Payout. Route paused
Duplicate control. Sign operation ID
Recovery. Rotate or migrate
Audit evidence. Signer, key, chain audit

C. Technology and asset decision records

The detailed records preserve comparable evaluation fields and mandatory diligence without interrupting the primary architecture narrative. Product, provider, legal, benchmark, and operating approvals remain organization-specific decisions.

Table 43. Evaluation summary for control/data-plane classes. Each status follows the common evaluation framework.

What / obligation	Why use	Why not / operational risk	Alternatives / change trigger	Performance and maintenance	Recommendation
Java 25 + Spring class; API/orchestration/integration	Strong typing, mature security/data/messaging/telemetry ecosystem, supportability	Framework/version complexity; blocking mistakes; not chain-native	Kotlin/Go/.NET; change if platform ownership or measured fit dominates	Measure request, event and DB contention; patch JDK/framework, contracts and dependencies	recommended
Relational database class; ledger/state local authority	ACID, constraints, mature backup/recovery and SQL reconciliation	Horizontal write scaling and cross-region single-writer complexity	Existing approved relational platform; specialized ledger only if invariants/export/operations proven	Benchmark account contention, journals, recovery; own schema, upgrades, PITR	recommended conditionally
Event-streaming broker; durable transport	Partition order, replay, ecosystem and decoupled consumers	Delivery not business exactly once; capacity/security/retention burden	Existing enterprise broker/queue; change on ordering/replay/operating needs	Measure partition skew, lag, recovery; manage schemas, ACLs, upgrades	recommended conditionally
Workflow engine; timers/retries/cases	Durable long waits, visibility and deterministic orchestration support	New critical platform, versioning/operating semantics; engine state not ledger	Explicit DB state/timer scheduler; managed or self-hosted products	Measure workflow history, task lag, failover; pin deterministic code	evaluate in pilot
OpenTelemetry class; cross-service telemetry	Vendor-neutral signals and context propagation	Cardinality/privacy/cost; not audit or reconciliation	Existing standards-compatible platform	Measure ingest/query cost; govern semantics and retention	recommended
Reconciliation SQL/data jobs; external truth	Transparent comparison, aging and evidence export	Bad data can automate false clearing; large history cost	Existing finance data/recon platform	Partition by source/period; rerunnable versioned rules, lineage and access	recommended conditionally

Table 44. DLT and SDK options illuminate strategic choices; none is preselected for the pilot.

What / obligation	Why use	Why not / operational risk	Alternatives / change trigger	Ecosys-tem/DX/performance	Recommen-dation
Web3j; Java EVM adapter	RPC, contract bindings, events and transaction support in Java	Version/network gaps, SDK is neither node nor custody	Native sidecar or another supported SDK; change on encoding/support gaps	Confor-mance/golden tests; upgrade with chain/EIP support	recom-mended condition-ally
Hyperledger Besu; EVM execution client	Apache-licensed Java client for public/private EVM networks; familiar operations for some teams	Public Ethereum also needs consensus client; self-operation burden; private network creates governance	Managed multi-provider RPC, other execution clients; change if independent node ownership is justified	Measure sync, RPC, reorg recovery, storage; patch proto-col/client	evaluate in pilot, not default
Hedera/Hiero; governed network/client option	Java SDK and Java node source; EVM-compatible surface plus distinct mirror/consensus roles	Governance, fees, finality, data and interoperability differ from EVM public-chain assumptions	Other approved public networks; change if asset/liquidity/corridor fit is superior	Benchmark SDK/node APIs and recovery; maintain network-specific adapter	evaluate in pilot
R3 Corda 5; permissioned application network	Identity, flows, UTXO options and notary uniqueness can fit consortium obligations	Not a public stablecoin network; consortium governance and deployment burden	Existing bank/tokenized-deposit network; public rail	Strong JVM DX for Corda teams; performance must be consortium-specific	not recom-mended for initial pilot
bitcoinj; Bitcoin protocol library	Mature Java library for Bitcoin-oriented integration	No EVM/token/issuer contract model; does not meet recommended stablecoin path by itself	Selected network SDK/node	Good bounded Bitcoin DX; irrelevant capabilities dominate	rejected for initial scope
Custom/private blockchain	Maximum protocol/governance control in theory	Creates consensus, membership, liquidity, interoperability, security, upgrade and exit obligations	Approved public network; existing permissioned consortium	Must bench-mark entire network and operate validator governance	rejected for initial pilot

Table 45: Common twelve-question technology dossiers, evaluated 2026-07-13.

Java SE 25; Spring Boot 4.0.6 / Framework 7.0.x; supported control-plane runtime; Linux/container assumption (Oracle 2025; Spring Project 2026)

Obligation / why use. APIs, domain model, orchestration, SQL, event and telemetry integration; typed JVM ecosystem and enterprise operability

Why not / alternatives. Not chain-native and framework/dependency graph is material; alternatives are supported Java LTS baseline, Kotlin, Go, or .NET

Change trigger / operational risk / maintenance. Change on platform support or measured workload mismatch. Risks: CVEs, dependency drift, blocking/pool exhaustion. Maintain JDK/framework, SBOM, compatibility and contract tests

Performance / developer experience / maturity. Measure request/event latency, concurrency, allocation, DB/pool contention. Strong local debugging/test tooling; mature governed releases, multiple vendors, portable bytecode; GPL+Classpath/Apache-2.0 licensing requires normal review

Recommendation / rationale / owner. **recommended**; control-plane fit without language uniformity. Owner: application/platform architecture

Relational database class; PostgreSQL 18 documentation as reference; single-writer ledger partitions; approved managed or self-operated deployment (PostgreSQL Global Development Group 2026)

Obligation / why use. ACID journals, constraints, version checks, SQL reconciliation, PITR and inspectable evidence

Why not / alternatives. Cross-region writes and hot-account contention; alternatives are existing approved RDBMS or a specialized ledger only after invariant/export proof

Change trigger / operational risk / maintenance. Change if recovery, contention or jurisdiction cannot meet policy. Risks: corruption, operator error, failover split-brain, lock pressure. Maintain schema, migrations, backups, restore drills and statistics

Performance / developer experience / maturity. Measure commit/tail latency, lock waits, partition skew, journal growth, replica lag and restore. Excellent local SQL/test tooling; mature ecosystem, permissive license and broad support/portability

Recommendation / rationale / owner. **recommended conditionally**; prefer the approved RDBMS that proves invariants. Owner: ledger and SRE

Apache Kafka 4.1 class; broker/schema service; self-managed or approved managed deployment (Apache Software Foundation 2026; Spring 2026)

Obligation / why use. Ordered partitions, durable replay and decoupled outbox/inbox consumers

Why not / alternatives. Broker transactions exclude ledger and provider effects; alternatives are existing enterprise broker/queue or log service

Change trigger / operational risk / maintenance. Change on ordering, replay, retention, data or operating fit. Risks: partition loss/skew, ACL error, backlog, correlated managed outage. Maintain brokers/clients, schemas, quotas, certificates, replay drills

Performance / developer experience / maturity. Measure produce/consume tail latency, lag, skew, recovery and retention cost. Strong JVM clients and test containers; mature Apache governance/licensing and portable protocol, but operating skill remains

Recommendation / rationale / owner. **recommended conditionally**; transport, never authority. Owner: messaging/platform

Durable workflow class; Temporal documentation reviewed 2026-07-13; managed/self-hosted product unselected (Temporal Technologies 2026)

Obligation / why use. Long timers, retries, visibility, versioned orchestration and case handoff

Why not / alternatives. New critical history/worker platform; engine state is not ledger. Alternatives: database state machine plus timer scheduler, or an existing approved engine

Change trigger / operational risk / maintenance. Change when history scale, determinism, export or operating proof fails. Risks: hidden retry, nondeterministic upgrade, backlog, lock-in. Maintain worker compatibility, history controls, failover and replay tests

Performance / developer experience / maturity. Measure task lag, schedule-to-start, history size, worker saturation and recovery. Good SDK testing if deterministic discipline is enforced; maturity/support/licensing are product-specific and must be reviewed

Recommendation / rationale / owner. **evaluate in pilot**; domain owns semantics. Owner: workflow/platform

OpenTelemetry specification, accessed 2026-07-13; SDK/collector to approved telemetry backend (OpenTelemetry 2026)

Obligation / why use. Portable traces, metrics, logs and propagation across adapters

Why not / alternatives. Not an audit ledger or reconciliation source; high-cardinality/privacy/cost risk. Alternative: existing standards-compatible telemetry pipeline

Change trigger / operational risk / maintenance. Change if backend or data policy lacks required signals/export. Risks: dropped data, secret/PII leakage, sampling blindness. Maintain semantic conventions, collectors, redaction and retention

Performance / developer experience / maturity. Measure collector loss, ingest/query latency, cardinality and cost. Broad language tooling; CNCF governance and Apache-2.0 components support portability, subject to backend features

Recommendation / rationale / owner. **recommended**; operational signal standard only. Owner: observability/SRE

Reconciliation/data infrastructure class; relational rules plus approved finance/data platform; versioned jobs and immutable source snapshots

Obligation / why use. Independent comparison, aging, lineage, reruns, cases and close evidence

Why not / alternatives. Bad normalization can falsely clear; warehouse latency may miss operational breaks. Alternatives: existing finance reconciliation product or SQL jobs

Change trigger / operational risk / maintenance. Change when scale, cutoff, lineage or segregation cannot be proven. Risks: source lateness, rule drift, access error, silent tolerance. Maintain mappings, versions, lineage, backfills and control totals

Performance / developer experience / maturity. Measure source freshness, unmatched volume/age, rerun time and close latency. SQL is inspectable/testable; platform maturity and license depend on selected internal/product stack; export is mandatory

Recommendation / rationale / owner. **recommended conditionally**; own semantics even if platform is bought. Owner: accounting/data/reconciliation

Web3j 6.0.0; library inside a bounded EVM adapter, not a service authority (LF Decentralized Trust Web3j Project 2026b; LF Decentralized Trust Web3j Project 2026a)

Obligation / why use. Java RPC, ABI/event decoding, contract bindings and transaction construction

Why not / alternatives. Network/EIP/encoding gaps; not a node or custody control. Alternatives: native sidecar, another supported SDK, direct typed RPC

Change trigger / operational risk / maintenance. Change on byte mismatch, unsupported transaction/network or maintenance lag. Risks: dependency/CVE, silent encoding mismatch. Maintain pinned release, golden vectors, simulation and native-tool conformance

Performance / developer experience / maturity. Measure encoding/decoding and RPC overhead only within end-to-end adapter tests. Good JVM DX; Apache-2.0 open source and portable, but ecosystem breadth is smaller than native EVM tooling

Recommendation / rationale / owner. **recommended conditionally**; only after conformance. Owner: chain integration

Hyperledger Besu 26.7.0; execution client, managed or self-operated; public use also needs consensus client (Besu Project 2026b; Besu Project 2026a)

Obligation / why use. Independent EVM execution/RPC and possible private-network evaluation

Why not / alternatives. Node sync/storage/patch burden; private deployment creates governance and liquidity obligations. Alternatives: managed multi-provider RPC or other supported execution client

Change trigger / operational risk / maintenance. Change if integrity, concentration, archival or economics justify ownership. Risks: consensus/execution compatibility, reorg handling, disk/state growth, incident staffing. Maintain protocol upgrades, client pair, snapshots and recovery drills

Performance / developer experience / maturity. Measure sync/recovery, RPC tail, storage growth and reorg observation under selected network. Java aids some teams; Apache-2.0/Linux Foundation project with protocol portability but client-specific operations

Recommendation / rationale / owner. **evaluate in pilot, not default**; evidence need must justify. Owner: blockchain platform/SRE

Hedera services plus Hiero consensus node 0.74.0; public governed network with Java SDK/node and mirror model (Hedera 2026; Hiero Project 2026)

Obligation / why use. Candidate network, contract execution and independent observation pattern

Why not / alternatives. Distinct governance, fee, finality, mirror and interoperability assumptions; asset/corridor liquidity unproved here. Alternatives: other approved public or permissioned networks

Change trigger / operational risk / maintenance. Change if legal, asset, liquidity and measured operational fit outrank alternatives. Risks: network governance/upgrades, provider/mirror dependence. Maintain network SDK, adapter, fee/finality policy and migration tests

Performance / developer experience / maturity. Measure submission/finality observation, mirror lag, fee behavior and recovery. Strong Java path; Apache-2.0 Hiero governance is open, while network support/portability remains protocol-specific

Recommendation / rationale / owner. **evaluate in pilot**; no preselection. Owner: architecture/treasury/legal

R3 Corda runtime 5.2.2.0; permissioned application network / UTXO and flow stack (R3 2026; R3 2024)

Obligation / why use. Strategic tokenized-deposit or consortium identity/uniqueness alternative

Why not / alternatives. Not a public stablecoin rail; membership, notary, deployment and consortium governance burden. Alternatives: existing approved bank network or public rail

Change trigger / operational risk / maintenance. Change if a live approved consortium supplies issuer, corridor, liquidity, governance and exit. Risks: member coordination, upgrade compatibility, notary/flow incidents. Maintain network, CorDapps, identities and governance

Performance / developer experience / maturity. Measure flow/notary latency, member throughput and recovery only in target consortium. Strong JVM DX; Apache-2.0 open-source runtime with commercial ecosystem, but portability is Corda-specific

Recommendation / rationale / owner. **not recommended for initial pilot**; strategic interoperability only. Owner: institutional architecture/legal

bitcoinj 0.17.1; Java Bitcoin protocol/wallet library in an adapter (bitcoinj Project 2026b; bitcoinj Project 2026a)

Obligation / why use. Illuminates SDK-versus-node boundary for Bitcoin-oriented products

Why not / alternatives. Does not provide EVM token/issuer contracts or selected stablecoin functionality. Alternatives: selected network SDK/node

Change trigger / operational risk / maintenance. Change only if approved route is Bitcoin-native and satisfies asset/legal design. Risks: wallet/key misuse and protocol/library drift. Maintain consensus vectors, peer/RPC behavior and signing isolation

Performance / developer experience / maturity. Measure sync/RPC/transaction handling for that route. Familiar Java DX; Apache-2.0 mature project, but capability portability to account/token chains is low

Recommendation / rationale / owner. **rejected for initial scope**; obligation mismatch. Owner: chain architecture

Institutional custody/signing class; FIPS 140-3 HSM or approved MPC/service; language-neutral policy API; no keys in application (National Institute of Standards and Technology 2019; National Institute of Standards and Technology 2020; Fireblocks 2026b; Fireblocks 2026a)

Obligation / why use. Segregated key use, quorum, policy, recovery, statements and specialist operations; legal custody classification is not inferred

Why not / alternatives. Concentration, opaque MPC/control claims, export/recovery and cost; alternatives: legally approved institutional custodian, cloud/on-prem HSM, or approved in-house key operation

Change trigger / operational risk / maintenance. Change if due diligence, recovery drill, policy fidelity, data or economics fail. Risks: key loss/compromise, policy bypass, provider outage. Maintain ceremonies, roles, firmware/service changes, allowlists and recovery exercises

Performance / developer experience / maturity. Measure authorization/sign tail, availability, capacity, recovery and audit export without exposing secrets. DX must be testable via sandbox; certification is scoped, vendor maturity/support/license and asset portability require contract review

Recommendation / rationale / owner. **recommended conditionally to buy for pilot**; controls stay owned. Owner: security/treasury

Managed submit node plus administratively independent read path; provider/network selected through route approval; 2026-07-13 service-class evaluation

Obligation / why use. Submission availability and independent chain evidence without immediate node operations

Why not / alternatives. Correlated upstream/cloud/admin failures and rate/history limits; alternatives: two truly independent providers or self-operated full/archive node

Change trigger / operational risk / maintenance. Change on integrity, concentration, archival, latency, cost or jurisdiction. Risks: false/stale response, outage, data loss. Maintain provenance, quorum policy, provider exit and optional node runbook

Performance / developer experience / maturity. Measure RPC tail/error, head divergence, reorg visibility, archive coverage and recovery. Easy API DX can hide failure; maturity/support/licensing are provider/client-specific; raw evidence export and endpoint portability required

Recommendation / rationale / owner. **recommended conditionally**; independence must be demonstrated. Owner: blockchain platform/SRE

Custom/private blockchain; new validator/member network and protocol governance; no implementation selected

Obligation / why use. Maximum theoretical membership, privacy and upgrade control

Why not / alternatives. Creates consensus security, validator operations, liquidity, interoperability and exit program. Alternatives: approved public rail or existing consortium

Change trigger / operational risk / maintenance. Change only if a legal/business need cannot be met elsewhere and funded governance exists. Risks: capture, outage, fork, upgrade, weak liquidity. Maintain protocol, clients, validators, membership and ecosystem indefinitely

Performance / developer experience / maturity. Must benchmark whole network, fault tolerance and recovery; DX/support/security ecosystem must be created or inherited; portability is lowest and licensing spans all chosen components

Recommendation / rationale / owner. **rejected for initial pilot**; burden dominates learning. Owner: executive architecture/risk

Table 46. Language and build/buy responsibility matrix.

Capability	Java role	Non-Java / provider role	Pilot posture	Rationale
Participant API and intent	Spring HTTP/security, validation, idempotency, domain handlers	Participant client in its own stack	Build	Enterprise contract and business authority
Workflow/state	Durable Java handlers and timers through an engine abstraction	Managed/self-hosted workflow engine may use another runtime	Build domain; evaluate engine	State semantics are ours; scheduling can be bought

Capability	Java role	Non-Java / provider role	Pilot posture	Rationale
Ledger-facing services	Java domain/API, SQL transactions, outbox	Relational database engine	Build authority on proven database	Strong local transactions and audit
Policy integration	Java orchestration, versioned decision contract	Fraud/sanctions/identity vendors and participant systems	Integrate	Roles and models are external, evidence contract is ours
Messaging	Java producers/consumers, schema and inbox/outbox	Broker/schema registry	Integrate	Mature ecosystem; delivery is not business authority
Transaction construction	Java adapter/SDK where complete and testable	Chain-native simulation/tooling	Conditional build	Keep business operation binding in control plane
Smart contracts	Interface, allowlist, events, risk metadata	Solidity for EVM; Rust/Move/other selected chain toolchain	Reuse audited standard contracts; minimal custom code	Native audit/testing ecosystem and bytecode fidelity
Node clients	JSON-RPC/gRPC clients and observation normalization	Go, Rust, Java, C++ or vendor-supported client per network	Managed + independent read path	Client language is subordinate to protocol fit and operations
Signing/custody	Language-neutral authorized request, canonical hash and evidence validation	Approved institutional custody/HSM/MPC implementation; legal classification remains separate	Buy for pilot	Keys outside application memory and specialist controls
Reconciliation/operations	Java comparison/case services, SQL/analytics jobs, telemetry	Provider statements/data exports	Build core; integrate sources	Obligation ownership cannot be outsourced

Table 47: Build/buy decisions for high-specialization boundaries.

Choice	Initial posture	Risks to control	Revisit trigger
Custody/signing	Buy an approved institutional custody/HSM/MPC service; retain platform authorization and reconciliation; confirm any legal classification separately	Concentration, outage, policy fidelity, data/export, key governance, exit, incident cooperation	Volume/economics, control gaps, recovery/data needs, approved in-house key-operation capability
Node access	Managed provider for submit plus administratively independent read/observation path	Correlated cloud/provider failure, stale/false responses, rate/retention limits, data jurisdiction	Integrity, latency, cost, concentration or archival needs justify self-operated node
Chain analytics	Integrate one or more approved providers as risk signal	False positives/negatives, opaque model, privacy, vendor concentration	Model evidence or legal policy requires second source/in-house analysis
Workflow engine	Evaluate against durable state contract; prefer existing approved platform if fit	Lock-in, nondeterministic upgrade, hidden retry, history growth	Scale/visibility/operating proof or a simpler database scheduler is sufficient
Reconciliation	Own comparison semantics and cases; may use approved data platform	Outsourced black box clearing discrepancies, weak lineage	Existing finance platform proves replay, evidence, segregation, and export

Table 48: Settlement-asset/control-model option records. Every option requires legal, issuer, custody, liquidity, accounting, and operating diligence.

One external regulated stablecoin**Strategic benefit.** Fastest path to test a liquid institutional rail; issuer handles token/reserve operation**Principal obligations/risks.** Issuer/redemption/concentration/admin power, network/contract/custody, eligibility**Time/complexity.** Lowest relative if approvals and corridor liquidity exist**Exit/migration.** Stop new use; redeem/transfer inventory; introduce new route version**Initial recommendation.** recommended conditionally through ADR**Multiple external stablecoins****Strategic benefit.** Counterparty/network choice and broader reach**Principal obligations/risks.** Multiplies contracts, decimals, controls, liquidity, reconciliation, accounting and support**Time/complexity.** High before one route is proven**Exit/migration.** Per-asset kill switch and inventory unwind, but routing behavior complex**Initial recommendation.** not recommended for initial pilot**Bank-issued tokenized deposits****Strategic benefit.** Direct bank liability; potential institutional identity/singleness benefits**Principal obligations/risks.** Issuing-bank claim, convertibility, network governance, interbank liquidity, legal/accounting interoperability**Time/complexity.** Corridor/consortium dependent**Exit/migration.** Convert/redeem under bank terms; adapter/network exit required**Initial recommendation.** evaluate as interoperability path

Future ZelleUSD

Strategic benefit. Strategic control, economics and product alignment may be possible

Principal obligations/risks. Every issuer/reserve/redemption/network/custody/admin/liquidity/legal fact currently unspecified publicly

Time/complexity. Unknown; issuance is a distinct regulated operating program

Exit/migration. Must design redemption, migration, contract upgrade and wind-down before launch

Initial recommendation. unknown; delivery discovery/strategic ADR only

Permissioned settlement token/network

Strategic benefit. Membership, privacy and governance can be tailored

Principal obligations/risks. New governance, validator/member operations, liquidity, interoperability, exit and shared incident response

Time/complexity. Highest unless an existing approved consortium fits

Exit/migration. Consortium and token redemption/migration terms critical

Initial recommendation. not recommended for initial pilot

Stablecoin only as invisible rail

Strategic benefit. Preserves bank channel and avoids consumer custody/gas while testing settlement

Principal obligations/risks. Operator/participants still bear treasury, provider, legal, customer and recovery obligations

Time/complexity. Lowest product disruption; substantial back-office rigor remains

Exit/migration. Replace route behind stable participant contract

Initial recommendation. recommended product posture

Table 49: Full settlement-asset and product-control option dossiers.

One external regulated stablecoin

Issuer / authorization / reserves / insolvency. **Issuer/counterparty:** approved external legal issuer. **Authorization:** issuer/operator/participant roles by jurisdiction. **Reserves:** issuer evidence and terms. **Insolvency:** direct and indirect holder treatment requires counsel

Custody / redemption / admin powers / chain-contract. **Custody:** institutional custody arrangement approved for the role; any legal classification is confirmed separately. **Redemption:** institutional access, limits, hours, fees and suspension proven. **Powers:** mint, burn, freeze, denylist, pause and upgrade authority mapped. **Chain:** one approved contract/network/version

Liquidity / interoperability / accounting / privacy. **Liquidity:** corridor capacity and rebalance stress. **Interop:** issuer-supported networks only; no generic bridge. **Accounting:** asset/valuation/impairment/fees approved. **Privacy:** public metadata minimized and assessed

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** lowest relative after approval. **Control:** low over issuer/network. **Concentration:** deliberately bounded to one. **Customer:** invisible familiar payment

Exit / migration / posture. Redeem/transfer inventory, disable route version, migrate adapter and disclosures. **Conditionally recommended** through ADR

Multiple external stablecoins

Issuer / authorization / reserves / insolvency. **Issuer/counterparty:** several independent issuers. **Authorization/reserves/insolvency:** separately approved; no equivalence inferred

Custody / redemption / admin powers / chain-contract. **Custody/redemption/powers:** per asset and provider. **Chain-contract:** decimals, addresses, upgrade and network combinations versioned; bridge use is separate ADR

Liquidity / interoperability / accounting / privacy. **Liquidity:** fragmented and route-dependent. **Interop:** conversion creates new counterparties. **Accounting:** per-asset classification/valuation. **Privacy:** multiplied network footprints

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** high. **Control:** routing optionality. **Concentration:** lower issuer concentration but higher integration/provider exposure. **Customer:** little benefit if rail remains invisible

Exit / migration / posture. Per-asset kill switch, inventory unwind and route migration; preserve operation history. **Not recommended initially**

Bank-issued tokenized deposits

Issuer / authorization / reserves / insolvency. **Issuer/counterparty:** issuing commercial bank/depository claim. **Authorization:** bank/network membership and corridor roles. **Reserves:** deposit balance-sheet/regime, not stablecoin reserve shorthand. **Insolvency:** account/deposit treatment requires exact terms

Custody / redemption / admin powers / chain-contract. **Custody:** bank/network account or token control. **Redemption:** convertibility to bank money under scheme rules. **Powers:** bank/network issuance, transfer and restriction controls. **Chain:** permissioned or public implementation-specific

Liquidity / interoperability / accounting / privacy. **Liquidity:** issuing-bank and interbank reach. **Interop:** bank/network agreements dominate. **Accounting:** deposit/interbank treatment. **Privacy:** membership may reduce public exposure but increases shared-data governance

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** consortium/corridor dependent. **Control:** shared with banks. **Concentration:** issuer-bank and network. **Customer:** potentially institutional singleness; bank channel can remain unchanged

Exit / migration / posture. Convert/redeem under scheme, migrate participant/network adapters and settle residual claims.

Evaluate as interoperability path

Future ZelleUSD

Issuer / authorization / reserves / insolvency. **Issuer/counterparty:** unknown. **Authorization:** unknown. **Reserves:** unknown. **Insolvency:** unknown. Public announcement establishes none of these operating facts

Custody / redemption / admin powers / chain-contract. **Custody/redemption/admin powers/chain-contract:** all unknown. Issuance, key authority, contract governance and holder eligibility require primary documents and approvals

Liquidity / interoperability / accounting / privacy. **Liquidity/interoperability/accounting/privacy:** all unknown. No corridor, market-maker, redemption, network, statement or metadata posture is assumed

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** unknown; issuance is a separate regulated operating program. **Strategic control:** potential, not proven. **Concentration/customer proposition:** unknown

Exit / migration / posture. Wind-down, redemption, contract/network migration and holder treatment must be designed before launch. **Strategic ADR only; not pilot-ready on public evidence**

Permissioned settlement token/network

Issuer / authorization / reserves / insolvency. **Issuer/counterparty:** consortium or designated issuer. **Authorization:** membership/governance. **Reserves/insolvency:** token and member rules must establish claim

Custody / redemption / admin powers / chain-contract. **Custody:** member/key model. **Redemption:** scheme-specific. **Powers:** governance/validator/member controls. **Chain-contract:** new or existing permissioned protocol with upgrade/fork rules

Liquidity / interoperability / accounting / privacy. **Liquidity:** must be created or proven. **Interop:** gateways add risk. **Accounting:** scheme-specific. **Privacy:** configurable but requires data-sharing, regulator and operator governance

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** highest unless approved consortium exists. **Control:** shared but tailorable. **Concentration:** governance/operators. **Customer:** invisible or institutional, not inherently better money

Exit / migration / posture. Consortium exit, redeem token, export evidence, remove validator/member keys, migrate obligations. **Not recommended initially**

Stablecoin only as invisible rail

Issuer / authorization / reserves / insolvency. **Issuer/counterparty/authorization/reserves/insolvency:** inherited from the separately selected asset; product does not erase them

Custody / redemption / admin powers / chain-contract. **Custody/redemption/powers/chain-contract:** institutional and hidden from customer; still fully controlled, disclosed where required and reconciled

Liquidity / interoperability / accounting / privacy. **Liquidity/interoperability/accounting/privacy:** owned in treasury/back office; no customer wallet, arbitrary address or gas. Domestic route remains independent

Complexity / time / strategic control / concentration / customer proposition. **Complexity/time:** lowest customer disruption but substantial operations. **Control:** participant contract stable. **Concentration:** selected rail. **Customer:** bank account, recipient and status experience

Exit / migration / posture. Replace or disable settlement adapter behind stable participant API; reconcile/unwind inventory. **Recommended product posture**

Table 50. No asset may enter a production route until every mandatory diligence question has approved evidence or an explicit stop decision.

Domain	Required evidence/question	Reject or pause signal
Issuer and authorization	Legal issuer/entity, applicable permission, jurisdictions, audited financial/operating responsibilities	Unclear issuer/role, missing required authorization, unacceptable jurisdiction
Reserve and redemption	Composition, segregation, attestations/audits, direct/indirect redemption rights, timing, fees, minimums, suspension, insolvency treatment	No viable institutional exit or reserve evidence/control below policy
Contract and admin powers	Network, addresses, decimals, proxy/code, mint/burn/freeze/denylist/pause/upgrade roles, delays, events, audits, incident contacts	Undisclosed or unacceptable privileged path; unmonitored upgrade
Custody and keys	Supported custody, address ownership, key policy, insurance/terms, recovery, statement/API/export and exit	Keys cannot meet segregation/recovery/audit or provider exit requirements
Liquidity and treasury	Corridor liquidity, market makers, redemption capacity, hours, spread/slippage, concentration, prefunding and stress behavior	Capacity below exposure; no credible rebalance/redemption under stress
Network and nodes	Finality/reorg model, fees, congestion/censorship, governance/upgrades, node diversity, data/analytics, contract/tool support	No independent observation or unacceptable finality/governance risk
Compliance/privacy	Address/party controls, sanctions/freeze cooperation, information exchange, data location, public-chain metadata exposure	Required controls/roles unresolved or data exposure impermissible
Accounting/tax/reporting	Asset classification, valuation, subledger/GL mapping, statements, fees, close and impairment treatment	Cannot close/reconcile or meet reporting policy
Operations and exit	SLAs, incident/change notice, kill switch, contact escalation, test environment, export, wind-down/migration rehearsal	Unbounded incident dependency or no tested exit

D. Roadmap and tradeoff decision records

The roadmap gates and tradeoff records make exposure, ownership, reversibility, and stop conditions reviewable. Elapsed time never substitutes for exit evidence.

Table 51: Seven-phase delivery and control gate records. The phase/gate visual is the lower panel of figure 18.

0 Discover and decide

Objective. Close facts that materially shape the system

Capabilities/deliverables. Legal/entity/role map; participant contract baseline; corridor, asset, network, custody, node, FX, payout and accounting ADRs; threat model; finality policy; success/loss measures

Measurable exit evidence. Mandatory unknowns have accountable disposition; source/contract diligence complete; failure and exit obligations accepted

Exposure / readiness. No customer funds; tabletop only

Stop conditions. Issuer/route/legal role unresolved; no viable redemption/exit; controls or economics unacceptable

1 Control-plane and ledger foundation

Objective. Prove owned financial truth independent of blockchain

Capabilities/deliverables. Intent/idempotency; ledger/journals/reservations; state machine; policy/config versions; outbox/inbox; audit; cases; domestic regression isolation

Measurable exit evidence. Balance/property/concurrency tests pass; duplicate/replay produce one effect; point-in-time recovery and audit reconstruction demonstrated

Exposure / readiness. Synthetic funds; production-like security; on-call/runbooks drafted

Stop conditions. Invariant, access, recovery, or participant compatibility failure

2 Sandbox adapters

Objective. Prove provider contracts and ambiguity handling

Capabilities/deliverables. Custody/signing, node/observer, asset, FX, payout adapters; simulation; capability registry; reconciliation sources; kill switches

Measurable exit evidence. Golden transaction vectors; duplicate/timeout/conflict/reorg simulations; provider export and inquiry proven; all external accounts covered by reconciliation

Exposure / readiness. Test networks/provider sandboxes only; no customer promise

Stop conditions. Provider cannot supply stable identity, inquiry, evidence, segregation, or exit

3 Bounded end-to-end proof

Objective. Exercise full lifecycle without open customer exposure

Capabilities/deliverables. Production-shaped route, treasury operations, policy/case integration, monitoring, close, incident and disaster drills

Measurable exit evidence. Complete 13-flow suite; reconciliation closes; signing/key/node/payout drills meet approved measures; no critical/high review findings

Exposure / readiness. Internal/tightly bounded value under explicit loss cap

Stop conditions. Unexplained break, unsafe retry, drill failure, policy bypass, material security finding

4 Production pilot

Objective. Validate one corridor/asset/network with selected institutions

Capabilities/deliverables. Participant onboarding, customer disclosures/support, prefunding, limits, staffed operations, regulatory reporting as applicable

Measurable exit evidence. Approved reliability, reconciliation aging, customer outcomes, loss/exposure, cost and support measures across required observation window

Exposure / readiness. One corridor, one asset, one network, bounded participants/value; no wallet/bridge

Stop conditions. Limit breach, unresolved material break, provider/issuer incident, legal/control change, adverse customer/loss threshold

5 Controlled expansion

Objective. Expand one dimension at a time

Capabilities/deliverables. Additional participants or value bands, then corridor/provider capacity; automation justified by evidence

Measurable exit evidence. Each expansion cohort passes Phase 4 measures; concentration and liquidity stress tests; recovery remains within limits

Exposure / readiness. Incremental approved exposure; rollback to prior route/version

Stop conditions. Metrics degrade, operations saturate, concentration/credit exceeds policy

6 Strategic options

Objective. Evaluate new assets/networks or proprietary capability

Capabilities/deliverables. Multi-asset only if needed; tokenized-deposit interoperability; possible ZelleUSD/permissioned decisions; self-operated node/custody only if justified

Measurable exit evidence. Separate business, legal, technology, governance, security, liquidity, migration and exit ADRs; independent pilot gates

Exposure / readiness. No automatic inheritance from Phase 5; each option bounded anew

Stop conditions. Optionality cost exceeds value; issuance/governance/exit unresolved; control regression

Table 52: Dependency and risk burn-down. Learning without customer funds precedes exposure.

Risk	Earliest learning	Evidence before customer funds	Evidence before expansion
Ledger/state correctness	Phase 1	Property, contention, duplicate, recovery and audit reconstruction	Production invariant and close history
Provider ambiguity	Phase 2	Timeout/lost-response/inquiry and duplicate test vectors	Measured ambiguity aging and successful cases
Chain/reorg/finality	Phase 2	Historical/simulated reorg, node disagreement and threshold evaluation	Network-health history, drills and loss/exposure performance
Key/custody incident	Phase 2–3	Signer refusal, outage, recovery and migration drill with synthetic assets	Contractual incident/export proof and periodic production drill
Liquidity/FX/payout	Phase 2–3	Capacity, quote expiry, shortfall, failed payout and reconciliation tests	Stress/concentration measures and operating history
Legal/compliance/customer	Delivery phases 0–3	Approved roles, terms, decision evidence, case and disclosure walkthrough	Pilot cases, regulatory change review, customer outcome measures
Operational load	Phase 3	On-call, dashboards, runbooks, chaos/tabletop and disaster recovery	Capacity/incident trend and staffing readiness
Exit/concentration	Phase 2–3	Data/key/record export, route disable, inventory unwind and alternate observation	Periodic provider/asset/network exit rehearsal

Table 53. Tradeoff ledger. Optionality has operating cost and every posture has a revisit signal.

Decision	Chosen posture / benefit	Cost and failure exposure	Revisit signal	Owner
Operational truth	Internal ledger/state; represents all obligations	More systems, journals, reconciliation and repair; drift risk	External shared ledger proves every required obligation/control and recovery	Accounting/architecture
Customer instrument	Invisible settlement; familiar channel and support	Operator absorbs treasury/integration complexity; limited token proposition	Approved strategy and controls for customer-held token	Product/risk
Network model	Evaluate public network for pilot	Public fees, privacy, governance, reorg/censorship risk	Existing permissioned consortium proves liquidity, governance, exit and corridor fit	Architecture/legal/treasury
Issuer	One external regulated asset selected by ADR	Concentration and issuer/admin/redemption dependency	Proprietary asset or alternate issuer proves legal, liquidity, control and cost advantage	Treasury/legal/product
Custody/nodes	Managed custody and node plus independent read path	Concentration, contracts, export and incident dependency	Scale, integrity, economics or control justify self-operation	Security/platform
Generality	One asset/network/corridor	Later adapters/migrations; initial concentration	One route passes controls and demand supports another dimension	Product/architecture
Liquidity	Prefunded bounded inventory	Capital inefficiency and rebalance burden	Measured flows and credit/legal approval support just-in-time/credit model	Treasury/risk
Customer speed	Evidence-gated completion; early credit only with explicit risk owner	May feel slower; or creates loss exposure when accelerated	Loss/latency data and disclosure support threshold change	Product/risk/legal
Provider abstraction	Stable operation/evidence contract plus capability-specific branches	More adapter design and tests	Common semantics demonstrably mature without hiding risk	Architecture/integration
Workflow	Durable asynchronous state	Operational engine/history/queue complexity	Route is provably local/atomic or simpler scheduler meets timers/recovery	Platform
Regional availability	Fenced single-writer ownership per ledger partition	Failover may pause writes; operational complexity	Proven multi-region consistency model maintains invariants and recovery objectives	Ledger/SRE

Decision	Chosen posture / benefit	Cost and failure exposure	Revisit signal	Owner
Build versus buy	Build financial authority; buy specialist custody/node capabilities	Vendor dependency plus integration; owned core maintenance	Provider gaps or scale make internal specialization justified	Engineering/security/procurement

Table 54. Operating burden and reversibility companion to the tradeoff ledger.

Decision	Explicit operating burden	Reversibility / exit mechanism
Operational truth	Journal/schema governance, partition ownership, reconciliation, correction, close, recovery and finance/SRE on-call	Export immutable history and migrate opening balances only through controlled parallel reconciliation; no chain switch alone replaces obligations
Customer instrument	Treasury, liquidity, disclosures, support and provider incidents remain behind the bank channel	Disable settlement route while retaining participant API and domestic product; unwind institutional inventory
Network model	Finality policy, node/provider provenance, fees, upgrades, reorg drills, chain analytics and incident response	Stop new submission, observe/settle in-flight work, move future route version to another approved network; assets may require redemption/transfer
Issuer	Reserve/redemption monitoring, admin-power changes, eligibility, concentration, statements and issuer incident escalation	Kill new use, redeem or transfer inventory, close breaks and introduce a separately approved issuer/asset version
Custody/nodes	Key ceremonies/policy, provider due diligence, statement export, RPC integrity, recovery and exit drills	Transfer assets/roles under quorum, preserve signed evidence, switch endpoints only after observation parity and recovery proof
Generality	Adapter capability registry, route policy, migrations and regression matrix even when only one route runs	Add one dimension behind a new ADR; remove it with a scoped kill switch and inventory/case closeout
Liquidity	Forecasting, prefund monitoring, rebalance, limits, stress tests, valuation and concentration reporting	Reduce/close inventory and route limits; adopt credit or just-in-time funding only under new legal/risk approval and journals
Customer speed	Threshold telemetry, interval-risk measurement, disclosure/status policy, loss allocation and complaints	Tighten completion gates immediately for new work; loosening requires versioned policy and explicit risk acceptance
Provider abstraction	Capability-specific adapter tests, version mapping, inquiry, error taxonomy, provider certification and provenance	Replace adapter by route version after conformance, dual observation and in-flight operation closure; contract is stable but migration is not free
Workflow	History/version management, timers, worker capacity, replay, cases, schema evolution and engine disaster recovery	Export domain events and rebuild projection; replace engine only after deterministic replay and active-work migration
Regional availability	Partition fencing, failover runbooks, replication/backup, pause semantics and recovery drills	Move ownership through fenced failover; adopt another consistency model only after invariant and split-brain testing

Decision	Explicit operating burden	Reversibility / exit mechanism
Build versus buy	Core engineering/SRE plus vendor due diligence, contract/SLA, integration, incident coordination, data export and procurement	Specialist provider can be replaced through tested export/transfer; owned financial authority migrates only through reconciled cutover

E. Detailed state transitions

This appendix expands table 19. Routes may omit inapplicable states, but may not bypass their equivalent guards. Every transition appends an event with previous/new state, expected/new aggregate version, business operation ID where applicable, actor/workload, time, reason, evidence references, and policy/configuration versions.

Table 55. Full canonical transition matrix.

From	To	Guard/evidence	Side effect and timeout
New	RECEIVED	Participant authenticated/autho- rized; key+hash accepted; intent committed	Publish validation command; ingress budget
RECEIVED	VALIDATED	Schema/contract, participant, amount/asset/corridor and identity prerequisites valid	Request policy facts/decisions
RECEIVED	REJECTED	Deterministic validation or autho- rization failure	Publish reason-safe participant status; terminal
VALIDATED	POLICY_PENDING	Required policy set and versions recorded	Start decision SLA; no financial effect
POLICY_PENDING	REJECTED	Binding reject with evidence	Release any preauthorization if present; terminal
POLICY_PENDING	MANUAL_REVIEW	Hold/enhanced-review obligation or conflicting facts	Open assigned case and pre- serve restricted posture
POLICY_PENDING	FUNDS_RESERVED	All decisions allow; unique reser- vation operation and obligations committed	Request quote; reservation- aging timer
FUNDS_RESERVED	QUOTED	Required inventory/capacity present; quote and route stored	Start quote-expiry timer
FUNDS_RESERVED	FAILED_RETRYABLE	Quote/liquidity transient with no external effect	Bounded re-quote/rebalance; reservation aging policy
QUOTED	LEDGER_POSTED	Quote unexpired; capacity and policy rechecked; authorization journal and outbox committed atomically	Dispatch signing or route com- mand from outbox
LEDGER_POSTED	SIGNING	Route requires chain; simulation and signing authorization pass	Send canonical bytes/hash using sign operation ID
LEDGER_POSTED	Route-specific settle- ment/payout	Non-chain route; authorized provider operation recorded	Execute through route adapter
SIGNING	SUBMITTED	Signature and transaction identity durably recorded; submission evidence adequate	Submit signed bytes once; ambi- guity timer
SIGNING	FAILED_RETRYABLE	Signer proves no signature/effect and outage/refusal is retryable	Retry same canonical operation within policy
SIGNING	MANUAL_REVIEW	Signature existence/authorization uncertain or security concern	Disable scope as required; case/security response
SUBMITTED	CONFIRMING	Independent node observes exact transaction and expected effects	Poll/subscribe through finality policy
SUBMITTED	MANUAL_REVIEW	Inquiry cannot prove absence or presence within ambiguity policy	No resubmission; exposure/case timer
CONFIRMING	ECONOMICALLY_FINAL	Canonical lineage, independent sources, threshold and network health pass	Apply chain-final journal classi- fication; evaluate payout

From	To	Guard/evidence	Side effect and timeout
CONFIRMING	FAILED_RETRYABLE	Transaction proven reverted/absent and retry policy authorizes correction	New authorized operation or compensation
CONFIRMING	MANUAL_REVIEW	Reorg/delay/conflicting observation exceeds policy	Contain affected route/value; case
ECONOMICALLY_FINAL	PAYOUT_PENDING	Payout obligation, policy approval, capacity, and unique payout operation are committed	Dispatch payout request with recorded operation ID
ECONOMICALLY_FINAL	RECIPIENT_CREDITED	Token-to-token route has sufficient destination ownership evidence	Publish eligible participant status
PAYOUT_PENDING	RECIPIENT_CREDITED	Contractually sufficient destination-credit evidence	Set customer completion gate if product policy allows
PAYOUT_PENDING	MANUAL_REVIEW	Timeout/terminal error/conflicting payout evidence	Retain payable; inquiry and corridor remedy
RECIPIENT_CREDITED	RECONCILING	All route effects have comparison coverage	Launch/associate reconciliation run
RECONCILING	RECONCILED	All required sources match within approved tolerance or disposition	Record close evidence; terminal route state
RECONCILING	MANUAL_REVIEW	Material, aged, or integrity break	Open break/case; narrow kill switch if needed
FAILED_RETRYABLE	Prior safe execution state	Route-specific retry safety proven; attempts, time, authorization, nonce, inventory, and policy valid	Reissue the same idempotent operation or create a separately approved new attempt
Any recoverable state	COMPENSATING	Original effects known; compensation plan/legal/ledger authority approved	Create separate idempotent recovery operations
COMPENSATING	COMPENSATED	Every recovery effect and journal proven/reconciled as required	Participant notification and terminal evidence
Any nonrecoverable state	FAILED_FINAL	Permanent result; no outstanding effect or recovery obligation	Terminal notification; retain audit and reconciliation duties

F. Financial and flow exactness

The complete illustrative chart of accounts is table 6; exact journals are tables 7 to 14 and 16; the successful closeout is table 15; invariants are table 17; and external-source coverage is table 18. Those tables remain in the main argument because ledger authority cannot be delegated to an appendix.

The complete normal/failure flow catalog is tables 21 and 22; the concern records are table 42; and the local/external success-failure quadrants are table 27. Together they form a release consistency check: a route may add a provider-specific branch, but every mutation must map to an operation ID, ledger posture, state guard, ambiguity rule, customer status, audit evidence, and reconciliation source.

F.1. Illustrative participant intent and event

Illustrative participant intent. Fields are schematic, not a production API.

```
POST /v1/payment-intents
Idempotency-Key: participant-generated-opaque-key
{
  "contractVersion": "1.0",
  "source": {"participantAccountRef": "opaque", "currency": "USD"},
  "amount": "100.00",
  "recipient": {"participantAliasRef": "opaque"},
  "destinationCurrency": "INR",
  "product": "INVISIBLE_CROSS_BORDER",
  "termsAcceptanceRef": "opaque"
}
```

The platform derives asset, chain, contract, custody address, FX/payout providers, and finality policy from an approved route. A participant cannot inject them through the generic intent.

Illustrative durable payment event envelope.

```
{
  "eventId": "01...",
  "type": "payment.transaction-observed.v1",
  "aggregateId": "PAY-123",
  "aggregateVersion": 11,
  "businessOperationId": "PAY-123/CHAIN-SUBMIT/1",
  "causationId": "...",
  "correlationId": "PAY-123",
  "occurredAt": "2026-07-13T17:00:00Z",
  "producer": "chain-observer",
  "policyVersion": "finality-policy-7",
  "configurationVersion": "route-12",
  "evidenceRefs": ["evidence://opaque/..."]
}
```

G. Evaluation and decision artifacts

Technology decisions are summarized in tables 43, 44 and 47; asset/control options and diligence are tables 48 and 50; and major architecture costs/change triggers are table 53. A production decision record must add evaluated product version/release/commit, deployment model, contract terms, benchmark plan/results, security assessment, support ownership, total operating cost, recovery/export test, and sign-offs. A numeric score may supplement but never override a mandatory constraint.

G.1. Required organization-specific validation questions

Table 56. Questions that cannot be answered from public evidence.

Domain	Question	Closure / accountable roles
Current platform	Which contracts and systems currently own participant intent, alias, account authorization, risk, settlement status, notification, case, reconciliation, and audit?	Internal discovery; product/platform/accounting
Product	Is stablecoin visible or entirely institutional; who is sender/recipient; what status, disclosure, error and remedy promises apply?	Product/legal/participant agreement
Legal entities/roles	Who issues, transmits, custodies, converts, pays out, guarantees, screens, reports, supports and absorbs loss in each jurisdiction?	Counsel/compliance/entity RACI
Asset	Who is issuer; what are reserve, redemption, eligibility, insolvency and admin-control terms; is ZelleUSD relevant and when?	Asset ADR; treasury/legal/compliance
Network/contract	Which network/contract versions; what finality, fees, governance, upgrade, privacy, node and incident risks are approved?	Network ADR; chain risk/security
Custody/signing	Who owns keys/addresses; what quorum, value bands, recovery, export, incident and exit model applies?	Custody ADR; security/treasury
FX/liquidity	Who provides quotes and inventory; what prefunding, credit, slippage, concentration, stress and rebalance limits apply?	Treasury/risk/finance
Payout	What proves recipient credit; is retry idempotent; what cancellation/return/remedy and statement evidence exists?	Payout ADR; operations/legal
Finality	What evidence separately establishes blockchain, legal, customer and accounting finality per route?	Four-owner policy template
Workload/reliability	Peak shape, value bands, latency/status expectations, availability, RPO/RTO, backlog, dispute and retention windows?	Product/SRE/accounting evidence
Accounting	What is the legal/accounting system of record; chart, valuation, revenue, loss, suspense, close and auditor evidence?	Controller-approved accounting design
Data/privacy	Which personal/transaction data may cross institutions, providers, chains and regions; retention/deletion/access rules?	Privacy/security/data governance
Operations	Staffing, on-call, case SLAs, four-eyes actions, limit/kill-switch ownership, restart authority, drills and evidence retention?	Operations/SRE/security

H. Evidence taxonomy and summary

Table 57: Evidence badges map exactly to the approved taxonomy.

Badge	Class	Permitted inference
PUBLIC	Validated from public source	Exact fact established by authoritative public source; no private topology inference
CODE	Validated from source code	Official repository behavior at cited release/commit; no deployment/suitability inference
DOC	Validated from documentation	Responsible organization's documented behavior; no production-use inference
INFERRED	Inferred architecture	Logical decomposition used to reason; explicitly not deployed topology
PROPOSED	Proposed target architecture	Recommendation and rationale owned by this paper
ASSUMPTION	Operational assumption	Planning posture requiring measurement, owner and approval
UNKNOWN	Unknown / needs validation	Material fact not established and retained as a gate

Table 58: Evidence summary by class. The complete claim-level ledger is delivered as a separate release companion.

Evidence class	Claims
Validated from public source	22
Validated from source code	5
Validated from documentation	32
Inferred architecture	3
Proposed target architecture	7
Operational assumption	2
Unknown / needs validation	3
Total	74

The evidence supports five durable conclusions: Zelle’s institution-mediated public distribution paths and participant-side vendor interfaces do not disclose private EWS topology or establish a general-purpose independent-application API; announced international and future stablecoin direction does not establish a production asset or route design; an internal ledger and evidence-gated state machine are proposed operational authority rather than asserted current behavior; Java is well suited to the enterprise control plane without determining chain-native implementations; and issuer, custody, network, legal-finality, liquidity, and provider choices remain explicit diligence gates. The companion claim ledger retains confidence, caveat, source mapping, and paper-location detail for all 74 claims.

I. Figure descriptions and visual reading order

Canonical figures are original SVGs with title and description elements except for the documented API system-design PNG. That complete raster source is retained byte-for-byte and the paper uses LaTeX crops of it rather than edited copies. Colors repeat semantics and are paired with labels, badges, border styles, lane labels, or shapes. The longer descriptions below carry relationships that may be difficult to infer from the visual alone.

1. **Context boundaries (figure 1).** The left PUBLIC boundary contains customers, participating institutions, documented network/alias roles and domestic rails. The right PROPOSED boundary retains those participants and adds a control plane. Custody, issuer, chain, FX and payout boxes remain external; UNKNOWN badges mark organization-specific selections.
2. **Layered architecture (figure 2).** Requests flow down from participants through intent/policy/orchestration to ledger/workflow authority, then through adapters and signing to external rails. Dashed observations and dotted reconciliation flow upward without granting providers mutation authority.
3. **Trust/deployment (figure 3).** Institution, operator, signer/custody, node, public network and provider boundaries have different administrators. Submission and observation use separable paths; operator action is a governed boundary.
4. **Components (figure 4).** Synchronous acceptance ends at durable intent. Commands coordinate policy, ledger, treasury and adapters. Observations enter workflow; reconciliation compares source records.
5. **API-centric system design (figure 5).** Institutional clients cross an authenticated participant boundary where requests are authorized, canonicalized, deduplicated, and durably accepted as payment intents. Ledger and workflow control financial effects; an outbox publishes work to provider-neutral adapters. Independent observation and reconciliation advance evidence-gated state, while queries, events, and signed webhooks expose monotonic status. Blue public facts and role signals, purple vendor examples, green proposals, and amber unknowns do not assert private EWS topology.
6. **Ledger (figure 6).** Internal accounts represent participant, treasury, in-transit, issuer, FX, payout, fee and suspense obligations. External statements reconcile to specific accounts rather than replacing them.
7. **Happy state (figure 7).** States progress only when entry evidence satisfies a guard. Economic chain finality, recipient credit and reconciliation are visibly separate.
8. **Exception state (figure 8).** Rejection, retryable/final failure, manual review and compensation branch from multiple lifecycle points. No generic failure or automatic loop exists.
9. **Cross-border sequence (figure 9).** Participant acceptance and policy precede reservation, signing and submission. Independent observation precedes payout authorization; notification and reconciliation follow their own evidence.
10. **Ambiguous submission (figure 10).** A lost response creates ambiguity. Independent inquiry and stable transaction identity determine whether the original effect is absent, reverted, still uncertain, or complete; another attempt requires route-specific proof of retry safety. A duplicate participant request returns the original intent.
11. **Payout failure (figure 11).** Chain economic finality is already achieved, so recovery retains a payout payable and chooses inquiry, retry, alternative payout, refund/redemption or compensation; it does not reverse the chain.
12. **Transaction boundaries (figure 12).** Local stores have atomic boundaries. Broker, signer, chain and payout edges are non-atomic gaps handled by operation identity, inquiry and reconciliation.
13. **Outbox/inbox (figure 13).** Producer state/outbox and consumer inbox/effect each commit locally. At-least-once broker delivery is safe because event/effect identities are unique and replay is governed.
14. **Containment (figure 14).** Scoped switches prevent new exposure while leaving inquiry, observation, reconciliation and support available. Domestic routes remain isolated from digital-asset failure.
15. **Four finalities (figure 15).** Chain, legal, customer and accounting lanes have different evidence and owners. Their order can vary; no vertical line declares universal completion.

16. **Key authority (figure 16).** Application authorization and signer policy are independent. Exact bytes and transaction fields enter HSM/MPC/custody; keys never enter application memory. Independent observation checks the signed result.
17. **Compliance overlay (figure 17).** Compliance decisions recur from enrollment through reconciliation. Holds and case obligations can occur before or after submission without rewriting history.
18. **Java boundary and roadmap (figure 18).** The top panel keeps domain/control modules in Java and specialist components behind language-neutral APIs. The bottom panel shows seven evidence gates; arrows denote earned promotion, not elapsed time.

J. Glossary

Table 59. Cross-domain vocabulary used by this paper.

Term	Definition in this architecture
ABI (application binary interface)	Contract-call and event encoding boundary used by EVM tooling; conformance tests must prove exact bytes rather than trust a generated wrapper.
ACL (access-control list)	Explicit set of principals and permitted actions on a broker, service or resource; financial authorization also requires purpose, object, value and state constraints.
ACH (Automated Clearing House)	Batch-oriented United States bank-account payment network used here only as a documented domestic-rail option; its rules and reversal model differ from real-time rails and blockchains.
ACID (atomicity, consistency, isolation, durability)	Database transaction properties relied on for one local journal, operation record, versions and outbox; they do not extend to a broker, signer, chain, FX or payout provider.
ADR (architecture decision record)	Versioned decision artifact stating context, options, evidence, decision, owners, consequences, change triggers and supersession.
AML (anti-money laundering)	Risk-based policies, monitoring, investigation, reporting and controls applied according to the organization's role, jurisdiction, counterparties, asset and product.
API (application programming interface)	Versioned machine contract for participant or component commands, queries and evidence; it does not by itself define authority or atomicity.
AWS (Amazon Web Services)	Cloud provider whose official idempotent-API guidance is cited as technical documentation; the citation does not select AWS for deployment.
Accounting finality	The point at which required journals, valuations, external reconciliations, break dispositions and close controls satisfy the accountable accounting policy. Later corrections are new entries.
Available balance	Settled, policy-eligible balance minus active reservations, restrictions and other amounts unavailable for a new authorized operation.
BIS (Bank for International Settlements)	International institution whose policy research is cited for tokenized-money and settlement distinctions; its analysis is not a Zelle requirement.
Blockchain finality	A chain-risk policy determination that canonical transaction evidence meets the selected network, route, value and health threshold; not automatically legal or accounting finality.
Bridge	A mechanism that transfers or represents value across networks, often through lock/mint or burn/mint and additional validators/attesters. No generic bridge is in the initial pilot.
Business operation ID	Stable identifier for one intended financial effect, reused across delivery retries and linked to authorization, provider records, journals and reconciliation.
Chain reorganization (re-org)	Replacement of previously observed canonical blocks by another canonical history, potentially removing or changing a transaction's observed status.
CI/CD (continuous integration and continuous delivery/deployment)	Automated build, test, provenance, approval and release path; production promotion remains evidence-gated rather than automatic.
CCTP (Cross-Chain Transfer Protocol)	Circle's issuer-mediated USDC burn, attestation and mint mechanism, used only as a documented interoperability example rather than a selected generic bridge.
CFT (countering the financing of terrorism)	Controls paired with AML in risk-based frameworks; exact obligations depend on role, jurisdiction, counterparty, product and transaction.
CNCF (Cloud Native Computing Foundation)	Open-source foundation governing projects including OpenTelemetry; foundation governance does not by itself prove product fitness or support.

Term	Definition in this architecture
Confirmation	Evidence that a transaction appears in a block or provider record at a particular depth/status; an input to, not a synonym for, finality.
Custody	Safekeeping and controlled use of digital-asset keys/assets under defined legal, security, authorization, recovery and reporting obligations.
CVE (Common Vulnerabilities and Exposures)	Public identifier for a disclosed security vulnerability used in dependency, patch and incident processes; a CVE count alone is not a maturity score.
Customer-visible completion	Product-policy decision that permits the participant/customer experience to present a payment as completed based on required evidence and any explicitly absorbed interval risk.
Double entry	Accounting method in which every journal has equal debit and credit totals in the relevant unit, preserving a testable balance invariant.
DB (database)	Persistent data system; workflow DB and ledger DB mean separately owned transactional authorities, not necessarily distinct products in the pilot.
DLT (distributed ledger technology)	Shared ledger/protocol family that includes public and permissioned networks; the term says nothing by itself about legal claim, finality, privacy or operating fitness.
DX (developer experience)	Practical quality of APIs, tooling, local tests, diagnostics, documentation and failure visibility for engineers who build and operate a component.
Early Warning Services (EWS)	Owner and operator of Zelle according to the cited public record; the paper does not infer its private topology or vendors.
EIP / ERC (Ethereum Improvement Proposal / Ethereum Request for Comments)	Numbered Ethereum standards process and application-interface conventions; ERC-20 supplies token calls/events but not issuer economics or legal rights.
EVM (Ethereum Virtual Machine)	Execution environment and bytecode/contract model used by Ethereum-compatible networks; compatible execution does not make networks or assets equivalent.
Economic finality	This paper's chain-risk gate represented by ECONOMICALLY_FINAL: evidence and policy make further reorganization sufficiently remote for the approved next action, without making a legal conclusion.
FATF (Financial Action Task Force)	Intergovernmental standard-setting body whose virtual-asset guidance informs role- and jurisdiction-specific AML/CFT decisions; it is not a product implementation specification.
FIPS (Federal Information Processing Standard)	United States government standard family; FIPS 140-3 defines scoped cryptographic-module security requirements, not end-to-end custody governance.
Finality threshold	Versioned, route-specific evidence rule for a finality decision, including sources, network conditions, value bands and escalation.
Foreign exchange (FX)	Conversion and associated quote, fill, settlement, exposure and reconciliation between currencies/assets.
General ledger (GL)	Organization's authoritative accounting book to which the payment subledger reconciles and exports controlled totals and adjustments.
gRPC	Remote-procedure-call framework often used by node or provider APIs; transport choice does not decide financial authority or retry safety.
Hardware security module (HSM)	Tamper-resistant cryptographic module used to generate/protect/use keys under controlled interfaces and policy.
HTTP / JSON-RPC	Common request and remote-procedure-call transports used by participant and blockchain APIs; a successful response is evidence from one boundary, not proof of end-to-end effect.

Term	Definition in this architecture
Idempotency	Property that repeating the same authorized request/operation identity produces the same business result without duplicating its effect; conflicting payload reuse is rejected.
Issuer	Legal entity that creates and governs an asset and its terms, including applicable reserve, redemption and administrative controls.
JDK / JVM (Java Development Kit / Java Virtual Machine)	Java toolchain and managed runtime. The evaluated control plane uses them without requiring node, contract or custody implementations to be Java.
LTS (long-term support)	Vendor or community maintenance designation for a release line. Support terms, patch horizon and provider must be verified rather than inferred from the acronym.
Ledger	Internal authoritative double-entry record of the platform's business obligations, reservations, assets, liabilities, income, expenses and corrections.
Legal settlement finality	Counsel/product-policy determination of when an obligation is legally discharged or irrevocable under applicable instruments, agreements, roles and law. This paper provides no opinion.
Multiparty computation (MPC)	Cryptographic technique that can distribute signing authority across parties/devices without reconstructing a full key in one place; governance remains necessary.
NIST (National Institute of Standards and Technology)	United States standards organization cited for cryptographic key management, secure development and module requirements.
Node	Software/service that participates in or exposes a blockchain network's transactions, blocks, state and protocol. A node response is one evidence source.
On/off ramp	Service/process converting between fiat/bank value and digital assets, potentially involving issuer redemption, exchange, treasury and payout.
OFAC (Office of Foreign Assets Control)	United States Treasury office administering economic sanctions; applicability and screening decisions require current role, party, asset and jurisdiction facts.
Outbox	Record committed atomically with local business state and later published to messaging, closing the local database-to-broker commit gap without making publication synchronous.
Payout rail	Destination mechanism/provider that credits recipient fiat or account value and returns evidence under its own contract and failure model.
Posting	Immutable debit or credit line belonging to a balanced journal and identified by account, unit, operation, time and audit metadata.
PII (personally identifiable information)	Data that identifies or can be linked to a person; the design minimizes it, keeps it off public chains, and governs purpose, region, access and retention.
PITR (point-in-time recovery)	Database recovery capability to restore state to an approved time; restored state must still reconcile with immutable external effects that occurred afterward.
PDF (Portable Document Format)	Final publication format; LaTeX, CSV, SVG and BibTeX remain canonical and the PDF is a reproducible derivative.
Prefunding	Holding approved inventory/cash before payment execution so settlement does not depend on unsecured real-time credit.
PYUSD	Paxos-issued PayPal USD stablecoin, considered only as an external-asset diligence example under its current issuer terms and controls.
RACI (responsible, accountable, consulted, informed)	Ownership matrix used to assign decisions, incidents, evidence and approvals; it does not replace technical authorization.
RBAC / ABAC (role-based / attribute-based access control)	Authorization models using roles or contextual attributes. Financial operations also require purpose, value, quorum and exact-object constraints.

Term	Definition in this architecture
RDBMS (relational database management system)	Database class providing relational schema, constraints, transactions and SQL; the paper recommends the class conditionally rather than a procurement choice.
Reconciliation	Versioned comparison of internal records with independent external evidence, including tolerance, break, owner, aging and disposition.
Reservation	Ledger/control record that reduces available capacity for an intent without claiming final external movement.
RPO / RTO (recovery point / recovery time objective)	Approved maximum data-loss interval and restoration time targets. Exact values are organizational assumptions to test, not public Zelle facts.
RPC (remote procedure call)	Request/response interface that invokes an operation on another process; timeouts and success responses must be mapped to authority and inquiry semantics.
RTP (Real-Time Payments)	The Clearing House real-time payment network; cited public material describes its own interbank settlement properties, not every participant's customer policy.
SBOM (software bill of materials)	Machine-readable inventory of build dependencies used for vulnerability, license and provenance review.
SDK (software development kit)	Library/tooling that helps construct protocol calls or decode results; it is not automatically a node, signer, ledger or reliability boundary.
SLA (service-level agreement)	Contracted service commitment and remedies; it must be distinguished from measured objectives, end-to-end product behavior and legal finality.
SQL (Structured Query Language)	Relational data definition/query language used for ledger constraints, reconciliation and inspection; correctness still depends on schema and transaction design.
SRE (site reliability engineering)	Engineering discipline owning reliability objectives, telemetry, capacity, incident response and recovery together with domain owners.
Stablecoin	Token intended to maintain a reference value under issuer, reserve/redemption, contract, network and legal terms; interface similarity does not imply economic equivalence.
Suspense	Governed temporary account for an unresolved difference; it requires an owner, evidence, aging and approved resolution, not use as a balancing dump.
Tokenized deposit	Digital representation of a commercial-bank deposit claim under the issuing bank/network arrangement, distinct from a nonbank-issued stablecoin.
Zelle alias / token	In cited participant-side documentation, an enrolled email address or mobile identifier; it is not a stablecoin, blockchain token, tokenized deposit, or cryptographic asset.
Transactional inbox	Consumer-owned record of processed event IDs/effects committed with the local effect to prevent duplicate application under at-least-once delivery.
Travel Rule	Risk-based regulatory information-sharing requirements that may apply to qualifying transfers depending on jurisdiction, role, counterparty and asset.
UI (user interface)	Customer or operator interaction surface; it presents derived status and governed actions but does not directly mutate ledger truth.
URI (Uniform Resource Identifier)	Identifier used for web and PDF links; the release verifies embedded URI schemes and spot-checks external destinations.
USDC	Circle-issued dollar stablecoin used as a documented external-asset example; selection still requires issuer, redemption, custody, network, liquidity and legal diligence.
UTXO (unspent transaction output)	Ledger model in which transactions consume and create uniquely referenced outputs; Corda and Bitcoin-related uses differ and do not imply EVM account semantics.

Term	Definition in this architecture
Wallet	Software/service and associated keys/address capable of controlling digital assets; the initial product does not expose one to consumers.
